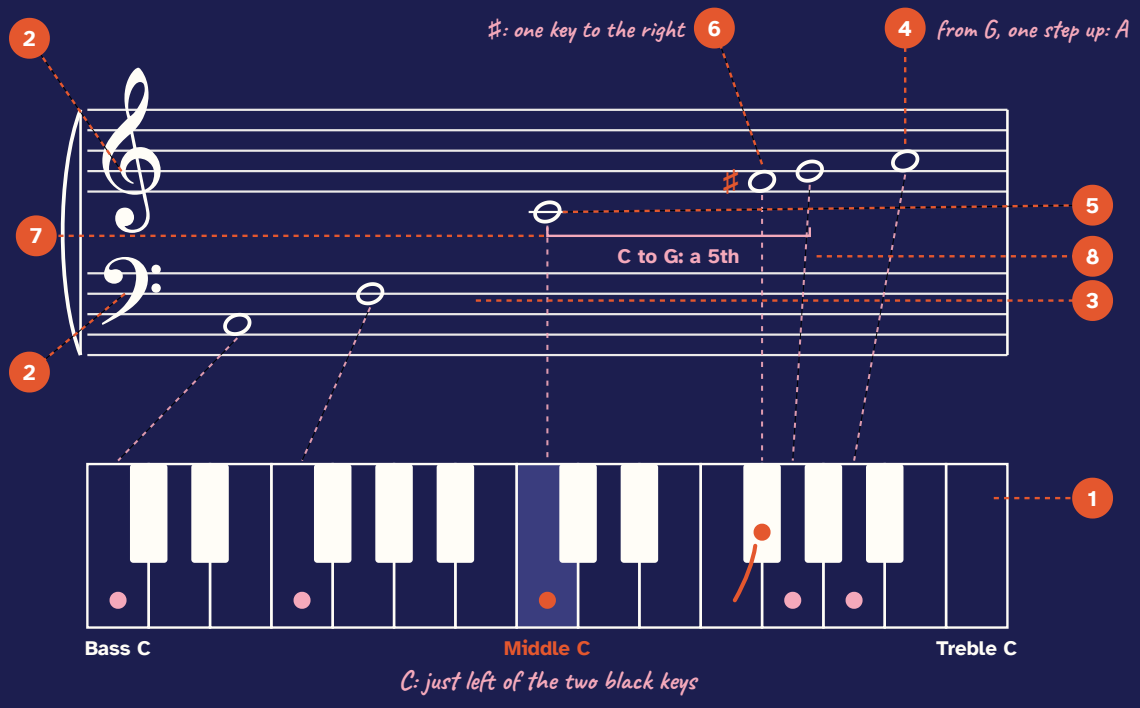
PART I · WHICH NOTE

# Pitch

High or low? That is the first question every note answers. Start with the keys under your fingers, find where they live on the staff, see how sharps and flats move them, then measure how far apart two notes are. Everything in this part is in this one picture.





1 **The keyboard** p. 22
two and three black keys; C is left of the two

2 **The staff and the clefs** p. 27
the treble clef names G, the bass clef F

3 **The staff and the keyboard** p. 29
every line and space is one white key

4 **Reading the notes** p. 32
start from a clef note, then step or skip

5 **Ledger lines** p. 36
step out from the staff; Middle C has its own

6 **Sharps, flats and naturals** p. 39
a sharp moves one key right, a flat one key left

7 **Counting intervals** p. 42
count the notes, both ends: C to G is a 5th

8 **Major, minor and perfect** p. 45
count the keys between: 6 make a perfect 5th

★ **Tips: when a note reads wrong** p. 48

PART I · PITCH · 1

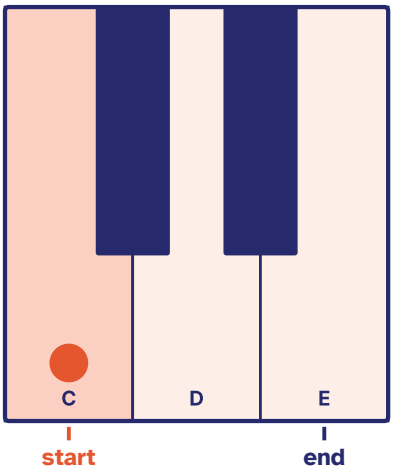
# The keyboard

Left is **low**, right is **high**. The black keys come in groups of **two** and **three**. Each group has a first white key, just to its left: **C** for the two, **F** for the three. Find the group, find its first key, then walk on.

**THE SECRET**

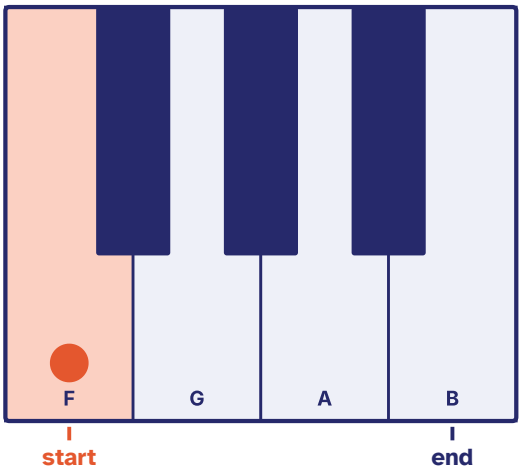
**Two black keys: start on C. Three black keys: start on F. Then walk on.**

**2 black keys: C D E**




*on the piano E and F touch*

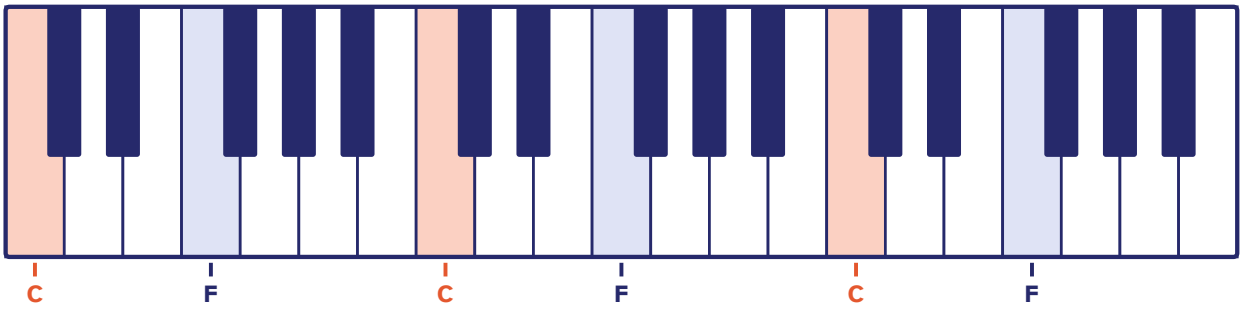
**3 black keys: F G A B**




*Pulled apart: C starts the two, F starts the three · E and B end them*

**THE RULE**

**Find the group, then its first key. Looking for D or E? Find two black keys and start on C. Looking for G, A or B? Find three black keys and start on F. You never have to count from far away.**

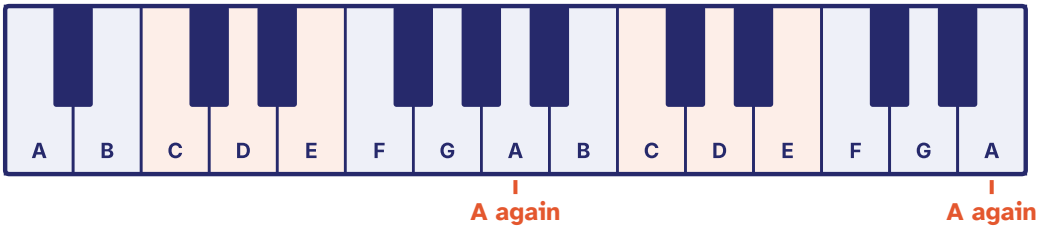



*The pattern repeats all the way along: every two starts on C, every three on F*

PART I · PITCH · 1 · CONTINUED

## Seven letters, round and round

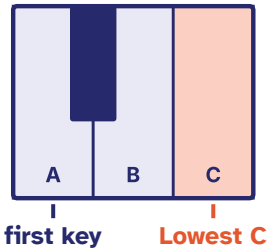
The white keys use only seven letters: **A B C D E F G**. There is no H. After G the next key is **A** again, and the seven letters go round and round, like the days of the week, all the way up the keyboard. **C D E** always sit round the two black keys, **F G A B** round the three. **E–F** and **B–C** have no black key between.




*A B C D E F G, A B C D E F G, A …: after G, back to A*

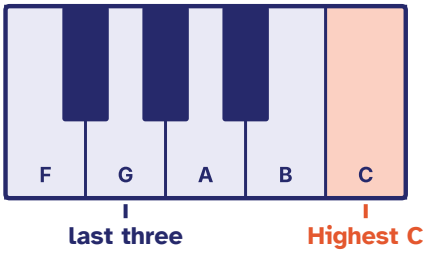
## Where the keyboard starts

The very first key, bottom left, is an **A**. Look at the black keys there: only **one**, on its own. It is a group of three that has been cut off: only its end is left. The end of F G A B is **A B**, so the piano starts on A. After A and B comes the first full group of two, and its first key is **Lowest C**.




## Where it ends

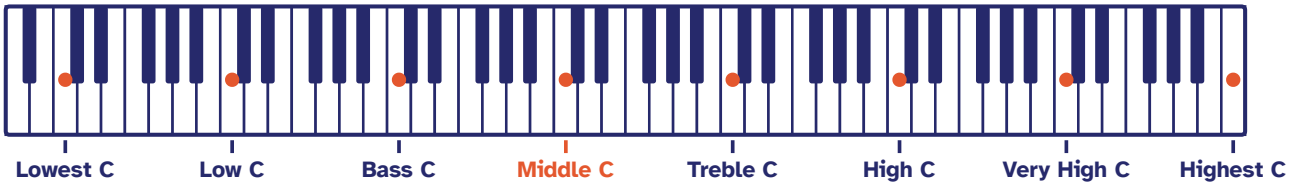
The very last key, top right, has **no black key** after it. Look left: the last group is **three** black keys, F G A B. After B comes C, the first key of a new group of two that was never built. So the last key is a **C**: **Highest C**.




## The eight Cs have names

Count the Cs from the bottom and there are eight. The C nearest the middle is **Middle C**. Either side are **Bass C** and **Treble C**; further out, **Low C** and **High C**. At the ends: **Lowest C** at the bottom; at the top, **Very High C**, then **Highest C**, the very last key. (No Very Low C: the piano starts on A.) From one C to the next is an **octave**: C D E F G A B C, eight notes, both Cs counted.

Theory books and apps number them from the bottom instead: C1 is Lowest C, C4 Middle C, C8 Highest C.

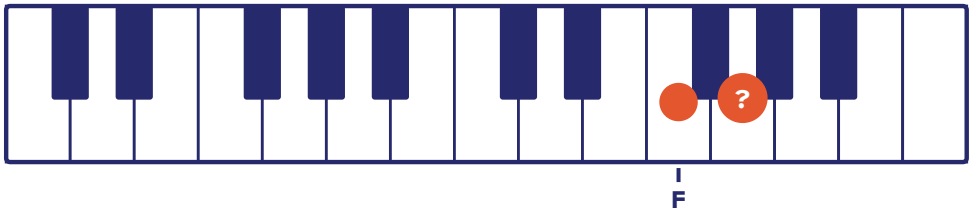



*All 88 keys, eight Cs: the five in the middle, Low C to High C, are the ones you will use most*

PART I · PITCH · 1 · CONTINUED

EXAMPLE 1

## Find G

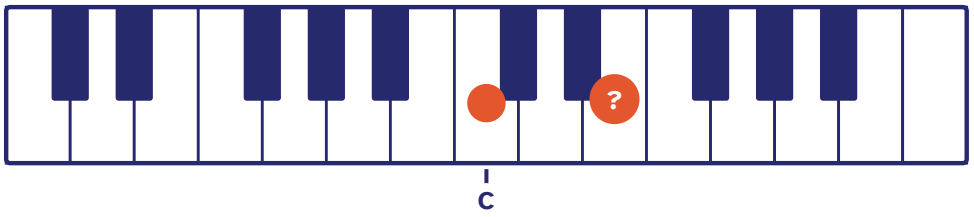



1. **Which family?** G is in F G A B: look for **three** black keys.
2. **Find its first key:** F, just left of the three (the red dot).
3. **Walk on one key.** That is the **?**.

The **?** is **G**: the key after F. No need to start from C and count five.

EXAMPLE 2

## Find E

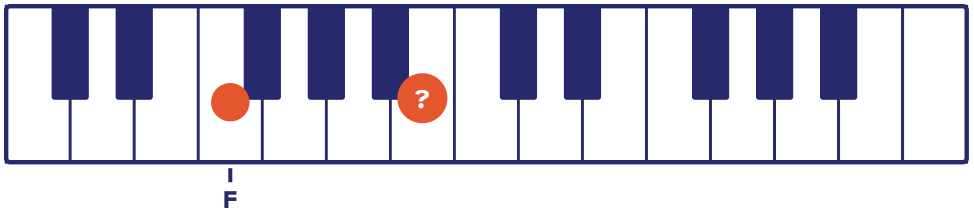



1. **Which family?** E is in C D E: look for **two** black keys.
2. **Find its first key:** C, just left of the two (the red dot).
3. **Walk on:** C, D … the third key is the **?**.

The **?** is **E**: the last of the C D E family, just right of the two black keys.

EXAMPLE 3

## Find B



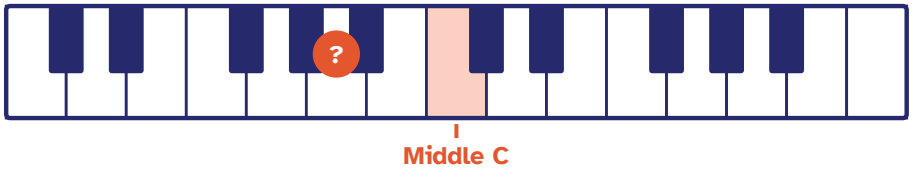

1. **Which family?** B is in F G A B: three black keys.
2. **Start on F** (the red dot).
3. **Walk on:** F, G, A … the last key of the family is the **?**.

The **?** is **B**: the last of F G A B, just right of the three black keys.

PART I · PITCH · 1 · CONTINUED

EXAMPLE 4

### Find the A below Middle C

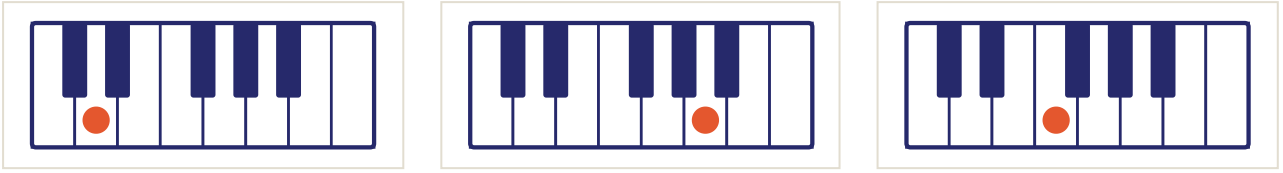



1. **Which family?** A is in F G A B.
2. **From Middle C, look left** for the nearest three black keys, and start on their F.
3. **Walk on:** F, G … the third key is the **?**.

The **?** is **A**: F, G, A. Find the family first, and every key is at most three steps away.

GAME

### Name the key



1 2 3

Find the group, then its first key, then walk on. Name each red key.

Answer: 1 D (C, then D) · 2 A (F, G, A) · 3 F (the first of the three).

GAME

### Which family?

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| D | A | F | E | B | G |

Does each white key live with the **two** black keys (start on C) or the **three** (start on F)?

Answer: 1 two · 2 three · 3 three (the first) · 4 two (the last) · 5 three (the last) · 6 three.

EXAMPLE 5

### Higher or lower?



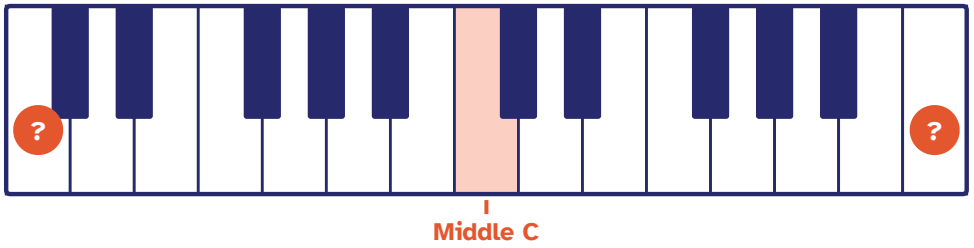

1. **Find the two red keys.**
2. **Which one is further right?**
3. **Right means higher:** play both and listen.

The second note is **higher**: it is to the right. Left is always lower, right always higher.

PART I · PITCH · 1 · CONTINUED

EXAMPLE 6

### Find Bass C and Treble C




1. **Start on Middle C.**
2. **Walk left** to the next pair of black keys. The C just left of it is one **?**.
3. **Walk right** the same way, to the next pair: the other **?**.

Left **?**: **Bass C**. Right **?**: **Treble C**. Each is one octave, eight notes, from Middle C.

TRY IT · 10 SECONDS

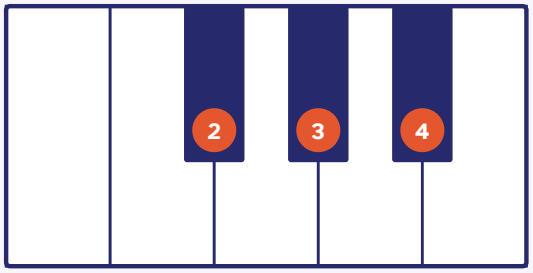



### Play every C, then every F.

Bottom to top, saying the names: Lowest C, Low C, Bass C, Middle C, Treble C, High C, Very High C, Highest C. Then every F: left of the three.

YOU KNOW THIS

### Hot Cross Buns, on three black keys




Right hand, fingers **2 3 4** on the three black keys (thumb is 1, little finger 5). Play from the top: **4 3 2**, 4 3 2, then quickly, twice as fast, **2 2 2 2, 3 3 3 3**, and 4 3 2 to finish. A whole tune, and not one white key.

**Try it.** Find another group of three, higher up, and play it again there.

THINK OF IT The black keys are bumps you can feel in the dark. Close your eyes and slide your hand along: two bumps, and the key to their left is C; three bumps, and the key to their left is F.

TRAP

- × Counting every key from C: find the nearest group and start from its C or F.
- × Mixing up the families: C is left of the **two** black keys, F left of the **three**.

SEE ALSO

The staff and the clefs p. 27 · The staff and the keyboard p. 29

PRACTISE IT

Workbook Pitch 1 · Online: Keyboard Patterns · The Musical Alphabet

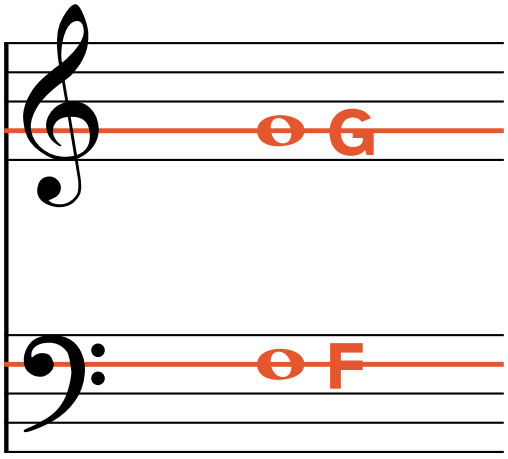
PART I · PITCH · 2

# The staff and the clefs

Music is written on a **staff**: five lines and four spaces, counted from the **bottom**. Higher on the staff means higher on the keyboard. The **clef** at the start is the key to the code: it names one line, and every other note follows from it.

THE SECRET

**The treble clef curls round G; the bass clef's two dots hug F.**






**Treble clef**
curls round line 2: G

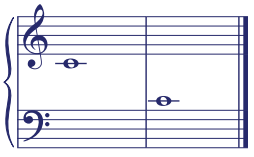


**Bass clef**
its dots hug line 4: F

*Look closely: the treble clef curls round G · the bass clef's dots hug F*

## The grand staff

Piano music uses two staves joined by a brace: the **treble** staff on top, usually for the right hand, and the **bass** staff below, usually for the left. **Middle C** sits between them on a short line of its own, and can be written in either staff.

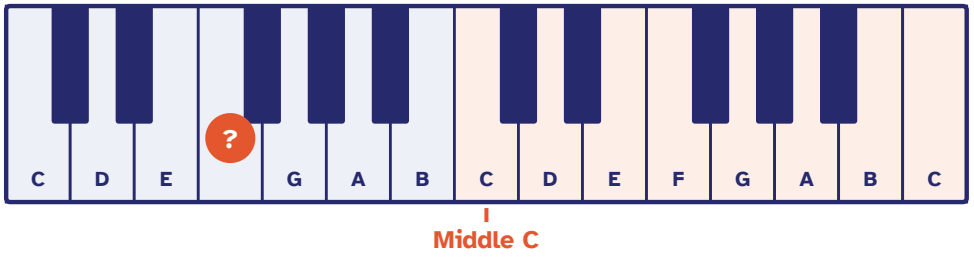


Middle C in the treble staff (left) and in the bass staff (right): the same key.

PART I · PITCH · 2 · CONTINUED

EXAMPLE

## Which staff is this key written on?




1. **Find Middle C.**
2. **Is the ? to its left or right?** Left is lower: the blue half.
3. **Low keys go on the bass staff**, high keys on the treble staff.

The **?** is below Middle C, so it is written on the **bass staff**, usually for the left hand.

GAME

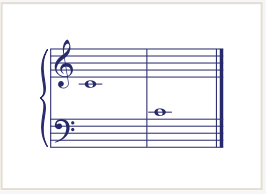
## Treble or bass?

1. the G above Middle C
2. the lowest key on the piano
3. Middle C
4. High C

Which staff would each note be written on?

Answer: 1 treble · 2 bass · 3 either: it sits between the two · 4 treble.

TRY IT · 10 SECONDS
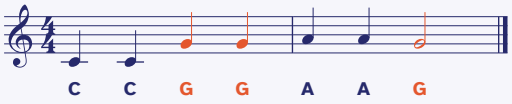


**Point, then play.**

Point to the line the treble clef curls round and play that G, the G just above Middle C. Point to the line between the bass clef's dots and play that F, the F just below Middle C.

YOU KNOW THIS

## Twinkle, Twinkle, Little Star




The treble clef curls round **line 2: G**, the very note Twinkle leaps up to. Read the round head, not the stem: the stem never changes the note.

**Try it.** Point to the G line, then play C C G G A A G.

THINK OF IT A clef is a pair of glasses: the same five lines read as different notes, depending on which pair you put on.

TRAP

× Reading bass-clef notes as if they were treble: check the clef on every line.

SEE ALSO
The staff and the keyboard p. 29 · Reading the notes p. 32

PRACTISE IT
Workbook Pitch 2 · Online: The Staff & the Two Clefs

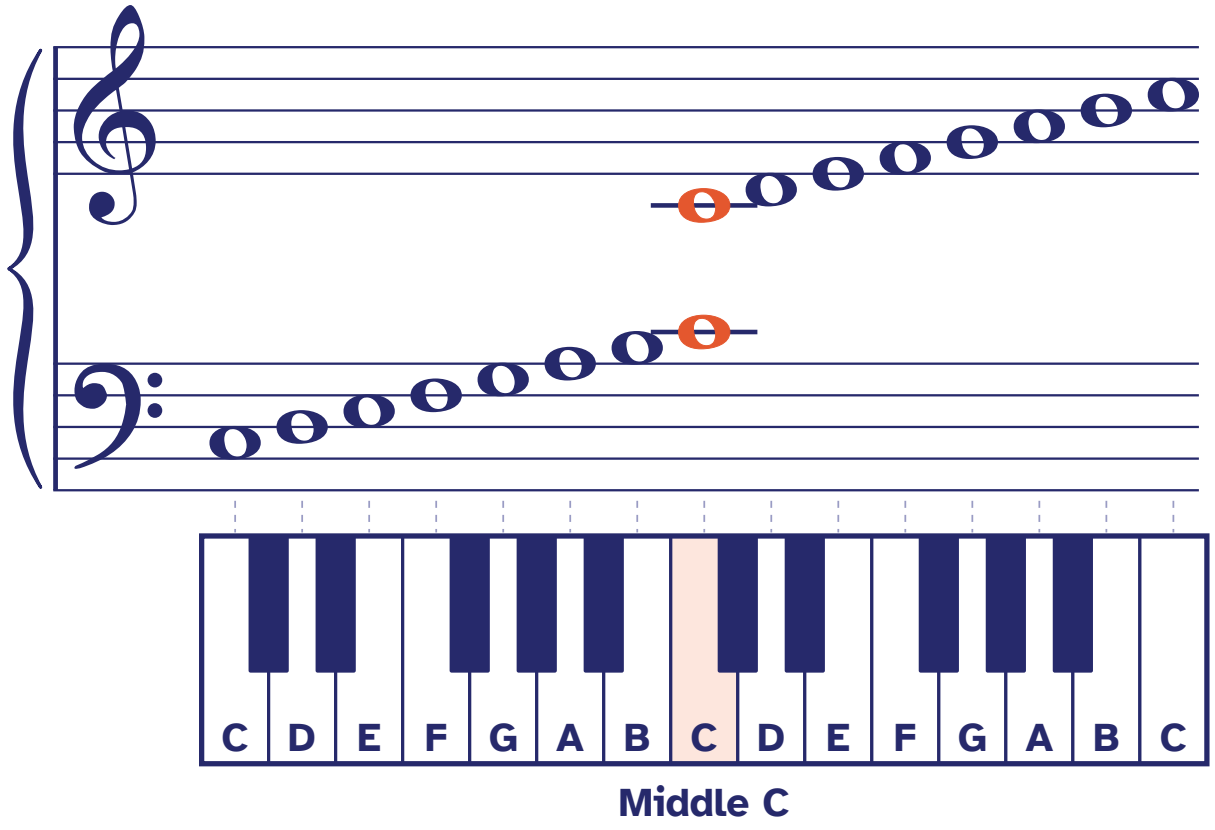
PART I · PITCH · 3

# The staff and the keyboard

Each line and each space of the grand staff is one white key. Climb the staff and you climb the keyboard: here are two octaves, from Bass C to Treble C, with Middle C in red where the two staves meet.

THE SECRET

**Every line and every space is one white key, in order.**




Between the two staves there is only one key: **Middle C**, written twice, below the treble staff or above the bass staff. No other key is missed out.

THE RULE

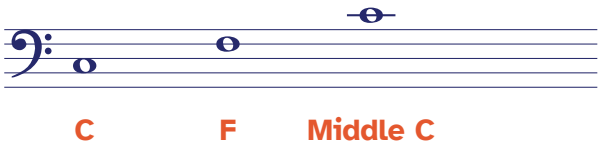
**One step on the staff (line to space, or space to line) is the very next white key. The black keys sit in between; how they are written is chapter 6 (p. 39).**

Up the staff is to the **right** on the keyboard; down the staff is to the **left**. A note on a line and the note in the space just above it are next-door white keys, so no key is ever missed out.

PART I · PITCH · 3 · CONTINUED

EXAMPLE 1

## From the page to the key

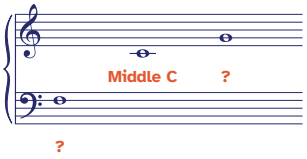



1. **Name the note:** line 4 of the bass staff, between the clef's dots, is F.
2. **On the piano,** find the three black keys just below Middle C.
3. **F is the first key** of that family: play it with the left hand.

The bass-clef F is the **F just below Middle C**: left of the three black keys under it.

EXAMPLE 2

## Mirror notes either side of Middle C

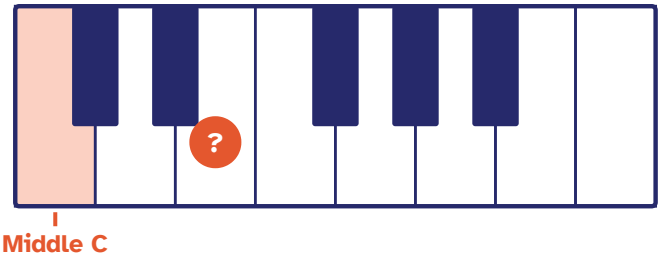



1. **Treble G:** the three black keys just above Middle C; G is the second key, after F.
2. **Bass F:** the three black keys just below Middle C; F is the first key.
3. **Now count each from Middle C:** C D E F G up, C B A G F down. Which number?

Both are **note 5**: treble G and bass F are mirror images, the same distance either side of Middle C.

EXAMPLE 3

## From the key to the page




1. **Find the ? key:** count up from Middle C, C 1, D 2 ...
2. **Do the same on the staff:** Middle C on its ledger line, D in the space under the staff ...
3. **Where do you land?**

The **?** is **E**, note 3, and it sits on the **bottom line** of the treble staff.

PART I · PITCH · 3 · CONTINUED

GAME

**Count the notes**

| 1 | 2 | 3 |
|---|---|---|
| Middle C up to A | Middle C down to E | G up to the next G |

Count the notes, both ends included. The staff climbs the same way: one line or space per note.

Answer: 1 six (C D E F G A) · 2 six (C B A G F E) · 3 eight: an octave.

TRY IT · 10 SECONDS

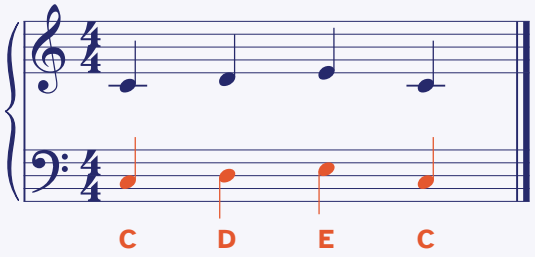



**Play Middle C, then walk up.**

Say each white key as you play it, C D E F G A B C, and picture each one climbing the staff: line, space, line, space.

YOU KNOW THIS

**Frère Jacques, in both hands**




The left hand plays the **same letters**, C D E C, one octave lower, written in the bass clef.

**Try it.** Play it right hand, then left hand, then both together.

THINK OF IT The staff is a ladder laid beside the keyboard: each rung, a line or a space, is the next white key.

TRAP

- × Skipping a key: every line and every space is used, one white key each.
- × Counting from 0: the note you start on is note 1.

SEE ALSO

The staff and the clefs p. 27 · Reading the notes p. 32

PRACTISE IT

Workbook Pitch 3 · Online: The Staff & the Two Clefs · Line and Space Names

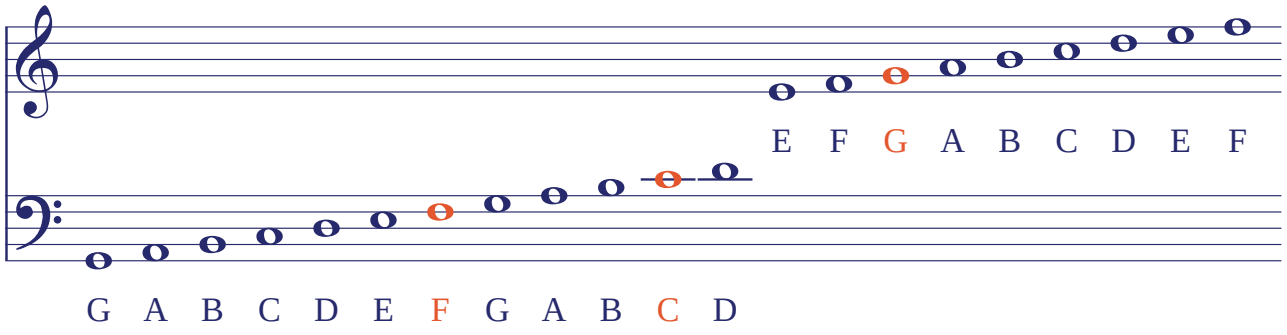
PART I · PITCH · 4

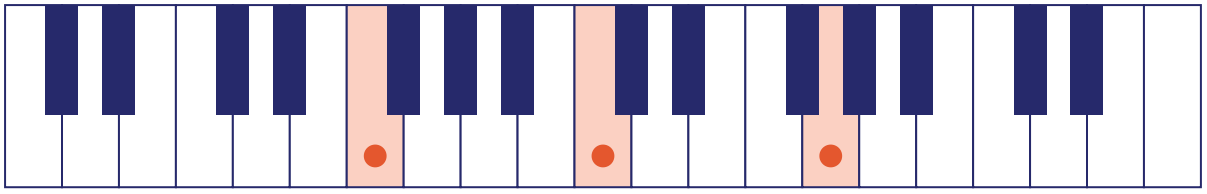
# Reading the notes

Every line and space has one letter. You do not have to learn them all at once: start from a note you always know, then **step** or **skip** to the one you want.

THE SECRET

**Start from a note you know (bass F, Middle C, treble G), then step or skip.**



*Middle C in the middle; treble G and bass F, the clef notes, either side. Every line and space is the next letter, with its own key underneath (the keyboard starts on G, so its first black keys belong to a group of three).*

THE RULE

**Step: to the very next line or space (one letter on). Skip: line to the next line, or space to the next space (two letters on).**

## Line note or space note?

A **line note** has the line running through its middle. A **space note** sits between two lines and touches both. Count both from the bottom: line 1 is the bottom line, space 1 the bottom space.

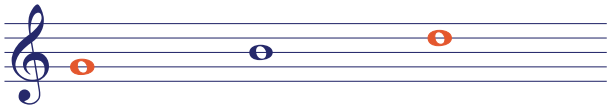
## Words to remember the lines and spaces

| STAFF | LINES (BOTTOM TO TOP) | SPACES (BOTTOM TO TOP) |
|---|---|---|
| Treble | **E G B D F**: *Every Good Boy Deserves Fun* | **F A C E**: they spell *FACE* |
| Bass | **G B D F A**: *Great Big Dragons Fly Away* | **A C E G**: *All Cows Eat Grass* |

PART I · PITCH · 4 · CONTINUED

EXAMPLE 1

### Read a note you are not sure of

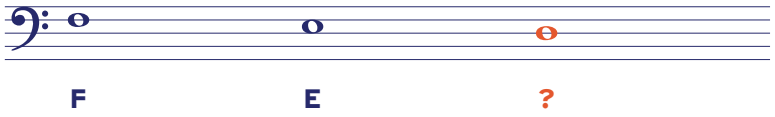


1. **Treble lines** from the bottom: *Every Good Boy Deserves Fun*.
2. **The other red note, on line 4:** the fourth word. Which word, which letter?
3. **Check it** from the clef note: G on line 2, skip to B, skip to …

Line 4 is *Deserves*: **D**. And from the clef: G, B, D, two skips up.

EXAMPLE 2

### Read a bass note from the clef




1. **Bass lines** from the bottom: *Great Big Dragons Fly Away*.
2. **The red note is on line 3:** the third word. Which letter?
3. **Check it** from the clef note: the dots hug F on line 4; step down to E, then …

Line 3 is *Dragons*: **D**. And from the clef: F, E, D, two steps down.

EXAMPLE 3

### One line, two notes



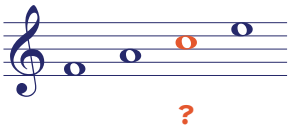

1. **Look at the red notes:** both sit on the bottom line.
2. **Now look at the clefs:** treble on the left, bass on the right.
3. **Use each clef's sentence:** the first word of the treble lines, *Every…*; the first word of the bass lines, *Great…*.

Treble bottom line: **E** (*Every*). Bass bottom line: **G** (*Great*). Same line, different note: always read the **clef** first.

PART I · PITCH · 4 · CONTINUED

EXAMPLE 4

### Find Treble C on the staff

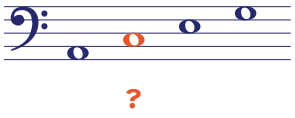



1. **These notes sit in the spaces.** The bottom space is F.
2. **Step up through the spaces:** F, A … they spell a word.
3. **The red note is space 3.** Which letter is it?

F A **C** E: space 3 is **C**, Treble C, the same C you found on the keyboard, one octave above Middle C.

EXAMPLE 5

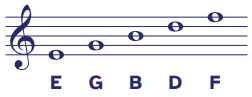
### Find Bass C on the staff




1. **These notes sit in the bass spaces.** The bottom space is A: *All.*
2. **Say the next word:** *All Cows …*
3. **The red note is space 2.** Which letter?

A **C** E G: space 2 of the bass staff is **C**, Bass C, one octave below Middle C.

TRY IT · 10 SECONDS




### Say the treble lines up and down.

"E G B D F", then "F D B G E", out loud, as fast as you can without a slip. Then the spaces: "F A C E".

GAME

### Name it from the clef note

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| treble line 3 | treble space 1 | bass line 3 | bass space 3 |

Start from the clef note (treble G on line 2, bass F on line 4), then step or skip.

Answer: 1 B (skip up from G) · 2 F (step down from G) · 3 D (two steps down from F) · 4 E (step down from F).

PART I · PITCH · 4 · CONTINUED

HOW TO

### Read a new note in three moves

1. **Look at the clef**: treble or bass? That decides every letter.
2. **Find the nearest note you know**: treble G, bass F or Middle C, or use the sentence for lines or spaces.
3. **Step or skip** to the new note, say its name, then play it.
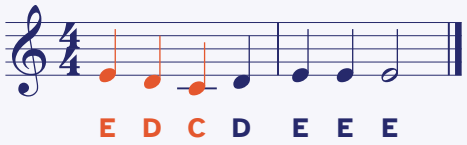
EXTRA · GOING FURTHER

### Read the shape, not just the notes

Good readers see the jumps before the letters. Notes that touch, line then space, are a **step**: the next letter. Notes with one line between them, line to line or space to space, are a **skip**: one letter missed out. Name the first note, then let the shape name the rest: up a step, up a skip, down a step.

YOU KNOW THIS

### Mary Had a Little Lamb




E D C: **line, space, Middle C's own little line**, one step down each time. Read the shape and the tune walks down with it.

**Try it.** Read the notes aloud before you play them.

THINK OF IT The clef notes are lamp posts on a dark road: walk from the nearest one and you are never lost.

TRAP

- × Naming bass notes with treble names: in the bass, the bottom line is G, not E.
- × Mixing up step and skip: a step changes line to space; a skip stays on lines (or on spaces).
- × Mistaking a space note for a line note: a line runs through a line note; a space note sits between two lines.
- × Counting lines from the top: line 1 is always the bottom line.

SEE ALSO

The staff and the keyboard p. 29 · Ledger lines p. 36

PRACTISE IT

Workbook Pitch 4 · Online: Line and Space Names · Steps & Skips

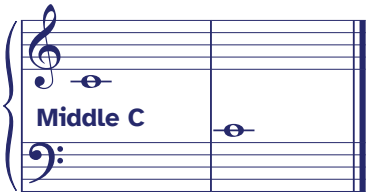
PART I · PITCH · 5

# Ledger lines

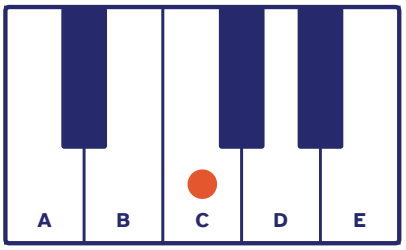
Some notes are too high or too low for the staff. They sit on short extra lines called **ledger lines**, spaced exactly like the staff lines. Read them the way you read the staff: step out from the nearest staff line, line, space, line.

THE SECRET

**Step out from the nearest staff line: line, space, line. The first ledger line is the one nearest the staff.**

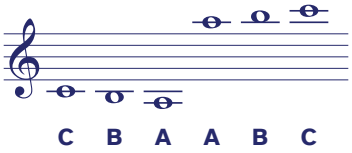



Middle C

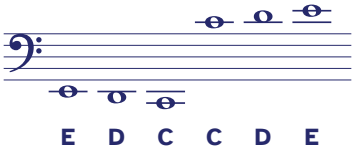



*Middle C sits on a ledger line of its own: below the treble staff, or above the bass staff. One key, two ways to write it.*

## Below and above the staff




Treble: C, B, A below; A, B, C above.




Bass: E, D, C below; C, D, E above.

Name ledger notes by stepping out from the staff, one letter at a time. Treble: below the bottom line E come D, then Middle C on the first ledger line, then B and A. Above the top line F come G, A on the first ledger line, B, then C on the second. Bass: below the bottom line G come F, E on the first ledger line, D, C; above the top line A come B, then Middle C. The few keys round Middle C, from A below it to E above it, can be written in either staff: the hand that plays them decides.

THE RULE

**Count ledger lines from the staff outwards. The first ledger line is the one nearest the staff. Step out line, space, line, and never skip one.**

PART I · PITCH · 5 · CONTINUED

EXAMPLE 1

## Read a high note on ledger lines




1. **The top line** of the treble staff is F.
2. **Step up** one note at a time: G sits on top of the staff, A on the first ledger line.
3. **Keep stepping** to the red note: B above it, then the second ledger line.

Two ledger lines above the treble staff: **High C**, two octaves above Middle C.

EXAMPLE 2

## Read a low note on ledger lines

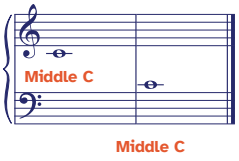


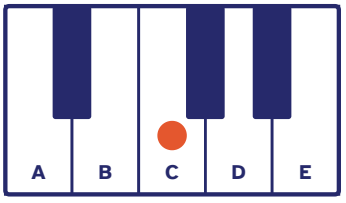

1. **The bottom line** of the bass staff is G.
2. **Step down** one note at a time: F in the space below, E on the first ledger line.
3. **Keep stepping** to the red note: D below it, then the second ledger line.

Two ledger lines below the bass staff: **Low C**, two octaves below Middle C.

EXAMPLE 3

## Middle C, two ways

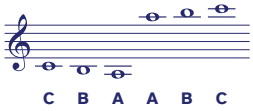



1. **Treble staff:** Middle C hangs on the first ledger line below.
2. **Bass staff:** Middle C sits on the first ledger line above.
3. **Find both on the keyboard.** Are they different keys?

The same key, **Middle C**: the right hand reads it in the treble, the left hand in the bass.

TRY IT · 10 SECONDS




### Say the ledger notes.

Treble: C, B, A below; A, B, C above. Point to each one, say it, then play it.

PART I · PITCH · 5 · CONTINUED

## The four first ledger lines

| WHERE | FIRST LEDGER LINE | NOTE |
|---|---|---|
| Treble, below the staff | just under line 1 | **Middle C** |
| Bass, above the staff | just over line 5 | **Middle C** |
| Treble, above the staff | just over line 5 | **A** |
| Bass, below the staff | just under line 1 | **E** |

Two of them are the same key, Middle C. The other two are not: always look at the clef first.

EXTRA · GOING FURTHER

### Too many ledger lines? 8va

Very high notes need lots of ledger lines, and lots of ledger lines are hard to read. So a high passage is often written lower, with **8va** and a dotted line above it: play those notes **one octave higher** (eight notes up, from a C to the next C, p. 23) until the dotted line stops. **8vb** under the notes means an octave lower. (On its own, 8ve is just short for octave.)

GAME

### Name the ledger note

1. treble, first ledger line below
2. bass, first ledger line above
3. treble, first ledger line above
4. bass, first ledger line below

Step out from the nearest staff line.

Answer: 1 Middle C · 2 Middle C · 3 A · 4 E.

THINK OF IT Ledger lines are the staff's ladder carried on past its ends: the same rungs, just shorter.

TRAP

- × Reading ledger lines from the wrong end: step out from the staff.
- × Counting from the outside in: the first ledger line is the one nearest the staff.

SEE ALSO
Reading the notes p. 32 · Sharps, flats and naturals p. 39

PRACTISE IT
Workbook Pitch 5 · Online: Ledger Lines

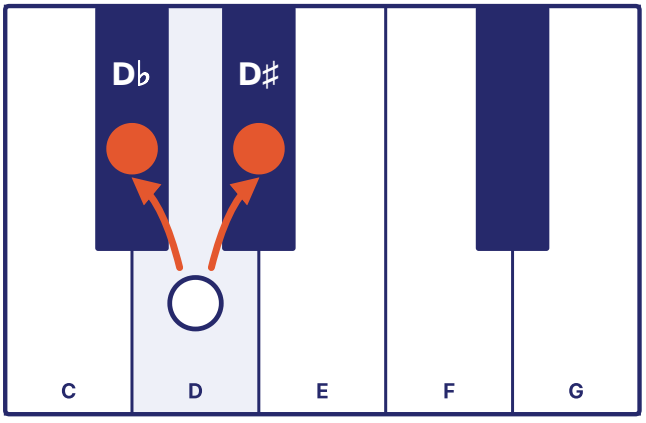
PART I · PITCH · 6

# Sharps, flats and naturals

A sign in front of a note moves it to the very next key. **Sharp**: one key to the **right**, a **semitone** higher. **Flat**: one key to the **left**, a semitone lower. Usually that next key is black, which is how the black keys get their names. These signs are called **accidentals**.

THE SECRET

**Sharp: the very next key to the right. Flat: the very next key to the left.**




♯ one key right · ♭ one key left

| SIGN | NAME | WHAT IT DOES | FIRST MET |
|---|---|---|---|
| ♯ | sharp | one key to the right (a semitone higher) | Level 0 |
| ♭ | flat | one key to the left (a semitone lower) | Level 0 |
| ♮ | natural | cancels a sharp or flat: back to the white key | Level ½ |

A **semitone** is the smallest step on the piano: from one key to the very next, black or white (American books call it a *half step*). More on p. 42. With no black key between (E–F, B–C), the next key is white: E♯ = F and B♯ = C; F♭ = E and C♭ = B. And every black key has two names: C♯ = D♭.

THE RULE

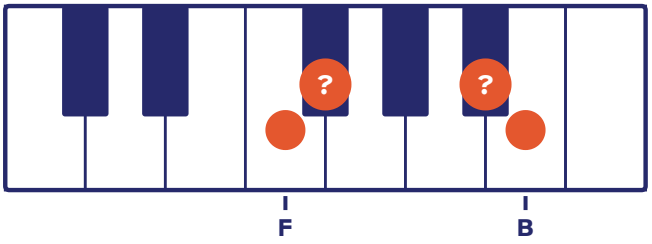
**A sign belongs to one line or space. It changes only that line or space, in that octave, and only until the bar line. The same letter in another octave, and every other note, stays as it is.**

A **bar** is the space between two **bar lines** (p. 50). A **key signature** at the start of every line is different: it lasts the whole piece (p. 99). And a natural cancels a sharp just as it cancels a flat.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 1

### Find F♯ and B♭
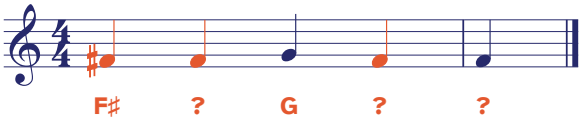


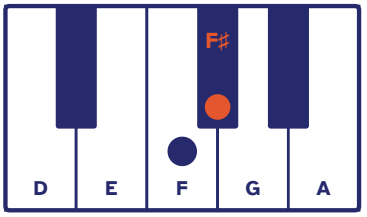

1. **Find F and B** (the red dots): F just left of the three black keys, B just right of them.
2. **Sharp means right:** F♯ is the very next key to the right of F.
3. **Flat means left:** B♭ is the very next key to the left of B. Black or white?

The left **?** is **F♯**, the first black key of the three; the right **?** is **B♭**, the highest one, on the right.

EXAMPLE 2

### How long does the sharp last?



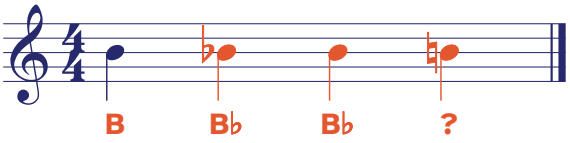



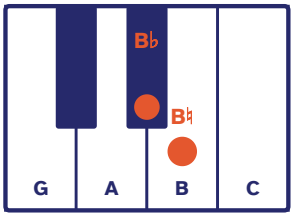

1. **The first note** has a sharp: F♯.
2. **Later in the same bar** there is no sign. Is it still sharp?
3. **After the bar line**: is it still sharp?

Still **F♯** until the bar line; after it, plain **F** again. A sign lasts **to the end of its bar**.

EXAMPLE 3

### A natural cancels the flat



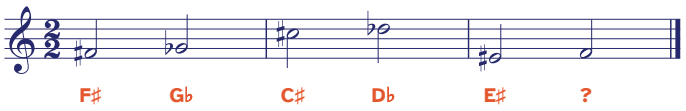

1. **B**: the white key.
2. **B♭**: one key left, and it stays flat for the rest of the bar.
3. **The last note** has a natural sign: what does it do?

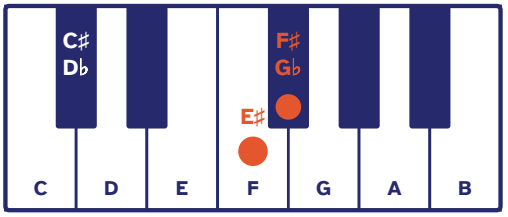
The natural **cancels** the flat: back to the white key B, until the bar line.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 4

## One key, two names




1 **The first bar:** F♯, then G♭. Find both on the keyboard.

2 **Are they different keys?**

3 **The last bar:** E♯. Which key is one right of E?

F♯ and G♭ are the **same black key**; E♯ is the white key **F**, so the **?** is plain F, the same key as E♯. Same sound, different names: **enharmonic**.

GAME

## Which key is it?
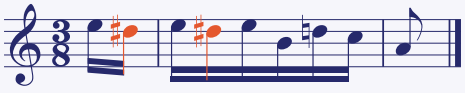
| 1 | 2 | 3 | 4 |
|---|---|---|---|
| C♯ | E♭ | B♯ | F♭ |

Sharp: one key to the right. Flat: one key to the left. Which key do you press?

Answer: 1 the black key right of C · 2 the black key left of E · 3 the white key C · 4 the white key E.

YOU KNOW THIS

## Für Elise (Beethoven)



E, **D♯**, E, **D♯**: the next key down and back. The bar line wipes the sharp out, so it is written again; later the D needs a **natural**, or it would still be D♯.

**Try it.** Play E and D♯ back and forth: the black key just left of E.

THINK OF IT A sharp or flat is a sticky note on one line of the staff: it stays stuck until the bar line, then it falls off.

TRAP

× Carrying a sharp or flat past the bar line: after it, the note is plain again.

SEE ALSO

Keys and key signatures p. 99

PRACTISE IT

Workbook Pitch 6 · Online: Accidentals · Tones & Semitones

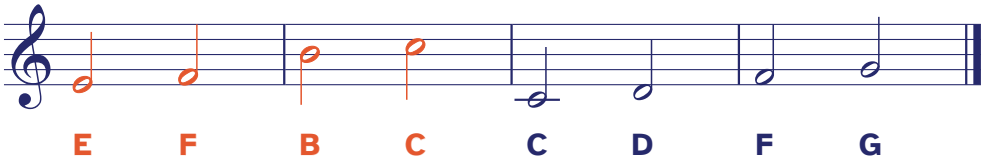
PART I · PITCH · 7

# Counting intervals

The smallest step on the piano is a **semitone**: from one key to the very next. Two semitones make a **tone**. Bigger distances are **intervals**: to give one a number, **count the notes**, both ends included.

THE SECRET

**The very next key is a semitone; two make a tone. For an interval, count the notes, both ends included.**




*E–F and B–C: next door, a semitone. C–D and F–G: a black key between, a tone.*

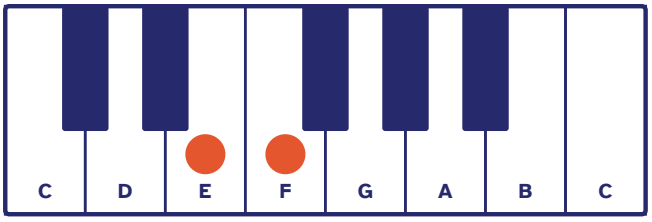
From any key to the **very next key**, black or white, is a **semitone**: the smallest step on the piano. Two semitones make a **tone**. On the white keys most steps are tones, but **E–F** and **B–C** have no black key between them: they are semitones.

THE RULE

**Semitone: the very next key, with nothing between: white to white (E–F, B–C), white to black, or black to white. Tone: exactly one key between.**

EXAMPLE 1

**Tone or semitone?**

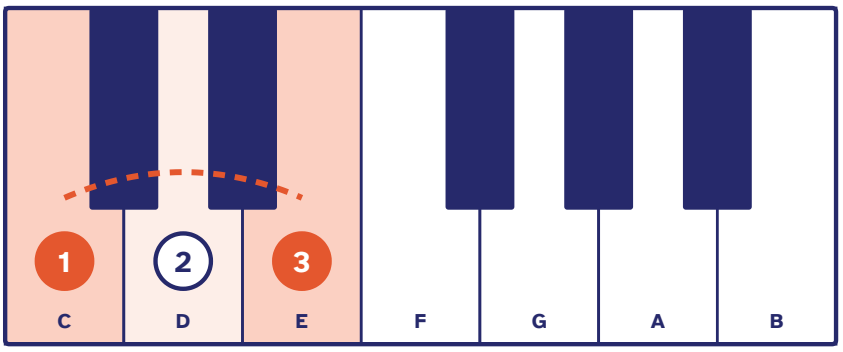



1. **Look between the red keys:** is there a black key?
2. **No black key** means they are next-door keys.
3. **Now try C and D**: is there a black key between?

E–F is a **semitone** (next door). C–D has C♯ between: a **tone**.

PART I · PITCH · 7 · CONTINUED

## Intervals: count the notes



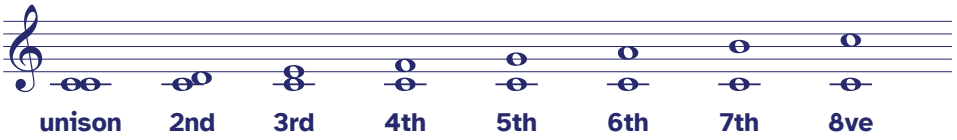

C to E: count the notes, 1 2 3: a 3rd

THE RULE

**Count the letters for the number: start at 1 on the first note and count white keys, or lines and spaces, never the black keys. C to E is C D E: 3 notes, a 3rd. Sharps and flats never change the number.**

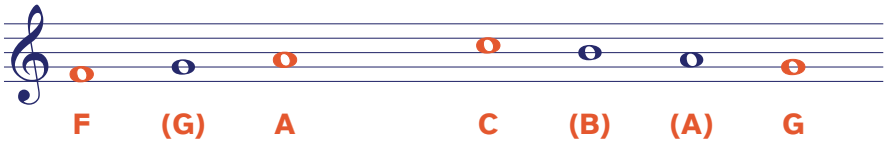
A step on the staff (the next letter) may be a tone (C–D) or a semitone (E–F); C to C♯ is a semitone with no new letter.

## Interval numbers from C




EXAMPLE 2

### How far apart are these notes?



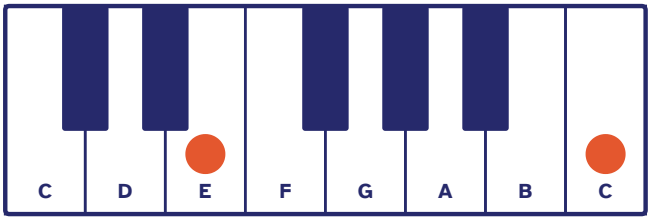

1. **Call the first note 1:** here F is 1.
2. **Count every letter** up to the second note.
3. **Going down works the same:** start at 1 on the top note and count down.

F G A is 3 notes: a **3rd**. C B A G is 4 notes: a **4th**.

PART I · PITCH · 7 · CONTINUED

EXAMPLE 3

### Name an interval on the keyboard

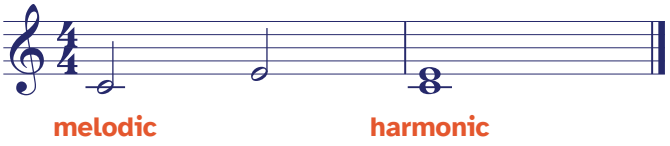



1. **Find both red keys:** E, and the C above it.
2. **Count the notes** from E: E is 1 …
3. **Stop on the C.** What number is it?

E F G A B C is 6 notes: a **6th**.

EXAMPLE 4

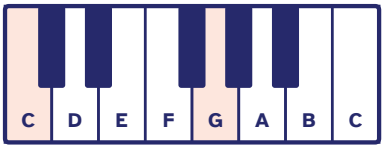
### Melodic or harmonic?




1. **The first bar:** C, then E, one after the other.
2. **The second bar:** C and E at the same time.
3. **Count the notes** in each.

Both are a **3rd**: **melodic** when played one after the other, **harmonic** when played together.

TRY IT · 10 SECONDS
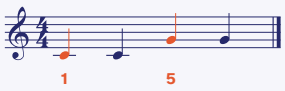



### Thumb on C, finger 5 on G.

Your hand is a ruler: thumb to little finger on white keys is a 5th. Play it, then thumb to finger 3: a 3rd.

YOU KNOW THIS

### Twinkle, Twinkle, Little Star




"Twinkle, twinkle": C up to G, C D E F G, **five notes, a 5th**.

**Try it.** Sing the first two notes of a song and count them, both included.

TRAP

× Counting only the gaps: C to E is 3 notes, not 2 gaps. And a sharp or flat never changes the number: C to G♯ is still a 5th.

SEE ALSO
Major, minor and perfect p. 45 · Major scales p. 93

PRACTISE IT
Workbook Pitch 7 · Online: Intervals · Tones & Semitones

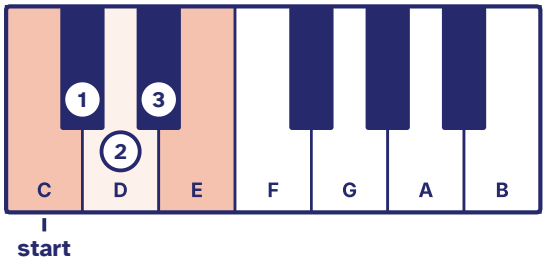
PART I · PITCH · 8

# Major, minor and perfect

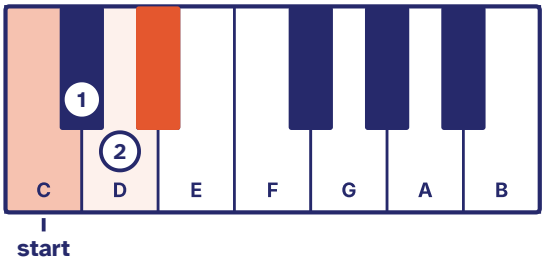
C to E and C to E♭ are both 3rds: three notes each. But they sound different, bright and dark. To tell them apart, **count the keys in between**, black and white: the keys you do not play.

THE SECRET

**Count the keys between: 3 make a major 3rd, 2 a minor 3rd.**




C to **E**: 3 keys between. A **major** 3rd: bright.




C to **E♭**: 2 keys between, one fewer. A **minor** 3rd: dark.

*One key fewer between: a major 3rd becomes minor*

THE RULE

**Count the notes for the number (a 3rd). Count the keys in between for the size (major or minor): black and white, not the two you play.**

**Number**: count the letters, both ends included. **Size**: count every key in between, black and white, leaving both ends out.

Why count the keys in between? Because the size is what you hear: one key more or less turns bright into dark, or makes a 5th sound hollow and strange.

| NUMBER | KEYS BETWEEN: SMALLER | MAIN SIZE |
|---|---|---|
| 3rd | 2 (minor) | 3 (major) |
| 4th | | 4 (perfect) |
| 5th | | 6 (perfect) |
| 6th | 7 (minor) | 8 (major) |
| octave | | 11 (perfect) |

**3rds and 6ths** come in two sizes: **major**, and **minor**, one key fewer between.

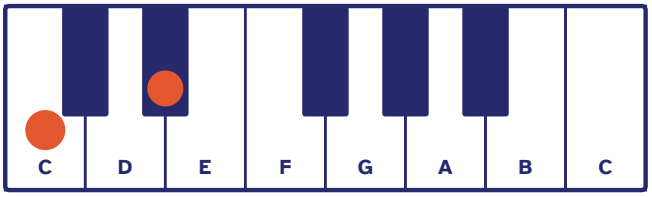
**4ths, 5ths and octaves** have one main size, called **perfect**. They are never major or minor.

2nds and 7ths have two sizes too (p. 93). F up to B has 5 keys between: not perfect; its name comes later. Semitones = keys between + 1: a major 3rd is 4.

PART I · PITCH · 8 · CONTINUED

EXAMPLE 1

## Major or minor 3rd?

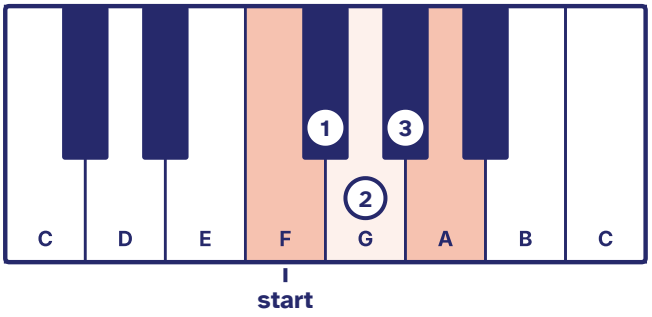



1. **Count the notes:** C D E, three letters, so it is a 3rd.
2. **Count the keys in between** C and the red black key, E♭: C♯, D …
3. **A major 3rd has 3 between.** How many here?

C to E♭ has **2 keys between**: a **minor 3rd**.

EXAMPLE 2

## Name F up to A

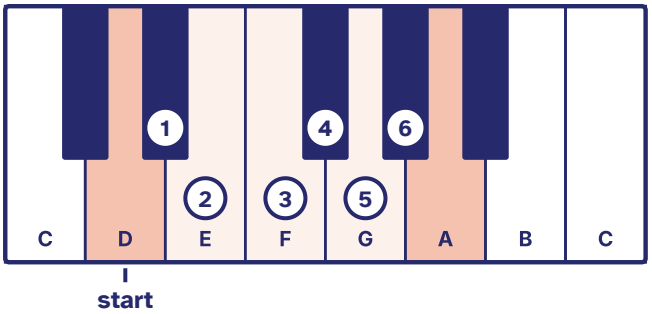



1. **Count the notes:** F G A is 3, so it is a 3rd.
2. **Count the keys in between:** they are numbered.
3. **3 between, or 2?**

F to A has 3 keys between: a **major 3rd**, the same size as C to E.

EXAMPLE 3

## Name D up to A



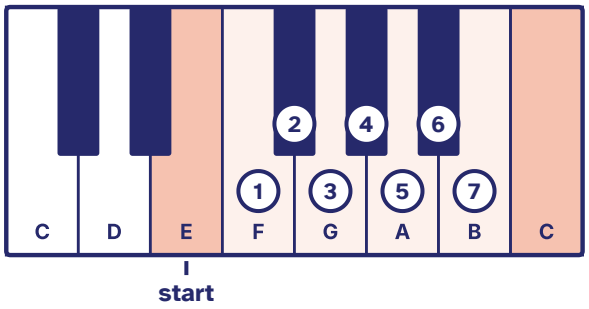

1. **The number:** D E F G A, five notes, a 5th.
2. **The size:** count the keys in between: the numbered keys.
3. **A 5th with 6 between** is its main size. What is that called?

D up to A is a **perfect 5th**: 6 keys between.

PART I · PITCH · 8 · CONTINUED

EXAMPLE 4

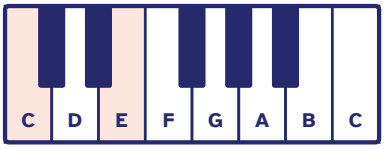
### Name E up to C




1. **The number:** E F G A B C, six notes, a 6th.
2. **The size:** count the keys in between.
3. **A major 6th** has 8 between. Is this one more or one fewer?

E up to C has 7 keys between, one fewer than major: a **minor 6th**.

TRY IT · 10 SECONDS
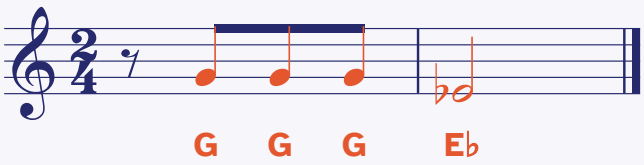



### Play C–E, then C–E♭.

Bright, then dark. One key makes the difference: the major 3rd has three keys between, the minor 3rd two.

YOU KNOW THIS

### Beethoven, Symphony No. 5




Da-da-da-DUM: G down to E♭ is three notes, a 3rd, with 3 keys between (F♯, F, E): a **major 3rd**. Counting down works the same way; bright and dark is clearest when you play the two 3rds side by side.

**Try it.** Count the keys between G and E♭ on the keyboard.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every key between, black and white.
- × Counting the two notes you play: count only the keys in between.
- × Calling a 4th or 5th “major”: they are perfect.
- × C–D♯ is a 2nd, C–E♭ a 3rd: the same keys, but the letters decide the number.

SEE ALSO
Counting intervals p. 42 · Chords and triads p. 110

PRACTISE IT
Workbook Pitch 8 · Online: Intervals (Major or minor 3rd)

PART I · PITCH · TIPS

# When a note reads wrong

Most wrong notes come from a handful of slips. Name the slip first, then fix only that note: playing the whole line again rarely finds it.

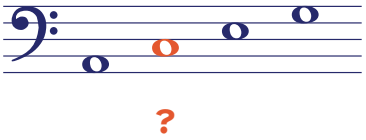
TIP

### What you notice, what to check

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| The left hand reads like the right | **the clef**: bass lines are G B D F A, not E G B D F | p. 32 |
| You land an octave too high or too low | **the Cs**: find Middle C first, then walk out to Bass C or Treble C | p. 23 |
| Notes above or below the staff slow you down | **the ledger lines**: step on from the nearest staff line | p. 36 |
| A sharp or flat vanishes, or stays too long | **the bar line**: a sign lasts only to the end of its bar | p. 40 |
| You play the black key on the wrong side | **the direction**: sharp goes right, flat goes left | p. 39 |
| Two notes sound too close or too far apart | **the interval**: count the notes, then the keys between | p. 45 |

EXAMPLE

### Spot the slip



?

*Sam played A. Wrong!*

1. **Check the clef first:** this is the bass clef, not the treble.
2. **Use the bass sentence:** the spaces are *All Cows Eat Grass*.
3. **The red note is in space 2:** the second word.

The note is **C** (*Cows*): Bass C. Sam read the bass staff with treble names (treble space 2 is A): the most common slip of all.

HOW TO

### Three checks before you read a new line

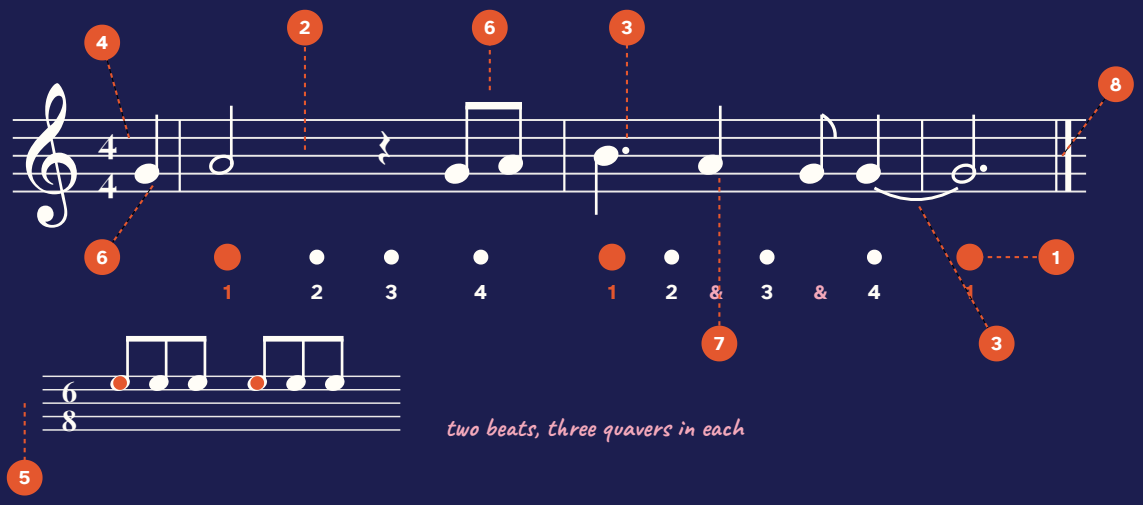
1. **The clefs:** which staff is treble, which is bass?
2. **The clef notes:** find treble G, bass F and Middle C, and read the other notes from them.
3. **The signs:** mark each sharp, flat or natural, and see where its bar ends.

THINK OF IT Like reading a map: find the big landmarks first, then the streets around them. A note far from every clef note is the one most often misread.

PART II · WHEN, AND FOR HOW LONG

# Time

Music never stands still. Underneath runs a steady **pulse**, packed into **bars**, and every note and rest gets its fair share. Everything in this part is in these two bars.




1 **Pulse, tempo and the bar** p. 50
the steady beat underneath; beat 1 is strongest

2 **Notes and rests** p. 53
the shape says how long; rests are counted too

3 **How long is a note?** p. 56
a dot adds half; a tie glues two notes into one

4 **Time signatures** p. 59
four seats in every bar, a crotchet in each

5 **Compound and irregular time** p. 62
beats in threes; bars of five or seven

6 **Rhythm: groups and upbeats** p. 65
beams show the beats; starting before bar 1

7 **Rhythm: off the beat** p. 68
a note between the beats that holds on

8 **Repeats and road signs** p. 71
repeat signs, D.C., D.S., Fine and Coda

★ **Tips: when the rhythm slips** p. 74

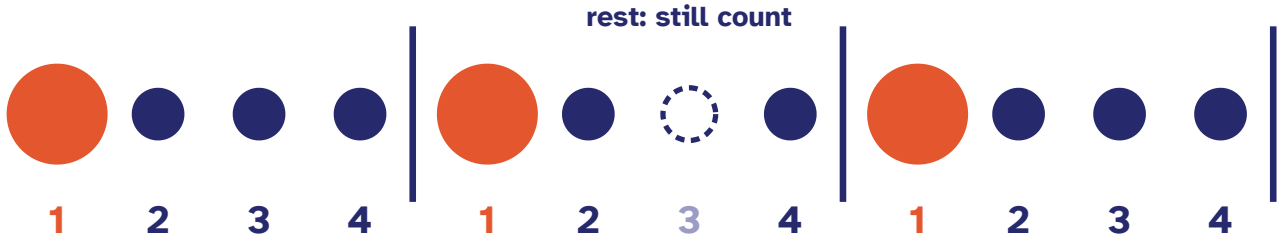
PART II · TIME · 1

# Pulse, tempo and the bar

Put your hand on your heart: that steady beat is a **pulse**, and almost every piece has one. Its speed is the **tempo**. Beats travel in groups called **bars**, and beat 1 is the strongest.

**THE SECRET**

**The beat never stops, not even in a rest. Feel it before you play.**




*Beat 1 of every bar is the strongest*

## Tempo

Tempo is measured in beats per minute. A **metronome mark** at the top of the music shows it: ♩ **= 60** means 60 crotchet beats a minute, one beat a second. ♩ **= 120** is twice as fast. Speed words such as *Andante* and *Allegro* are on p. 82.

## Strong and weak beats

| BEATS | PATTERN | FEELS LIKE |
|---|---|---|
| 2 | **Strong** weak | a march |
| 3 | **Strong** weak weak | a waltz |
| 4 | **Strong** weak **medium** weak | most songs |

## Bars and bar lines

| SIGN | NAME | MEANING |
|---|---|---|
| | bar line | divides the music into bars |
| | double bar line | marks the end of a section, or a change of time signature or tempo |
| | final bar line | marks the end of the piece |
| | repeat signs | play the passage between them again; if there is no start sign, go back to the beginning |

PART II · TIME · 1 · CONTINUED

## The beat in tunes you know

| TUNE | BEATS IN A BAR | PATTERN |
|---|---|---|
| Twinkle, Twinkle, Little Star | 4 | **Strong** weak **medium** weak |
| Happy Birthday | 3 | **Strong** weak weak |
| The Blue Danube (a waltz) | 3 | **Strong** weak weak |
| Radetzky March | 2 | **Strong** weak |

## Counting

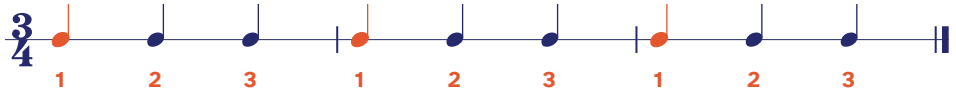



Three bars of 3/4 (the sign at the start: three beats in each bar, p. 59). Every beat is counted, even inside a long note: the red number is where you play, the grey numbers are the beats you hold.

Count aloud at one speed, a little louder on beat 1. With a metronome, start slower than you think and speed up only when the passage is steady.

EXAMPLE 1

### Is this a march or a waltz?

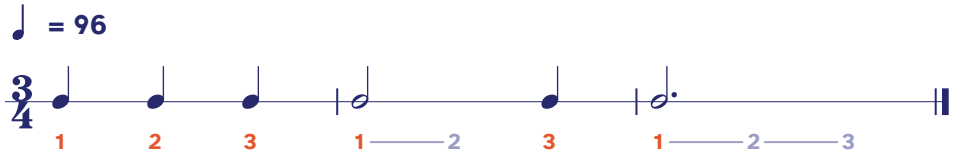



1. **Tap a steady beat** as you listen.
2. **Which beats feel heavier?** The red notes help.
3. **Count from one heavy beat to the next.** How many?

Heavy every three: 1 2 3, 1 2 3. A **waltz**. Every two would be a march.

EXAMPLE 2

### Feel the beat before you play




1. **Read the mark:** ♩ = 96, 96 crotchet beats a minute.
2. **Listen first:** set the metronome and let one full bar go by, counting “1 2 3” in your head.
3. **Step in** on the next beat 1, the way you step onto a moving escalator.

The first note lands **on the beat**, not a guess.

PART II · TIME · 1 · CONTINUED

GAME

## How many beats in a bar?

1 2 3 4

The big dot is the strong beat. Count the beats in each bar: 2, 3 or 4?

Answer: 1 three (a waltz) · 2 four · 3 two (a march) · 4 three, twice.

TRY IT · 10 SECONDS




### Hand on your heart. Tap the beat.

Say “1 2 3 4” with your taps, 1 a little louder. Keep going for ten seconds without slowing down.
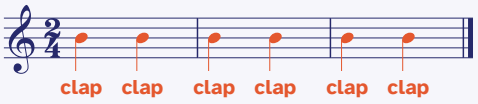
HOW TO

### Find the beat of a new piece

1. **Read the time signature:** how many beats in each bar.
2. **Tap the beat** on your knee before you play, counting one bar aloud.
3. **Start slower** than the marked speed, and keep it steady to the end.

YOU KNOW THIS

### Radetzky March (Johann Strauss I)




At the New Year concert the audience claps along on **every beat**. That steady clap is the pulse; the orchestra’s tune moves freely over it.

**Try it.** Put on any march and clap only the beat, not the tune.

THINK OF IT The beat is the stage and the notes are the actors: the stage is there before anyone walks on. So step in the way you step onto an escalator or take a plate off a sushi train: watch it move for a bar, then join in.

TRAP

- × Speeding up in easy bars and slowing in hard ones: the pulse never changes.
- × Counting only the notes you play: the beat goes on through long notes and rests.

SEE ALSO
Time signatures p. 59 · Speed words p. 82

PRACTISE IT
Workbook Level 0 · Pulse · Online: Metronome

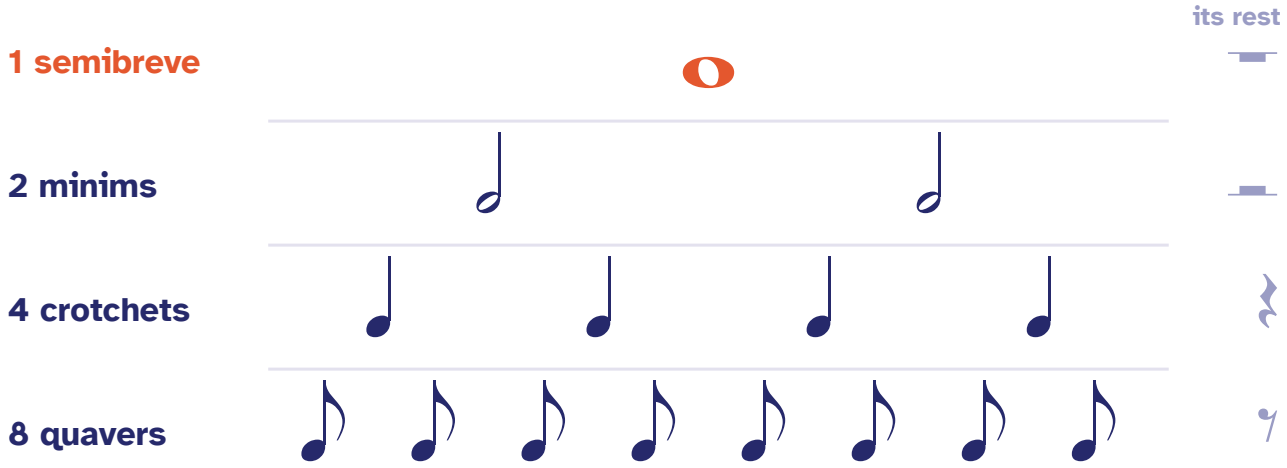
PART II · TIME · 2

# Notes and rests

You can see how long a note lasts from its shape: each plain note is **half** the one above it. And every note has a twin **rest**, a silence you count just as carefully.

**THE SECRET**

**Each note is half the one above it, and every note has a rest twin.**



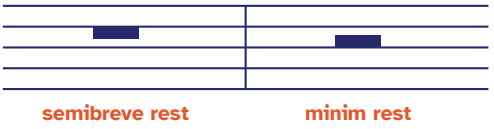

*Each note splits into two of the next*

**Counting quavers:** say the beat's number, then "&" halfway to the next beat: "1 & 2 & 3 & 4 &".

## Head, stem, flag

The **head** sits on a line or in a space. A **stem** goes up on the right of the head or down on the left. **Flags** on the stem halve the value each time; groups of quavers are joined by **beams** instead of flags.

## Which rest is which?

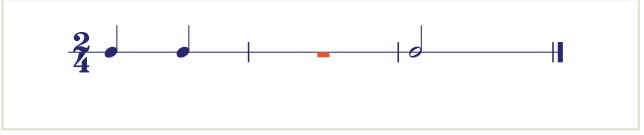



The semibreve rest **hangs** below the fourth line; the minim rest **sits** on the middle line.

**THE SEMIBREVE REST'S SPECIAL POWER**

Most rests last a fixed number of beats. The **semibreve rest** can also mean **a whole bar of silence** in any time signature: 4 beats in 4/4, but 3 in 3/4 and 2 in 2/4. It is then centred in the bar. No other rest can do this.





PART II · TIME · 2 · CONTINUED

| NOTE | REST | NAME | ALSO CALLED | BEATS* | FIRST MET |
|---|---|---|---|---|---|
|  |  | **semibreve** | whole note | **4** | Level 0 |
|  |  | **minim** | half note | **2** | Level 0 |
|  |  | **crotchet** | quarter note | **1** | Level 0 |
|  |  | **quaver** | eighth note | **½** | Level 0 |
|  |  | **semiquaver** | sixteenth note | **¼** | Level 2 |

### Why whole, half and quarter?

Think of each beat as a pizza. A bar of 4/4 is four pizzas. The note that fills the whole bar is the **whole note**; the others are a half, a quarter, an eighth and a sixteenth of it. Two quavers beamed together fill one pizza. A rest is the same share left empty: shaded with lines.

| | | | |
|---|---|---|---|
| **semibreve** |  | whole note | 4 pizzas · 4 beats |
| semibreve rest |  | silence | 4 empty pizzas · 4 beats |
| **minim** |  | half note | 2 pizzas · 2 beats |
| minim rest |  | silence | 2 empty pizzas · 2 beats |
| **crotchet** |  | quarter note | 1 pizza · 1 beat |
| crotchet rest |  | silence | 1 empty pizza · 1 beat |
| **quaver** |  | eighth note | half a pizza · ½ beat |
| quaver rest |  | silence | half an empty pizza · ½ beat |
| **semiquaver** |  | sixteenth note | a quarter of a pizza · ¼ beat |
| semiquaver rest |  | silence | a quarter of an empty pizza · ¼ beat |

TRY IT · 10 SECONDS



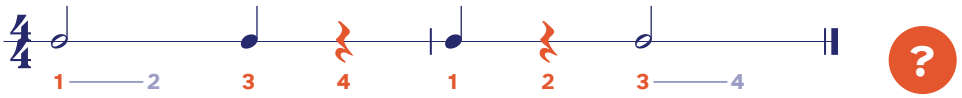
### Clap once and hold for 2: a minim.

Now clap four crotchets (1 2 3 4), then one semibreve (clap, and hold all four beats). Say the counts out loud.

PART II · TIME · 2 · CONTINUED

EXAMPLE 1

**How long is each sound, and each silence?**

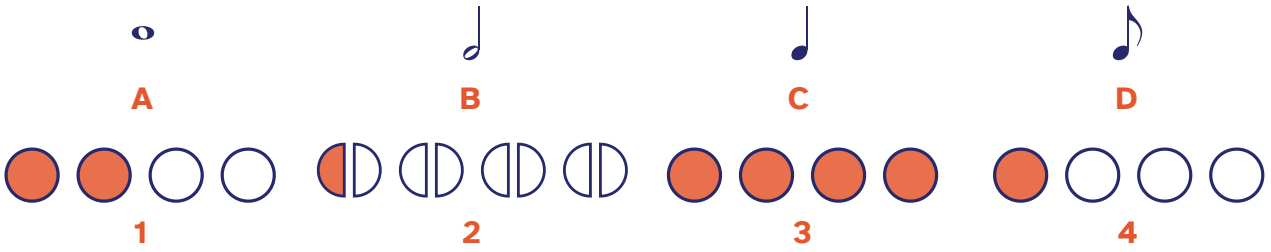



1. **Name each shape** in the bar.
2. **How many beats** does each one last?
3. **Add them up,** rests included. What is the total?

Minim 2, crotchet 1, crotchet rest 1: every bar adds up to **4 beats**, silences included.

GAME

**Pizza match**

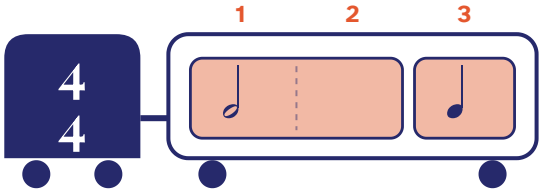



Match each note (A to D) with its pizzas (1 to 4). One pizza is one beat.

Answer: A 3, B 1, C 4, D 2.

EXAMPLE 2

**Fill a bar of 4/4: what goes in the last seat?**
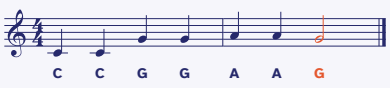



1. **How many seats** does the minim take?
2. **And the crotchet?**
3. **How many seats are left** for a silence, and which rest fills one seat?

2 + 1 = 3, one seat left: a **crotchet rest**. The bar is full.

YOU KNOW THIS

**Twinkle, Twinkle, Little Star**




Six **crotchets**, then a **minim** at the end of the line: one beat, one beat … and two beats to breathe.

SEE ALSO
Pulse and the bar p. 50 · How long is a note? p. 56

PRACTISE IT
Workbook Level 0 · Note values · Online: Library › Notes

PART II · TIME · 3

# How long is a note?

Need a note longer than its shape allows? Two signs stretch it: the **dot**, which adds half, and the **tie**, which glues two notes into one sound.

**THE SECRET**
**A dot adds half. A tie glues two notes into one.**
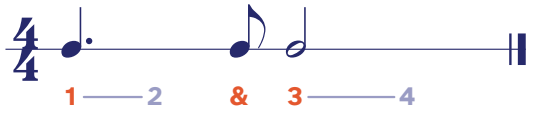


**2 beats** **2 + 1 = 3 beats**

*The dot adds half*

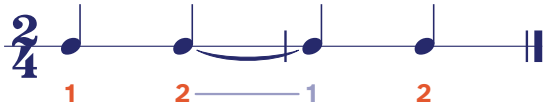
## The dot

A dot adds **half the note's value**: a dotted minim lasts 2 + 1 = 3 beats; a dotted crotchet 1 + ½ = 1½.




## The tie

A curve joining two notes **of the same pitch** makes one longer sound: play the first, hold through the second. Ties can cross a bar line, where a dot cannot.




### The dot, in pizzas

The dot is a helper that adds **half of whatever its note is worth**. Dark slices: the note. Lighter: what the dot adds.

| | | |
|---|---|---|
| dotted minim |  | 2 + **1** (half of 2) = **3 beats** |
| dotted crotchet |  | 1 + **½** (half of 1) = **1½ beats** |
| dotted quaver |  | ½ + **¼** (half of ½) = **¾ beat** |
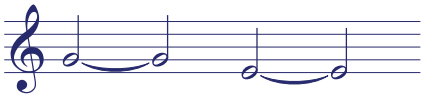
EXTRA · GOING FURTHER

### The double dot

A second dot adds half of the first dot: a double-dotted minim lasts 2 + 1 + ½ = **3½ beats**. Rare before the higher Levels.

PART II · TIME · 3 · CONTINUED

## Tie or slur?



**Tie**: the same pitch. Play once and hold: G for 4 beats, then E for 4.

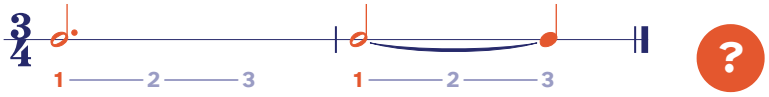


**Slur**: different pitches. Play every note, smoothly joined (legato).

**TWO-MINUTE DRILL** Clap a dotted minim ("1 2 3"), a dotted crotchet and quaver ("1 & 2"), then two tied crotchets ("1 2"). Say the counts aloud and never re-clap a tied note.

EXAMPLE 1

### Dot or tie: which is longer?
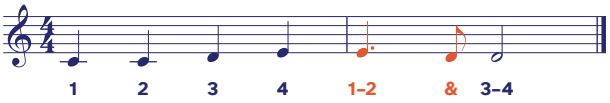



1. **The dotted minim:** a dot adds half. How many beats?
2. **The tied notes:** add the two notes together.
3. **Compare** the two totals.

Both last **3 beats**: the same length. A tie can also cross a bar line, which a dot cannot.

EXAMPLE 2

### A dotted crotchet you already know: Ode to Joy




1. **Bars 3–4 of Beethoven's Ode to Joy.** Sing it: "C C D E, E — D D".
2. **The red E is a dotted crotchet:** 1 + ½ = 1½ beats, held through "1 & 2".
3. **The quaver D** falls on the "&" of beat 2; the last D takes beats 3 and 4.

Long–short, then long: the dot is why the tune **leans** before it lands.

EXAMPLE 3

### Which is longer: a dotted minim or a minim tied to a quaver?




1. **Dotted minim:** add half of 2. How many pizzas?
2. **Minim tied to a quaver:** add the two.
3. **Compare** the totals.

3 against 2½: the **dotted minim** is longer, by half a beat.

PART II · TIME · 3 · CONTINUED

GAME

### How many beats?




With a crotchet as one beat: how long does each one last?

Answer: 1 three (2 + 1) · 2 one and a half (1 + ½) · 3 three (2 tied to 1).

TRY IT · 10 SECONDS

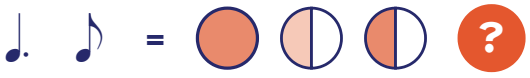


### Clap once, hold for three.

Say “1 2 3” while you hold: that is a dotted minim. Then a dotted crotchet: clap on “1”, hold through “& 2”, and let go on the “&” after 2.

EXAMPLE 4

### How many beats: a dotted crotchet and a quaver?

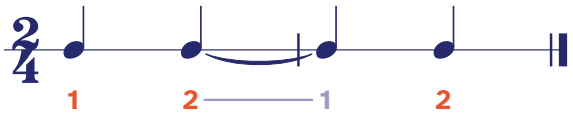



1. **The dotted crotchet:** 1 beat plus half of 1.
2. **The quaver:** half a beat.
3. **Add them.** Where does the quaver fall?

1½ + ½ = **2 beats**. The quaver lands on the “&” of beat 2: “1 (&) 2 **&**”.

EXAMPLE 5

### How long do you hold a tie?




1. **Play the first note** and count its beats.
2. **Keep holding** through the tie. Do you play the second note again?
3. **Add the beats** of both notes.

One sound, never played twice: a tie **adds the beats** of both notes.

TRAP

- × Putting the dot on the wrong note: a dot adds half of **its own** note.
- × Tying two different pitches: that curve is a slur, not a tie.

SEE ALSO

Time signatures p. 59 · Rhythm p. 65

PRACTISE IT

Workbook Level 0 · Note values · Online: Library › Notes and rests

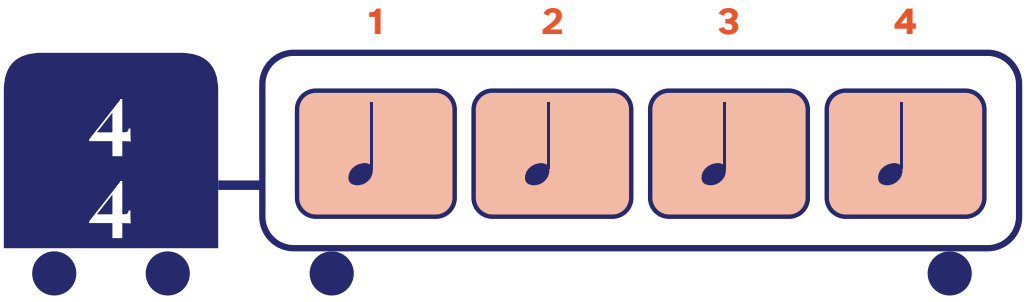
PART II · TIME · 4

# Time signatures

Two numbers, like a train ticket: the **top number** says how many seats (counts) each bar has; the **bottom number** says which note fills one seat.

**THE SECRET**

**Top number: how many beats. Bottom number (in simple time): which note gets one beat.**




*Top: how many seats · bottom: which note sits in one*

## The bottom number

| BOTTOM | NOTE | ONE COUNT IS A | FIRST MET |
|---|---|---|---|
| 2 | 𝅗𝅥 | minim | Level 3 |
| 4 | ♩ | crotchet | Level 0 |
| 8 | ♪ | quaver | Level 1 |
| 16 | 𝅘𝅥𝅯 | semiquaver | *Extra* |

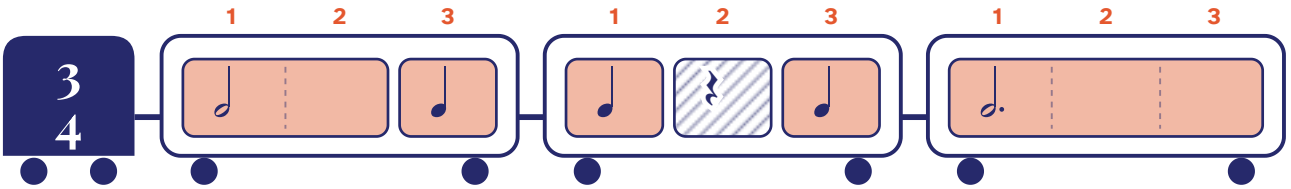
## Simple time: each beat divides in two

| TIME SIGNATURE | EACH BAR HOLDS | FIRST MET |
|---|---|---|
| 2/4 | 2 crotchets | Level 0 |
| 3/4 | 3 crotchets | Level 0 |
| 4/4 𝄴 | 4 crotchets (the C sign: common time) | Level 0 |
| 2/2 𝄵 | 2 minims (the cut C: cut common time) | Level 3 |
| 3/2 | 3 minims | Level 3 |
| 4/2 | 4 minims | *Extra* |
| 3/8 | 3 quavers | Level 3 |

PART II · TIME · 4 · CONTINUED

### Think of a train

In 3/4 every carriage (bar) has exactly **3 seats**; each seat is one crotchet beat. Notes book seats: a minim takes 2, a crotchet 1. A rest books a seat and leaves it **empty** (shaded). Every seat in every carriage must be booked.



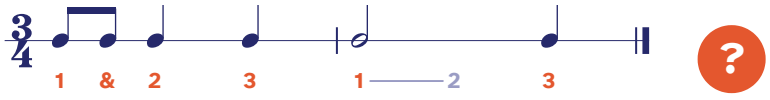

minim + crotchet · crotchet, crotchet rest, crotchet · dotted minim: 3 seats each

## Changing time signature

A new time signature can appear in the middle of a piece, right after a bar line; it holds until the next one. A time signature is written only once, not on every line.

EXAMPLE 1

### What does 3/4 mean?




1. **The top number:** how many counts in each bar?
2. **The bottom number:** 4 stands for a crotchet. Which note gets one count?
3. **Check a bar:** does it add up?

3/4: **three crotchet beats** in every bar (two quavers + crotchet + crotchet = 3).

GAME

### Which ticket?

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| 3 crotchets | 4 crotchets | 3 quavers | 2 minims |

Each bar holds these notes. Write its time signature.

Answer: 1 3/4 · 2 4/4 · 3 3/8 · 4 2/2 (or 4/4).

TRY IT · 10 SECONDS

4
4

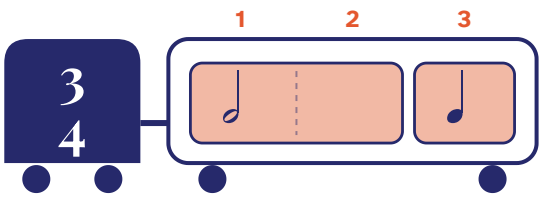
### Open a piece you are playing.

Find the time signature, then count the seats in bar 1: do the notes add up to the top number? Try bar 2 as well.

PART II · TIME · 4 · CONTINUED

EXAMPLE 2

**How do you count this bar of 3/4?**

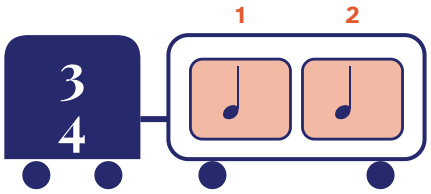



1. **How many seats** does 3/4 have?
2. **The minim**: which seats does it book?
3. **The crotchet**: which seat is left for it?

Count **1 2 3**: the minim holds through 1 and 2, the crotchet takes 3.

EXAMPLE 3

**Where does the rest go?**
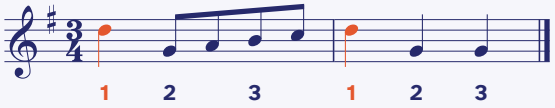



1. **How many seats** in 3/4?
2. **The two crotchets** fill which seats?
3. **Which seat is empty,** and what goes there?

Seat 3 is empty: a **crotchet rest**. Count it: 1 2 (3).

YOU KNOW THIS

**Minuet (from the Notebook for Anna Magdalena Bach)**




Every bar holds **three crotchet beats**: 1 2 3, 1 2 3, the step of the dance. That is 3/4.

**Try it.** Count 1 2 3 aloud while someone plays it.

TRAP

- × Reading the bottom number as a count: it names the note that gets one beat.
- × Reading 6/8 as six beats: at most speeds it moves in two.
- × Mixing up 3/4 and 6/8: both hold six quavers, grouped differently (p. 62).

SEE ALSO

Pulse and the bar p. 50 · Compound time p. 62

PRACTISE IT

Workbook Level 0 · Time signatures · Online: Library › Time signatures

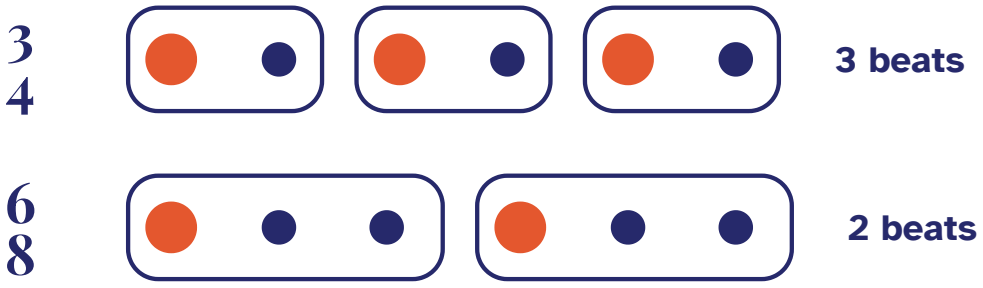
PART II · TIME · 5

# Compound and irregular time

Compound time is a skipping rope: each dotted beat splits into **three**. Irregular time keeps you guessing, with **five or seven** counts in uneven groups. (9/4, 12/4 and /16 signatures are rare.)

**THE SECRET**

**Dotted beats split in three: 6/8 is two big beats.**




*Same six quavers, new groups*

**SIMPLE OR COMPOUND?**

In **simple time** each beat divides into **two**. In **compound time** each beat is a dotted note and divides into **three**.

**Quick test:** a top number of 6, 9 or 12 means compound. Divide it by 3 to find the beats: 6/8 has 2, 9/8 has 3, 12/8 has 4.

## Compound time

| TIME SIGNATURE | EACH BAR HOLDS | FIRST MET |
|---|---|---|
| 6/8 | 2 dotted crotchets | Level 1 |
| 9/8 | 3 dotted crotchets | Level 3 |
| 12/8 | 4 dotted crotchets | Level 3 |
| 6/4 | 2 dotted minims | Level 4 |

## Counting 6/8




Slowly, count all six quavers. At speed, feel two beats, on 1 and 4: “1 & a, 2 & a”.

PART II · TIME · 5 · CONTINUED

EXTRA · GOING FURTHER

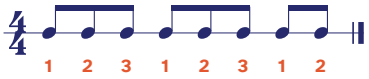
## Irregular time

Five or seven counts, grouped unevenly. Met from Level 5.

| TIME SIGNATURE | EACH BAR HOLDS | FIRST MET |
|---|---|---|
| 5/4 | 5 crotchets: 3 + 2 or 2 + 3 | Level 5 |
| 7/4 | 7 crotchets: 4 + 3 or 3 + 4 | *Extra* |
| 5/8 | 5 quavers: 3 + 2 or 2 + 3 | *Extra* |
| 7/8 | 7 quavers: 2 + 2 + 3 or 3 + 2 + 2 | *Extra* |

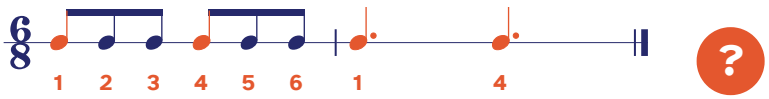
### UNEVEN GROUPS IN 4/4

Many dances keep 4/4 but group the eight quavers 3 + 3 + 2: “1 2 3, 1 2 3, 1 2”.




EXAMPLE 1

### How many beats in a bar of 6/8?




1. **Slowly, count all six quavers.**
2. **Which two counts** feel strong? (the red notes)
3. **Faster, count only the strong ones.** How many?

6/8 has **two beats** in a bar (1 and 4), each split into three quavers.

GAME

### Two beats or three?

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| 6/8 | 3/4 | 9/8 | 2/4 |

How many beats in a bar, and does each beat split in two or in three?

Answer: 1 two, in threes · 2 three, in twos · 3 three, in threes · 4 two, in twos.

TRY IT · 10 SECONDS

6/8

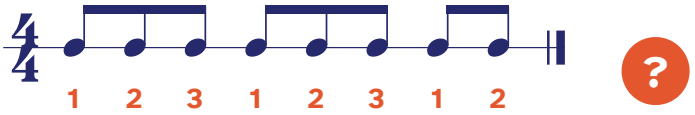
### Sway left, sway right.

Say “1 2 3, 4 5 6” and sway once on 1 and once on 4. That is 6/8: six quavers, two big beats.

PART II · TIME · 5 · CONTINUED

EXAMPLE 2

**Is this still 4/4?**




1. **Count eight quavers** in the bar.
2. **Group them 3 + 3 + 2:** where do the accents fall?
3. **Keep your foot on the beat.** How many foot taps?

Accents on 1, 4 and 7, but four foot taps: the bar is still **4/4**, only the accents are uneven.

EXAMPLE 3

**How many beats in 9/8?**



1. **The top number, 9:** does it divide by 3?
2. **How many groups of three** fit in 9?
3. **What is each beat**, in quavers?

9 ÷ 3 = **three beats**, each a dotted crotchet: “1 2 3, 4 5 6, 7 8 9”.
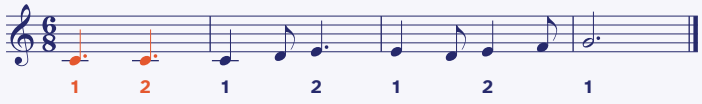
HOW TO

**Feel compound time**

1. **Say “pine-ap-ple”** on every beat and step only on “pine”.
2. **Clap the three syllables** while you step.

YOU KNOW THIS

**Row, Row, Row Your Boat**




Two big beats in every bar, each a **dotted crotchet**: “Row, row”. Each big beat is worth three quavers: “row your” is long-short, a crotchet and a quaver, filling one big beat. That is 6/8.

**Try it.** Sway twice per bar while you sing it.

THINK OF IT Like a skipping rope: two big swings, each with three little steps inside.

TRAP

- × Counting 6/8 as six slow beats at speed: feel two big ones.
- × Mixing up 3/4 and 6/8: the same six quavers, grouped differently.

SEE ALSO

Time signatures p. 59 · Rhythm: groups and upbeats p. 65

PRACTISE IT

Workbook Levels 1–4 · Compound time · Online: Rhythm clock

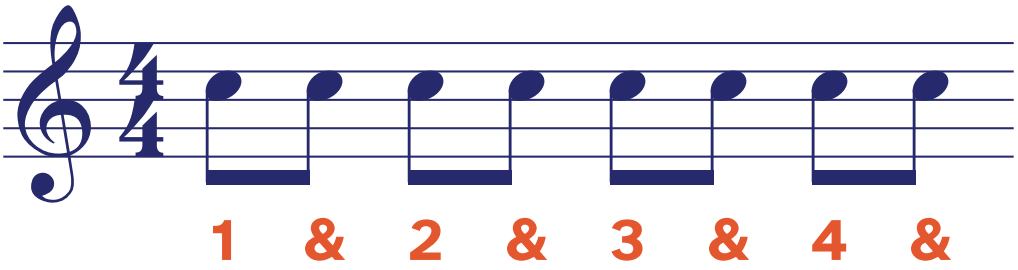
PART II · TIME · 6

# Rhythm: groups and upbeats

Say your own name: long and short sounds, that's **rhythm**. The pulse marches steadily on; the rhythm dances around it. How the notes are grouped on the page shows where the beats fall.

THE SECRET

**Beams group notes into beats; an upbeat borrows from the last bar.**




*One beam, one beat*

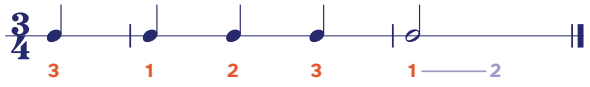
## Grouping with beams

Beams join the notes of **one beat**, so the beats stay visible. In 4/4 a beam may cover beats 1–2 or 3–4, but never cross the middle of the bar (except in a deliberate grouping such as 3 + 3 + 2). In compound time, quavers are beamed in threes.

× A beam across beats 2 and 3 would hide the middle of the bar.
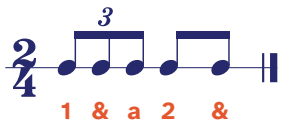
## The upbeat (anacrusis)

A piece can begin **before** the first full bar, with one or more notes called an upbeat or anacrusis. The last bar is then short by the same amount, so the two together make one full bar. Count the missing beats in silence before you start.




Count "1 2" silently, then play on 3.
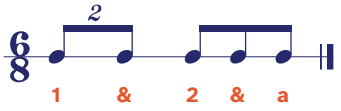
## Triplets




Triplet: three exactly equal notes in the time of two. First met: Level 2.

EXTRA · GOING FURTHER

### Duplet




Two notes in the time of three, in compound time: the triplet turned inside out.

PART II · TIME · 6 · CONTINUED




**Two quavers**: one beat cut in 2




A **triplet**: one beat cut in 3




**Four semiquavers**: one beat cut in 4

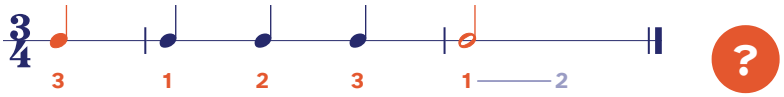
## Counting smaller values




Crotchet, two quavers, four semiquavers and a triplet (“trip-o-let”): one beat each.

EXAMPLE 1

**Does the upbeat balance the last bar?**




1. **The upbeat:** how many beats before the first full bar?
2. **The last bar:** how many beats?
3. **Add them:** do they make a whole bar of 3/4?

1 + 2 = 3: the upbeat and the last bar **make one full bar** together.

GAME

**May the beam join them?**

| 1 | 2 | 3 |
|---|---|---|
| beats 1 and 2 | beats 2 and 3 | beats 3 and 4 |

In 4/4, quavers are beamed so the middle of the bar stays visible. Which pairs may share one beam?

Answer: 1 yes · 2 no, that hides the middle of the bar · 3 yes.

TRY IT · 10 SECONDS



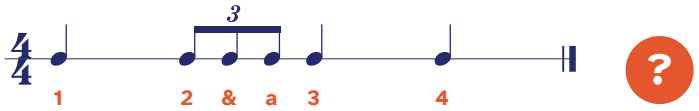
**Clap “1 & 2 & 3 & 4 &”.**

Now clap only the numbers, then only the &s, while your foot keeps the beat. The beat never moves; your claps do.

PART II · TIME · 6 · CONTINUED

EXAMPLE 2

**Where is the triplet, and how do you count it?**

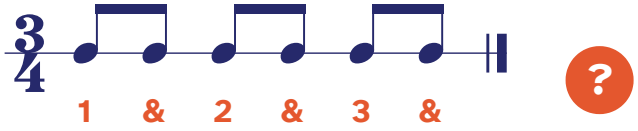



1. **Count the bar:** four crotchet beats.
2. **Find the little 3.** Which beat do the three quavers share?
3. **Say it** with "trip-o-let" on that beat.

Beat 2: "1, **trip-o-let**, 3, 4". Three notes squeeze into one beat; the beat never moves.

EXAMPLE 3

**Why are six quavers beamed in twos in 3/4?**




1. **How many beats** does 3/4 have, and what kind?
2. **How many quavers** fit in one beat?
3. **So how many** go under each beam?

Three crotchet beats, two quavers each: **2 + 2 + 2**. Beamed 3 + 3 they would look like 6/8.
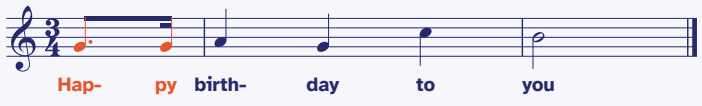
HOW TO

**Practise triplets**

1. **Say "trip-o-let"** on every beat, evenly.
2. **Clap the three syllables** while your foot keeps the beat.
3. **Switch** between two quavers ("1 &") and a triplet ("trip-o-let") every bar.

YOU KNOW THIS

**Happy Birthday**




"Hap-py" comes **before** the first strong beat: an upbeat. The strong beat lands on "birth".

**Try it.** Sing it and clap only on "birth", "day" and "you": you feel where bar 1 starts.

THINK OF IT An upbeat is the breath before you speak: "and-ONE". You lift on the upbeat so the strong beat can land.

SEE ALSO
Notes and rests p. 53 · Rhythm: off the beat p. 68

PRACTISE IT
Workbook Levels ½–4 · Rhythm · Online: Rhythm clock

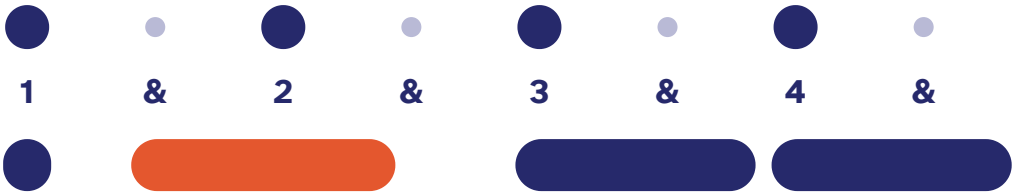
PART II · TIME · 7

# Rhythm: off the beat

Now the fun part: rhythm that teases the beat. It only works if the beat underneath stays rock-steady.

**THE SECRET**

**Keep the beat rock-steady and let the rhythm tease it.**




**starts on “&”, held over beat 2**

*A note between the beats*

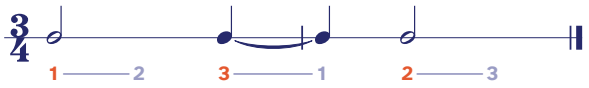
## Syncopation




The crotchet starts on “&” and is held through beat 2.

Level 4. Feel beat 2 even though nothing is played on it.

## Hemiola




Two bars of 3/4 felt as three minims.

Level 4. Common at the ends of phrases in Baroque dances.

## Swing Level ½

Common from the first pieces: jazz, blues and many easy modern pieces are marked *swing*. Pairs of quavers are written evenly but played **long–short**, like the first and last notes of a triplet.



Written



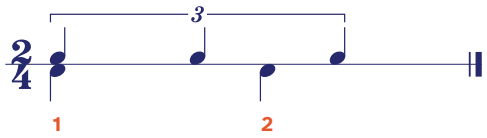

Played

PART II · TIME · 7 · CONTINUED

EXTRA · GOING FURTHER

## Polyrhythm and rubato

**Polyrhythm** (Level 7): two rhythms at once.

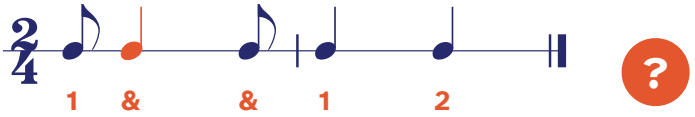



Three against two: a triplet above, two crotchets below, starting together.

**Rubato**: Time is stretched and then given back, for expression: the music leans into a phrase and catches up after it. The pulse always returns. Marked *rubato*, or simply part of the style in much music from the 1800s.

EXAMPLE 1

### Where does the long note start?



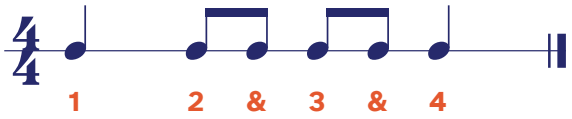

1. **Tap the beats:** 1, 2.
2. **Find the red crotchet.** Does it start on a beat, or on the "&"?
3. **Hold it:** through which beat?

It starts on the "&" and holds through beat 2: the long note starts **off the beat**. That is syncopation.

EXAMPLE 2

### Which notes swing?

**Swing**




1. **Find the quaver pairs.** On which beats?
2. **Play each pair long-short:** "doo-ba".
3. **What about the crotchets?**

Only the **pairs** (beats 2 and 3) swing; the crotchets and the beat underneath stay even.

TRY IT · 10 SECONDS




### Foot on the beat, hands on the &.

Tap a steady beat with your foot. Clap only between the taps. If the foot starts to follow the hands, slow down.

PART II · TIME · 7 · CONTINUED

GAME

**On the beat or off it?**

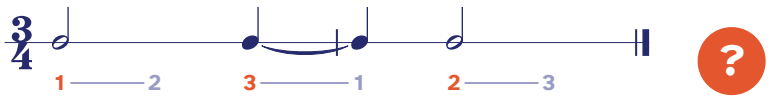
| 1 | 2 | 3 | 4 |
|---|---|---|---|
| starts on 1 | starts on & | starts on 2 | starts on the & of 3 |

A note that starts on a number is on the beat; on an "&" it is off the beat. Which are off?

Answer: 2 and 4 are off the beat; 1 and 3 are on it.

EXAMPLE 3

**Two bars of 3, or one big bar?**
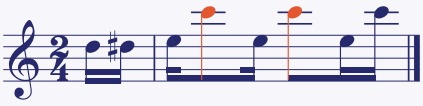



1. **Count as written:** "1 2 3, 1 2 3".
2. **Look at the notes:** how many beats does each minim last?
3. **Feel each minim** as one big beat. How many big beats?

Three minims of two beats: the six beats are felt as **three big ones**. That is a hemiola.

YOU KNOW THIS

**The Entertainer (Scott Joplin)**



The high **C** keeps landing just after the beat, not on it: syncopation, the bounce of ragtime.

**Try it.** Tap the beat with your foot and hum the tune: hear the C slip between the taps.

THINK OF IT Syncopation is a skip in your walk: a step that lands between the beats. Swing is walking with a spring in your step, long-short, long-short. Rubato is borrowing time like pocket money: whatever you spend, you pay back.

TRAP

- × Rushing the short notes after a long one: keep counting inside the long note.
- × Playing a triplet unevenly: three equal notes, not two quick and one long.
- × Letting syncopation drag the pulse: tap the beat with your foot.

SEE ALSO

Rhythm: groups and upbeats p. 65 · Words for how to play: Expression (later)

PRACTISE IT

Workbook Levels ½–4 · Rhythm · Online: Rhythm clock

PART II · TIME · 8

# Repeats and road signs

Music often says “play that again”. Instead of writing the same bars twice, it uses **repeat signs** and **road signs**. Read them like a map: they tell you which bar comes next.

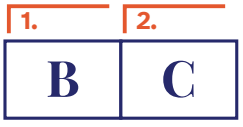
THE SECRET

**Follow the map: jump back at the sign, stop at Fine, or jump to the Coda.**




**You play: A → B → A → B → C**

*Two dots facing back mean: go back and play it again*

| SIGN | NAME | WHAT YOU DO |
|---|---|---|
|  | **repeat signs** | play the bars between them twice (with only the end sign: go back to the start) |
|  | **1st and 2nd time bars** | first time play the 1. bar; after the repeat skip it and play the 2. bar |
| **D.C.** | **da capo** | “from the head”: go back to the very beginning |
|  | **D.S.** (dal segno) | “from the sign”: go back to the segno |
| **Fine** | **fine** | “the end”: after a D.C. or D.S., stop here |
|  | **coda** | “tail”: at “To Coda”, jump to the coda sign and play the ending |

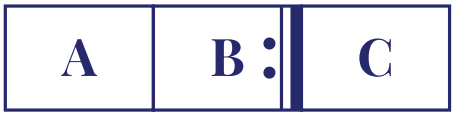
THE RULE

**Follow the map in order. At a jump (D.C., D.S.) go back; then the first time you meet Fine, stop; or at To Coda, jump to the coda.**

PART II · TIME · 8 · CONTINUED

EXAMPLE 1

**In what order do you play these bars?**



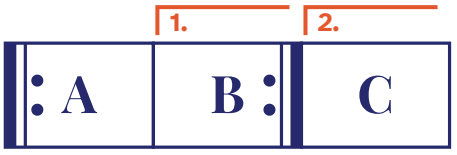

**You play:** ? ? ? ? ?

1. **Play A and B.**
2. **B ends with a repeat sign** facing back. There is no start sign: where do you go back to?
3. **Play it again,** then carry on.

Back to the very start: **A B A B C**.

EXAMPLE 2

**First and second time**




**You play:** ? ? ? ?

1. **First time through:** play A, then the bar marked 1.
2. **The repeat sign** sends you back to A.
3. **Second time:** which bar do you skip, and which do you play?

A B, A C: **A B A C**. The 1. bar is skipped the second time.

GAME

**How many bars do you play?**

| 1 | 2 | 3 |
|---|---|---|
| \|: A B :\| | A \|: B C :\| D | A B (Fine) C D.C. |

Each letter is one bar. Follow the signs and count every bar you play.

Answer: 1 four (A B A B) · 2 six (A B C B C D) · 3 five (A B C A B).

PART II · TIME · 8 · CONTINUED

EXAMPLE 3

### D.C. al Fine



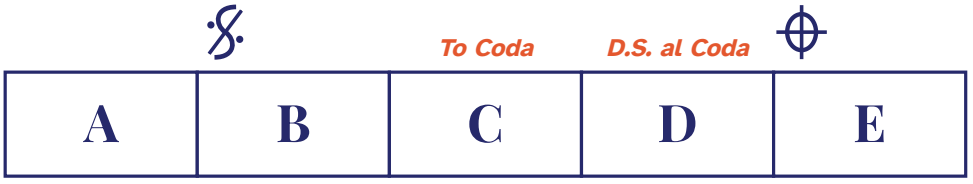

**You play:** ? ? ? ? ? ?

1. **Play straight through** to D.
2. **D.C.** means go back: to where?
3. **al Fine:** where do you stop the second time?

A B C D, then from the head to Fine: **A B C D A B**.

EXAMPLE 4

### D.S. al Coda




**You play:** ? ? ? ? ? ? ?

1. **Play through** to D. (Ignore "To Coda" the first time.)
2. **D.S.** sends you back to the sign: which bar has it?
3. **Play on to "To Coda",** then jump. Where?

A B C D, back to the sign, B C, jump to the coda: **A B C D B C E**.

TRY IT · 10 SECONDS




**Clap it, then play it.**

Clap A (four beats), clap B (four beats), back to A, B again. Say the letters as you go.

TRAP

- × Taking the repeat again after a D.C.: on the way back, skip the repeats (unless the music says otherwise).
- × Missing "To Coda" the first time through: it only works after the D.S. or D.C.

SEE ALSO

Pulse and the bar p. 50 · Time signatures p. 59

PRACTISE IT

Online: Library › Signs

PART II · TIME · TIPS

# When the rhythm slips

A rhythm that slips is nearly always a beat that was not counted. Find which beat, count it out loud, and the rest of the bar usually falls into place.

TIP

### What you notice, what to check

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| It speeds up or slows down | **the pulse**: count one bar aloud before you start | p. 50 |
| A rest gets skipped | **the rest**: it is counted just like a note | p. 53 |
| Long notes come out short | **the hold**: count every beat of a minim, dot or tie | p. 56 |
| The bar feels too long or too short | **the time signature**: how many counts, which note is one | p. 59 |
| An off-beat note lands on the beat | **the "&"**: keep the beat in your foot and count it | p. 68 |

## Four ways to steady a rhythm

1. **Count out loud**
   Say the counts while you play, not in your head: a slip you can hear is a slip you can fix.

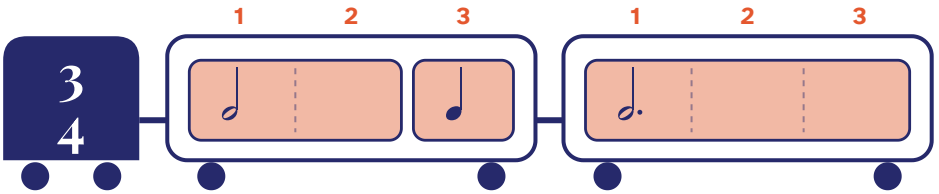
2. **Tap, then play**
   Tap the rhythm on the lid and say the counts; play only when the tapping is steady.

3. **Walk the beat**
   Step the pulse while you clap the rhythm, so the beat lives in your feet.

4. **Split the beat**
   For dots and quavers, count "1 & 2 &": the extra "&" shows exactly where each note goes.

EXAMPLE

### Count the long notes




1. **Say 1 2 3** once through before you play, to feel the beat.
2. **The minim books seats 1 and 2:** keep saying "2" while it sounds.
3. **The dotted minim books all three:** let go on the next "1", not before.

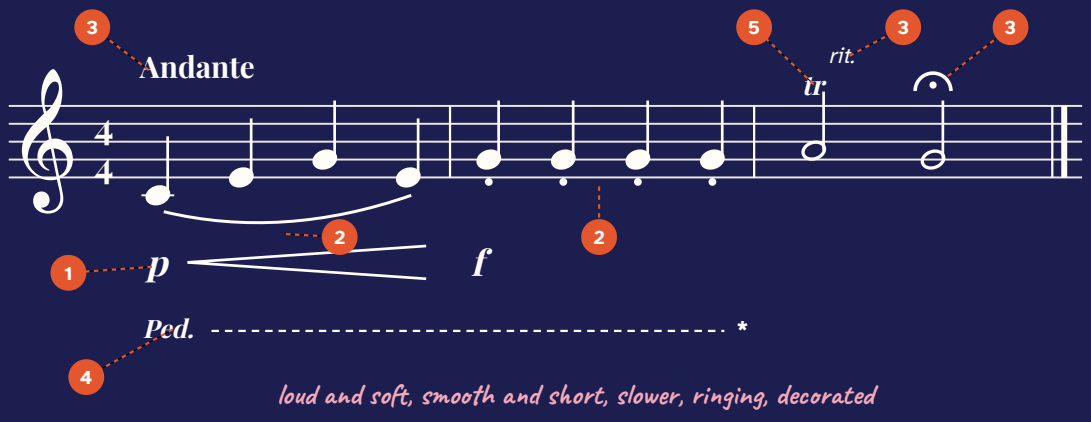
Long notes are where counting slips: **say every seat** they book.

THINK OF IT Like a dance partner: the beat leads, the rhythm follows.

PART III · HOW IT SHOULD SOUND

# Expression

The notes say **what** to play; expression says **how**: loud or soft, smooth or short, faster or slower, ringing or dry. Everything in this part is in these three bars.




1 **Loud and soft** p. 76
p, f and the steps between; hairpins grow and fade

2 **Touch: smooth, short, strong** p. 79
slurs, staccato dots, tenuto lines and accents

3 **Speed words** p. 82
Andante, Allegro; rit. slows down, a fermata waits

4 **The pedals** p. 85
Ped. down, * up: the right pedal keeps notes ringing

5 **Ornaments** p. 88
trills, mordents, turns and grace notes

★ **Tips: when your playing sounds flat** p. 91

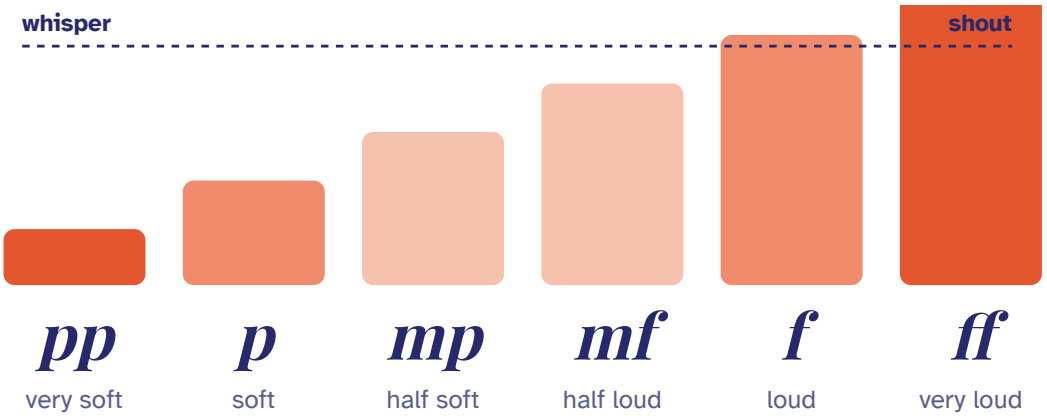
PART III · EXPRESSION · 1

# Loud and soft

A voice can whisper, chat or shout. Music does the same, and small Italian words tell you which: the **dynamics**. They sit under the notes, shortened to one or two letters.

**THE SECRET**

**Six steps, from a whisper to a shout: pp p mp mf f ff. m means half.**




*Six steps from a whisper to a shout*
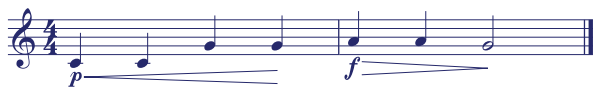
| SIGN | SAY IT | HOW IT SOUNDS |
|---|---|---|
| *pp* | **pianissimo** | very soft |
| *p* | **piano** | soft |
| *mp* | **mezzo-piano** | half soft: a little louder than *p* |
| *mf* | **mezzo-forte** | half loud: a little softer than *f* |
| *f* | **forte** | loud |
| *ff* | **fortissimo** | very loud |

**THE RULE**

**p is soft, f is loud; m (mezzo) means half; a second letter means very. A sign lasts until the next one.**

PART III · EXPRESSION · 1 · CONTINUED

## Growing and fading




A hairpin that opens grows louder; one that closes fades. From *p* to *f* and back.




The same idea in words: *cresc.* and dashes, then *ff*.

| SIGN | NAME | WHAT YOU DO |
|---|---|---|
|  | **crescendo** (*cresc.*) | getting louder, little by little |
|  | **diminuendo** (*dim.*, *decresc.*) | getting softer, little by little |
| ***sfz* / *sf*** | **sforzando** | one note made suddenly strong |
| ***fp*** | **forte-piano** | loud, then soft at once |
| *subito p* | **subito** | “suddenly”: *subito p*, suddenly soft |

EXAMPLE 1

**Which is louder: mp or mf?**
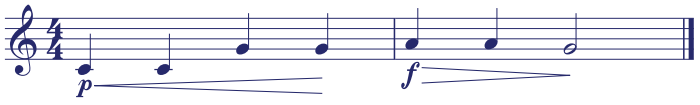
***mp*** ***mf*** ?

1. **m means half.** Half of what?
2. **p is soft, f is loud.**
3. **So which half** is louder?

**mf**, half loud, is louder than **mp**, half soft. Both sit in the middle of the ladder.

EXAMPLE 2

**What happens across the hairpins?**




?

1. **Find the start:** which sign?
2. **The first hairpin opens** to which sign?
3. **The second hairpin closes.** What happens?

Start *p*, grow through bar 1 into *f*, then **fade** through bar 2.

TRY IT · 10 SECONDS

***pp*** → ***ff***
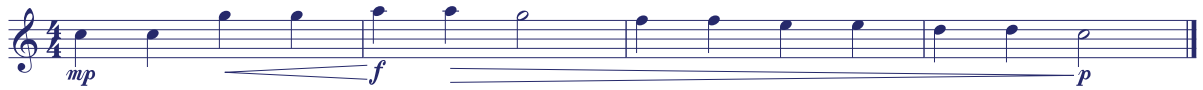
### One note, six ways.

Play Middle C *pp*, then *p*, *mp*, *mf*, *f*, *ff*. Six different steps? Then come back down.

PART III · EXPRESSION · 1 · CONTINUED

EXAMPLE 3

### Where is the loudest note?




?

1. **Start:** *mp*.
2. **Follow the hairpin** to its sign.
3. **Then the long hairpin closes** to which sign?

The **first A in bar 2**, at *f*. From the next note the tune fades all the way to *p*.

GAME

### Soft to loud

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| *mf* | *pp* | *f* | *p* | *ff* | *mp* |

Put the six signs in order from the softest to the loudest.

Answer: pp · p · mp · mf · f · ff.
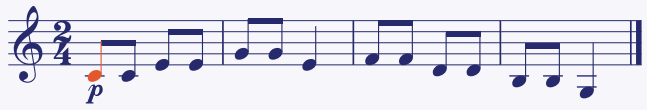
HOW TO

### Practise a crescendo

1. **Play five notes up**, C to G, all *p*.
2. **Now grow** one step on each note: *p* to *f*.
3. **Come back down** the same notes, fading to *p*, without slowing down.

YOU KNOW THIS

### Haydn's "Surprise" Symphony




The tune tiptoes along softly, then comes again even softer, *pp*. Just as everyone relaxes, the whole orchestra crashes in with one **ff** chord: the surprise that gave the symphony its name.

**Try it.** Hum it very softly, then clap once, as loud as you can.

THINK OF IT Dynamics are a volume knob with six clicks. Turn it with the weight of your arm, not by hitting harder.

TRAP

- × Playing everything *mf*: each sign is a different step.
- × Speeding up in a crescendo: louder is not faster.

SEE ALSO

Touch p. 79 · Pulse and tempo p. 50

PRACTISE IT

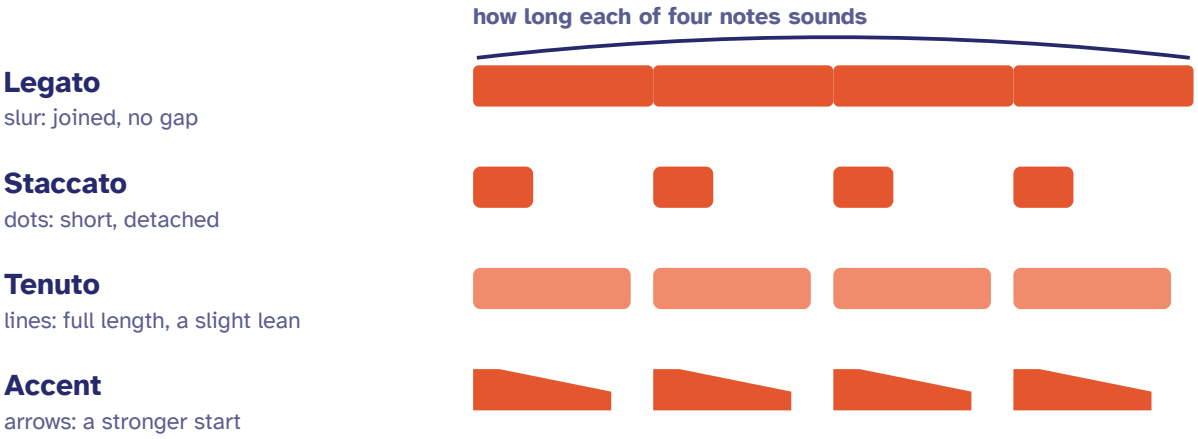
Online: Library › Dynamics

PART III · EXPRESSION · 2
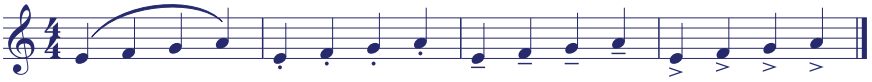
# Touch: smooth, short, strong

The same four notes can sound smooth or bouncy, gentle or strong. Small marks above or below the notes tell your fingers how to touch the keys: this is **articulation**.

**THE SECRET**
**Curve: join. Dot: short.**
**Line: hold. Arrow: strong.**
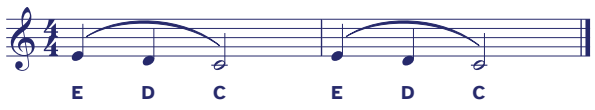



*Same notes, four touches*
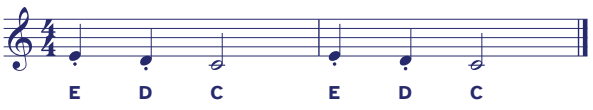


The same four notes: slurred, staccato, tenuto, accented.

| MARK | NAME | HOW TO PLAY |
|---|---|---|
| ⌒ | **slur** (*legato*) | join the notes smoothly, no gap |
| ● | **staccato** | short and detached: let go early |
| — | **tenuto** | hold the full length, leaning a little |
| > | **accent** | a stronger start to the note |
| ▼ | **staccatissimo** | very short *Extra* |
| ^ | **marcato** | very strong, marked *Extra* |

## Slurs and phrases




Slurred: join E D C, then a tiny lift.
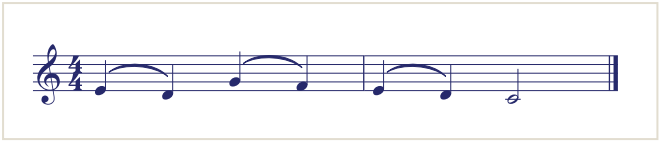



Staccato: every E and D let go early.

PART III · EXPRESSION · 2 · CONTINUED

A curve over **different** notes is a slur: join them, then lift at its end, like a breath. A long slur over a whole line of tune is a **phrase mark**: one musical sentence. (A curve joining the **same** note twice is a tie, p. 56.)

## Two-note slurs



A slur over just two notes is a little sigh: **drop** into the first note, then **lift** off the second, lighter and a little shorter. Drop, lift.
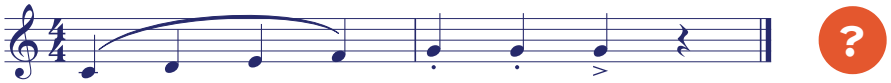
## Mood words

Some words describe the mood, not the speed.

| WORD | MEANING | WORD | MEANING |
|---|---|---|---|
| *dolce* | sweetly | *leggiero* | lightly |
| *cantabile* | in a singing style | *espressivo* | with feeling |
| *legato* | smoothly | *grazioso* | gracefully |
| *maestoso* | majestically | *con moto* | with movement |

EXAMPLE 1

**Which notes are short?**
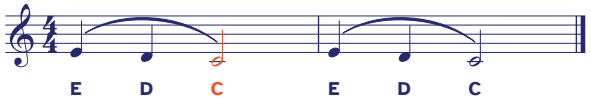


1. **Bar 1:** what does the curve ask?
2. **Bar 2:** find the dots.
3. **The last G** has a different mark: which?

Bar 1 is joined; the **two dotted Gs are short**; the third G is accented, stronger but full length.

EXAMPLE 2

**How do you play the end of a slur?**




1. **Join E D C** under the first curve.
2. **At the end of the curve:** hold, or lift?
3. **The next curve** starts on E again.

Join the notes, then **lift after the C**: a small breath before the next E, like a singer.

HOW TO
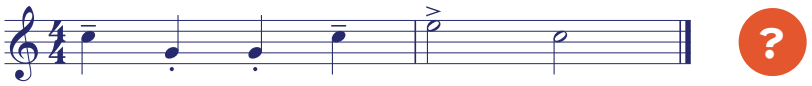
**Practise touch**

1. **Play one bar** of a piece three ways: all legato, all staccato, then as written.
2. **Listen:** can someone else hear which way was which?

PART III · EXPRESSION · 2 · CONTINUED

EXAMPLE 3

**Line, dot or arrow?**



1. **The first two Cs have lines:** how long?
2. **The Gs have dots.**
3. **The E in bar 2** has an arrow.

The first two Cs held their **full length**, the Gs short, the E strong, and the last C plain, held for its two beats.

GAME
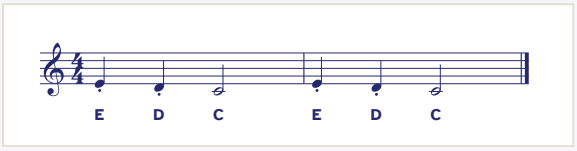
**Name the touch**

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| a dot | a line | > | a long curve |

Name each mark and say how the note sounds.

Answer: 1 staccato, short · 2 tenuto, full and leaning · 3 accent, a strong start · 4 slur, joined (over different notes).

TRY IT · 10 SECONDS
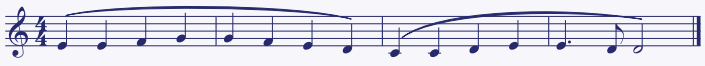



**Hot Cross Buns, two ways.**

Play E D C joined (fingers 3 2 1), then short and bouncy. Same notes, same fingers: two different tunes.

YOU KNOW THIS

**Ode to Joy (Beethoven)**



Two long slurs: two musical sentences. Lift your hand **only** at the end of each one, like a singer taking a breath.

**Try it.** Sing it and breathe only where the slurs end.

THINK OF IT Legato is walking: one foot leaves the ground as the other lands. Staccato is hopping on hot sand.

TRAP

- × Making staccato loud: short is not the same as strong.
- × Lifting in the middle of a slur: keep every note joined until the curve ends.

SEE ALSO

Loud and soft p. 76 · Tie or slur? p. 57

PRACTISE IT

Online: Library › Articulation

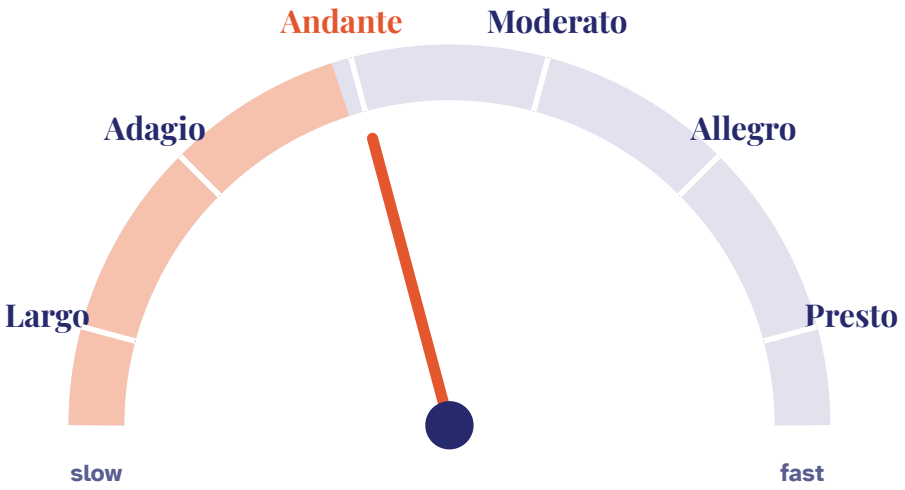
PART III · EXPRESSION · 3

# Speed words

How fast is the beat? A word at the top of the music tells you, usually in Italian, and often a number too. Other words inside the music ask you to slow down, speed up, or wait.

**THE SECRET**

**The word at the top sets the speed; rit. brakes gently, a tempo drives off again.**



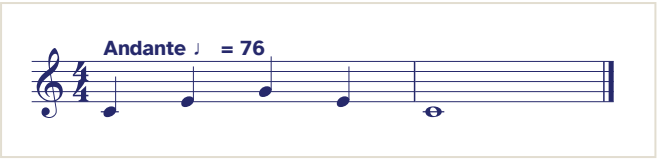

*Andante: the speed of walking*

| WORD | MEANING | ABOUT (BEATS A MINUTE) |
|---|---|---|
| **Largo** | very slow and broad | 40–60 |
| **Lento** | slow | 45–60 |
| **Adagio** | slow, at ease | 60–76 |
| **Andante** | at a walking pace | 76–108 |
| **Moderato** | at a moderate speed | 108–120 |
| **Allegretto** | fairly fast, a little slower than Allegro | 100–120 |
| **Allegro** | fast and lively | 120–156 |
| **Vivace** | lively and brisk | 152–168 |
| **Presto** | very fast | 168–200 |

The numbers are a rough guide: the music and the room decide the rest.

PART III · EXPRESSION · 3 · CONTINUED

## The metronome mark
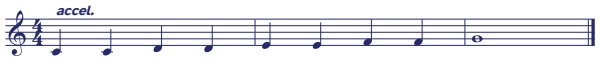



♩ **= 76** means 76 crotchet beats in one minute. Set a metronome to 76: each click is one beat. The word tells you the character; the number pins down the speed.

## Changing speed

| WORD OR SIGN | WHAT YOU DO |
|---|---|
| *rit.* (ritardando), *rall.* (rallentando) | gradually slower |
| *accel.* (accelerando) | gradually faster |
| *a tempo* | back to the speed before |
| 𝄐 fermata (pause) | hold longer than written, then go on |
| *poco a poco* | little by little |
| *molto* · *poco* · *più* · *meno* | very · a little · more · less (*più mosso*: faster; *meno mosso*: slower) |
| *rubato* | time borrowed and paid back (p. 69) *Extra* |




*rit.*: slow down into the end, then wait on the fermata.




*accel.*: speed up, little by little, to the end.

EXAMPLE 1

**How fast is ♩ = 60?**
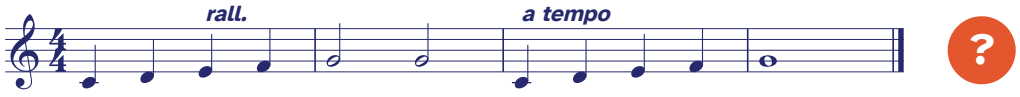
♩ = 60 ?

1. **60 beats** in one minute.
2. **How many seconds** are there in a minute?
3. **So one beat** lasts how long?

One beat every **second**: tick, tick, like a clock.

EXAMPLE 2

**Where does it slow down, and where does it come back?**
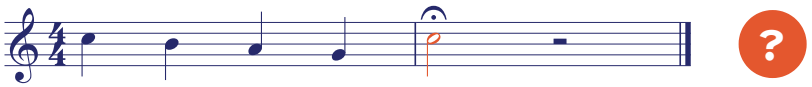



?

1. **Find rall.** Which notes does it slow?
2. **Find a tempo.**
3. **What speed** does a tempo mean?

Bar 1 slows into the two Gs; at **a tempo** (bar 3) the first speed comes straight back.

PART III · EXPRESSION · 3 · CONTINUED

EXAMPLE 3

**How long do you hold the fermata?**



1. **The minim** is worth how many beats?
2. **The fermata** asks for more. How much?
3. **Then what?**

Longer than 2 beats: hold until it **feels finished**, often half as long again or more, then go on.

GAME

**Slow to fast**

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| Allegro | Largo | Andante | Presto |

Put the four words in order from the slowest to the fastest.

Answer: Largo · Andante · Allegro · Presto.

TRY IT · 10 SECONDS
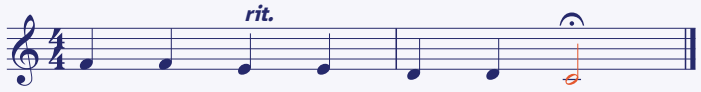
**Andante**

**Walk it.**

Walk across the room at a calm pace, saying 1 2 3 4 with your steps: that is Andante. Now jog on the spot: Allegro.

YOU KNOW THIS

**Twinkle, Twinkle, Little Star**




Players often slow down into the last “are”: a **rit.**, then a fermata on the final C. The ending feels like coming home.

**Try it.** Sing the last line twice: once in time, once with a rit.

THINK OF IT A rit. is a car braking gently at the lights; a tempo is driving off again at the same speed as before.

TRAP

- × Taking Andante as slow: it is a walking pace.
- × Slowing down all at once at rit.: slow little by little.
- × Forgetting a tempo: after a rit., the first speed comes back exactly.

SEE ALSO

Pulse and tempo p. 50 · Rubato p. 69

PRACTISE IT

Online: Metronome

PART III · EXPRESSION · 4

# The pedals

Most pianos have three pedals. The right one, the **sustain pedal**, lifts every damper off the strings, so notes keep ringing after your fingers let go. The left one makes the sound softer.

THE SECRET

**Play first, then change the pedal: up and straight down again.**



**soft pedal**
*una corda*
softer, gentler

**middle pedal**
*sostenuto*
holds only some notes

**sustain pedal**
*damper pedal*
notes keep ringing

*Right pedal: the notes keep ringing*

| PEDAL | NAME AND SIGN | WHAT IT DOES |
|---|---|---|
| **left** | soft pedal: *una corda*, written *u.c.* | a softer, gentler sound; lift it at *t.c.* (*tre corde*) |
| **middle** | *sostenuto* *Extra* | holds only the notes already down (on many uprights: a practice mute) |
| **right** | sustain (damper) pedal: *Ped.* | every note keeps ringing until you lift it |

## Writing the sustain pedal

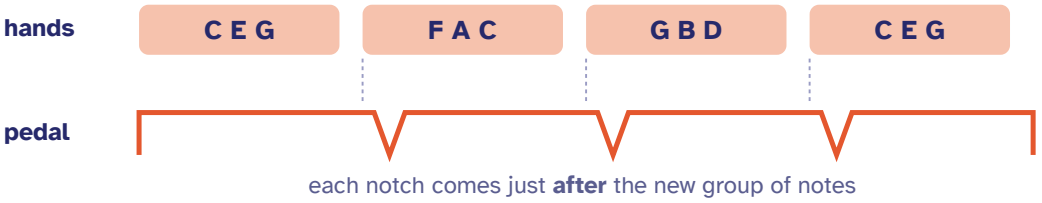


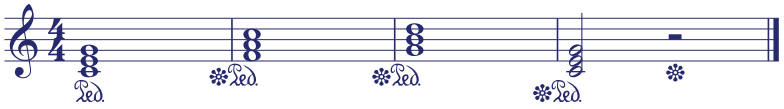

THE RULE

**Heel on the floor, press with the ball of your foot. Play first, then pedal: new notes, then up and straight down again.**

PART III · EXPRESSION · 4 · CONTINUED

## Changing the pedal



A chord is three or more notes played together. The ✻Ped. sits under each new chord: play it first, then change straight away.
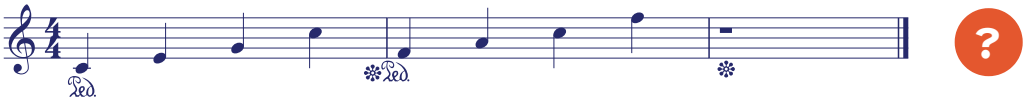
EXTRA · GOING FURTHER

### Half pedal and the middle pedal

**Half pedal:** lift the pedal only part of the way, to clear some of the sound but not all. **Sostenuto** (the middle pedal on grand pianos): press it while notes are held, and only those notes keep ringing.

EXAMPLE 1

**Where does the pedal go down, and where does it come up?**
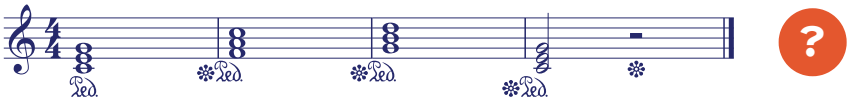


1. **Find Ped.** under the first C.
2. **Find the next sign:** a star with Ped.
3. **Find the last star.**

Down on the first C, **changed** at the first F (up and down again), and up at the end: each bar rings as one sound.

EXAMPLE 2

**Why change at every new chord?**



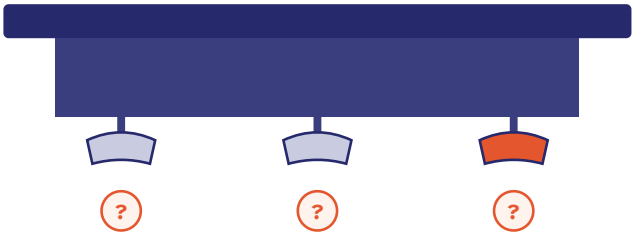
1. **Hold the pedal** from C-E-G into F-A-C. What do you hear?
2. **Now change** just after the F.
3. **Compare** the two.

Without a change, the two groups **blur** into a muddle. Change on each new chord: clean, and still joined.

PART III · EXPRESSION · 4 · CONTINUED

EXAMPLE 3

### Which pedal?




1 **Ped.**: which foot pedal? 2 **u.c.**: which one? 3 **t.c.**: what do you do?

Ped. is the **right** pedal; *u.c.* the **left** one; *t.c.* means let the left pedal up.

GAME
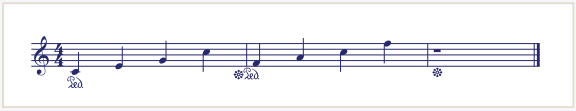
### Up or down?

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| Ped. | * | u.c. | t.c. |

Which pedal, and up or down?

Answer: 1 right pedal down · 2 right pedal up · 3 left pedal down · 4 left pedal up.

TRY IT · 10 SECONDS



### Hear the ringing.

Play C, E and G together, press the right pedal and take your hands off: the chord keeps sounding. Lift the pedal: silence.

HOW TO

### Practise the pedal change

1. **Without playing**, say “play, up-down” and move your foot on “up-down”.
2. **Play C-E-G, then F-A-C, then G-B-D**, slowly. Change just after each one.
3. **Listen** for a clean change with no gap of silence.

THINK OF IT The sustain pedal holds a door open: every note that walks in stays in the room until you let the door shut.

TRAP

- × Lifting the pedal before the new notes: a gap of silence. Play first, then change.
- × Keeping the pedal down through a new chord: muddy.

SEE ALSO
Touch p. 79 · Chords p. 110

PRACTISE IT
Online: Library › Pedal

PART III · EXPRESSION · 5

# Ornaments
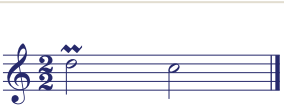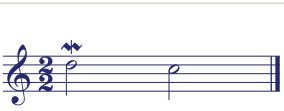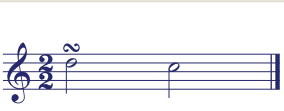
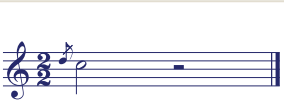
Ornaments are decorations: a small sign that stands for a quick burst of notes around the main note. They borrow the notes **next door**, one letter name up or down: on D, the note above is E and the note below is C.

**THE SECRET**

**An ornament is a necklace on the note: it borrows the next-door notes, and the main note still lands on time.**
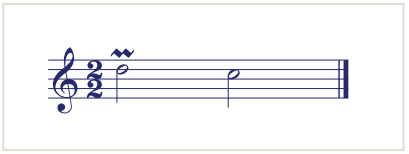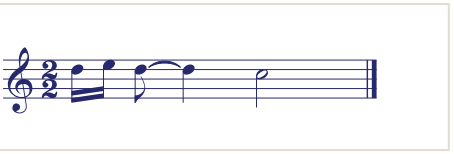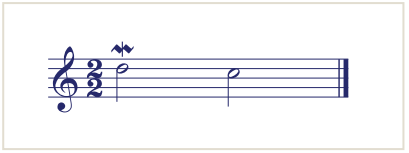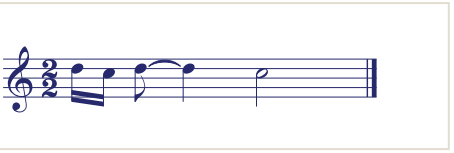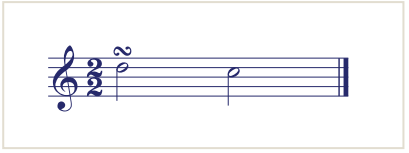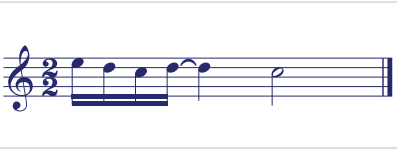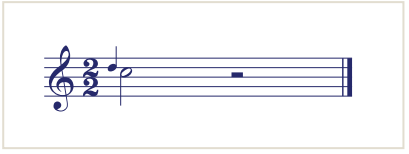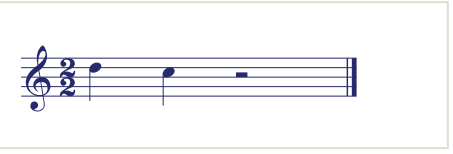



*A small sign, a burst of notes*

| WRITTEN | NAME | HOW TO PLAY |
|---|---|---|
|  | **trill** | the note and the one above, back and forth quickly |
|  | **upper mordent** | the note, the one above, the note: one quick flick |
|  | **lower mordent** | the note, the one below, the note |
|  | **turn** | above, the note, below, the note |
|  | **acciaccatura** | a tiny note with a slash: crushed in, as fast as you can |
|  | **appoggiatura** | a tiny note, no slash: leans, taking half the main note's time |

PART III · EXPRESSION · 5 · CONTINUED

## Written and played

| | WRITTEN | | PLAYED |
|---|---|---|---|
| **upper mordent** |  | → |  |
| **lower mordent** |  | → |  |
| **turn** |  | → |  |
| **appoggiatura** |  | → |  |

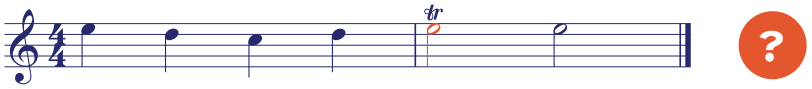
EXTRA · GOING FURTHER

### Where does a trill start?

In music from the 1700s (Bach, Handel), a trill usually starts on the note **above**: F E F E on an E. In later music it starts on the main note. Your teacher or the edition will tell you.

EXAMPLE 1

### How do you play this trill?




**?**

1. **Which note** has the tr?
2. **Its next-door note above** is...?
3. **Go back and forth** for how long?

On the E: **E F E F...** quickly for its two beats, then the last E on time.

GAME

### Above or below?

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| trill | upper mordent | lower mordent | turn |

Which next-door note does each one use: above, below, or both?

Answer: 1 above · 2 above · 3 below · 4 both.

PART III · EXPRESSION · 5 · CONTINUED

EXAMPLE 2

### Crushed, or leaning?
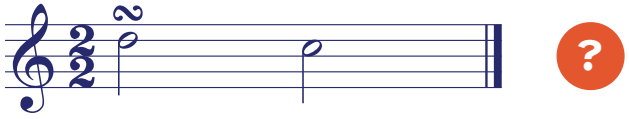


1. **Look for the slash** through the tiny note.
2. **With a slash:** how fast?
3. **Without:** how much time does it take?

Slash (acciaccatura): **crushed in**, almost at once. No slash (appoggiatura): **D for a crotchet**, then C.

EXAMPLE 3

### Which notes does the turn use?



1. **The main note** is D.
2. **Next door above**, and **next door below**?
3. **Order:** above, main, below, main.

**E D C D**, then hold the D: four quick notes, ending on the main note.

TRY IT · 10 SECONDS



### A slow trill.

Play D and E with fingers 2 and 3, slowly, eight times. Then a little faster. A trill is just that, at speed, with the beat still steady.

HOW TO

### Practise an ornament

1. **Play the main notes** alone first, in time.
2. **Add the ornament slowly**, as written-out notes.
3. **Speed it up** bit by bit, keeping the main note on the beat.

TRAP

- × Adding a sharp or flat that is not there: use the plain next-door letter.
- × Starting the ornament late: it begins on the beat (a crushed note just before it).

SEE ALSO
The keyboard p. 22 · Sharps and flats p. 39

PRACTISE IT
Online: Library › Ornaments

PART III · EXPRESSION · TIPS

# When your playing sounds flat

Every note can be right and the music still sound flat. The fix is almost always one sign you read but did not play. Find what you notice, then check that one thing.

TIP

### What you notice, what to check

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| Everything is the same loudness | **the dynamics:** is every sign a different step? | p. 76 |
| The tune sounds choppy | **the slurs:** join every note inside the curve | p. 79 |
| Short notes sound heavy | **staccato** is short, not loud | p. 79 |
| The end feels rushed | **rit. and fermata:** slow little by little, then wait | p. 82 |
| The sound is muddy | **the pedal:** change on each new chord | p. 85 |
| The ornament trips the beat | **the main note** lands on time; practise it slowly | p. 88 |

## Four ways to bring music to life

1. **Sing it first**
   Sing the phrase: your voice shapes it naturally. Then copy your voice on the keys.
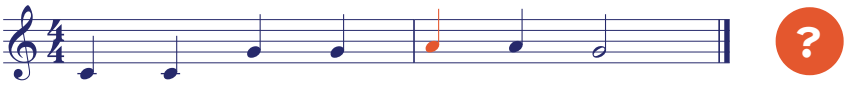
2. **Exaggerate**
   Play the dynamics twice as big as you think. It always sounds smaller to the listener.

3. **Record yourself**
   Listen back with the music: did each sign really happen?

4. **Shape every phrase**
   Grow towards the most important note, then let the phrase fall away.

EXAMPLE

### Give this phrase a shape



?

1. **Sing it:** where does it want to go?
2. **Which note** feels like the top of the hill?
3. **Where** would you start soft and end soft?

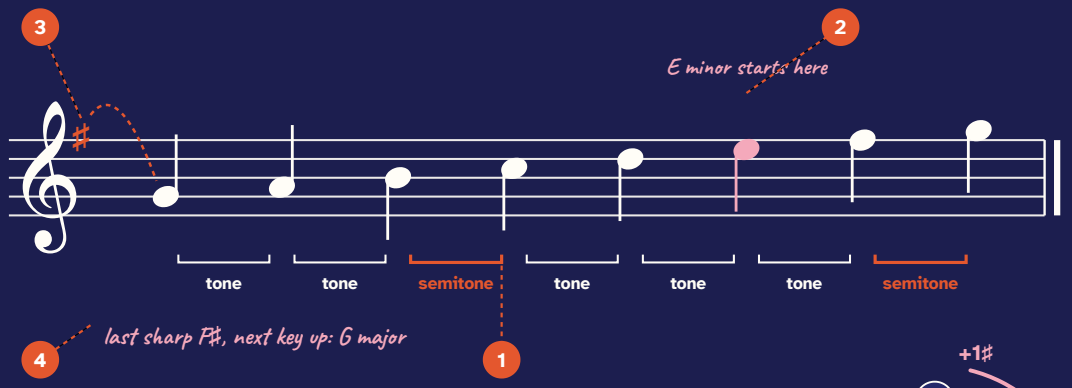
Start soft, grow to the first A, then fade: **a phrase is a hill**, up and down.

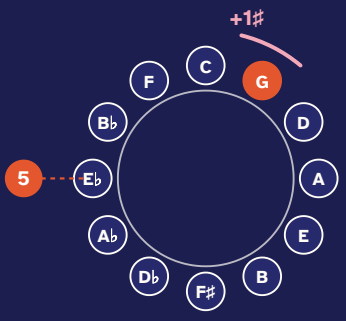
THINK OF IT Expression is the difference between reading a story aloud and telling it.

PART IV · NOTES IN FAMILIES

# Scales and keys

Notes travel in families. Build the scales that whole pieces are made from, then learn to read the key a piece lives in: its signature, its home note, its place on the circle. All of it is in one scale.




1 **Major scales** p. 93
T T S T T T S: the two short steps

2 **Minor scales** p. 96
the major with 3, 6 and 7 one key lower

3 **Keys and key signatures** p. 99
the sharps at the start apply all the way

4 **Finding the key** p. 102
last sharp, next key up; second-last flat

5 **The circle of fifths** p. 105
go right: one more sharp each step

★ **Tips: when a note sounds wrong** p. 108

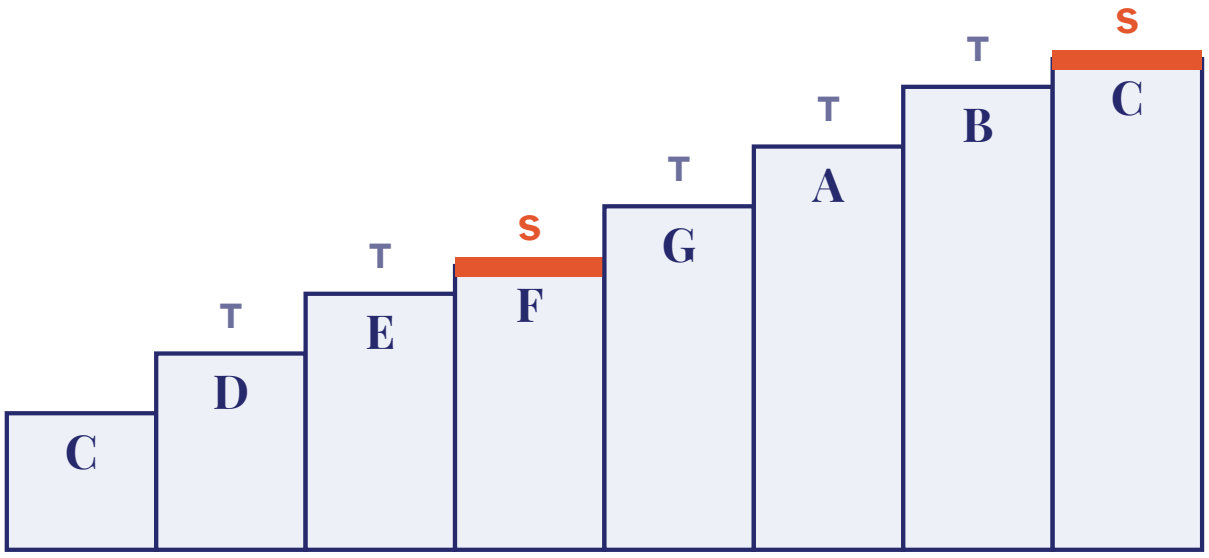
PART IV · SCALES AND KEYS · 1

# Major scales

A **scale** climbs from a home note, the **key note**, to the next note with the same name, using every letter once. A **major** scale always takes the same steps, whatever note it starts on.

THE SECRET

**Every major scale: tone, tone, semitone, tone, tone, tone, semitone.**



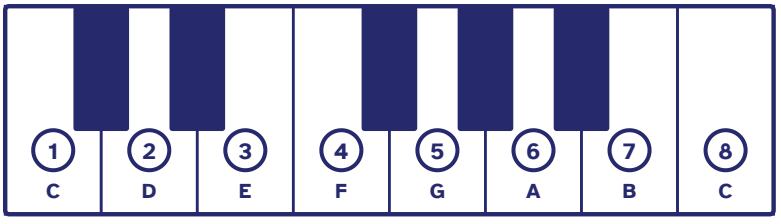

*Tone, tone, semitone, tone, tone, tone, semitone*

THE RULE

**T = tone = one key between. S = semitone = no key between: the very next key, best friends. Every major scale: T T S T T T S.**

EXAMPLE 1

**Where are the short steps?**




?

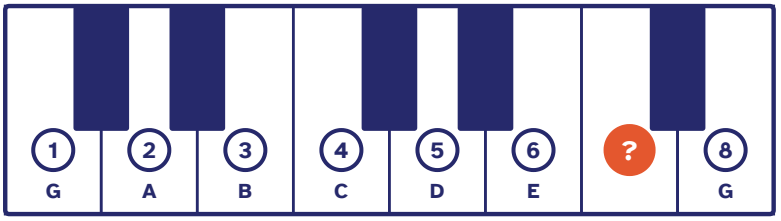
1. **Play C up to C** on the white keys, numbering the notes 1 to 8.
2. **Between each pair,** look for a black key. A key between means a tone.
3. **Which pairs have no black key** between them?

Only **3–4** (E–F) and **7–8** (B–C): nothing between. Those are the two short steps of every major scale.

PART IV · SCALES AND KEYS · 1 · CONTINUED

EXAMPLE 2

### Build G major

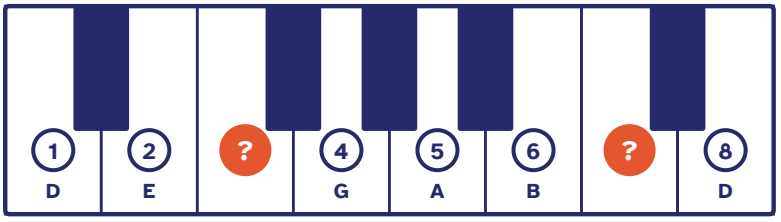



1. **Write the letters** G to G: G A B C D E F G.
2. **Check the top:** 6 to 7 must be T (one key between), 7 to 8 must be S (none between). E to F has nothing between.
3. **Move the ? one key right.** What is it called now?

The **?** becomes **F♯**. G major: G A B C D E F♯ G, one sharp.

EXAMPLE 3

### Build D major

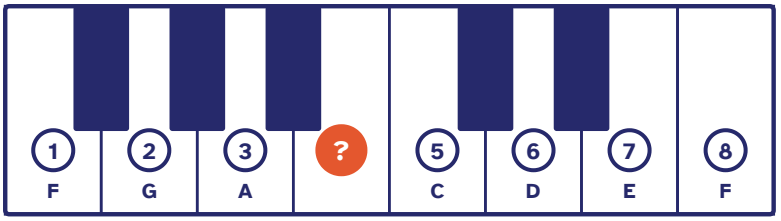



1. **Write the letters** D to D.
2. **Check each step** against T T S T T T S: is there a key between?
3. **Two steps are too short.** Move each ? one key right.

D major: D E **F♯** G A B **C♯** D, two sharps.

EXAMPLE 4

### Build F major



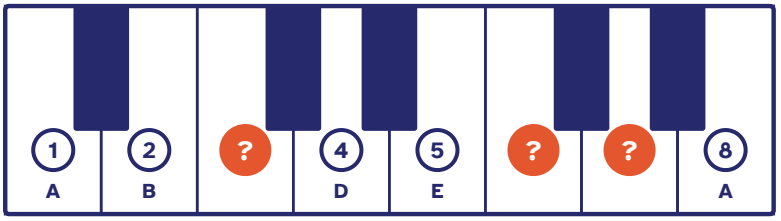

1. **Write the letters** F to F: F G A B C D E F.
2. **Check 3 to 4:** it must be S, nothing between. A to B has a key between: too big.
3. **Move the ? one key left.** What is it called now?

The **?** becomes **B♭**. F major: F G A B♭ C D E F, one flat.

PART IV · SCALES AND KEYS · 1 · CONTINUED

EXAMPLE 5

### Build A major




1. **Write the letters** A to A.
2. **Check each step:** three of them have nothing between where they need a key.
3. **Move each ? one key right.**

A major: A B **C♯** D E **F♯** **G♯** A, three sharps.

GAME

### Spot the wrong note

**1** G A B C D E F G

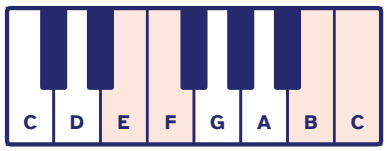
**2** D E F♯ G A B C D

**3** F G A B C D E F

Each of these major scales has one note that breaks T T S T T T S. Which one, and what should it be?

Answer: 1 F should be F♯ · 2 C should be C♯ · 3 B should be B♭.

TRY IT · 10 SECONDS
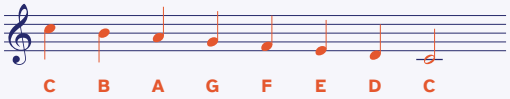



### One finger, C to C.

Play the white keys from C up to C and say “tone, tone, semitone, tone, tone, tone, semitone”. Feel the two short steps.

YOU KNOW THIS

### Joy to the World




The first line is the **C major scale**, top to bottom: C B A G F E D C.

**Try it.** Play C major down from the top C and sing “Joy to the world” over it.

TRAP

- × Using the same letter twice: every letter appears once (F♯ in G major, never G♭).
- × Forgetting the short step at the top: 7 to 8 never has a key between.

SEE ALSO

Minor scales p. 96 · Keys and key signatures p. 99

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Scales

PART IV · SCALES AND KEYS · 2

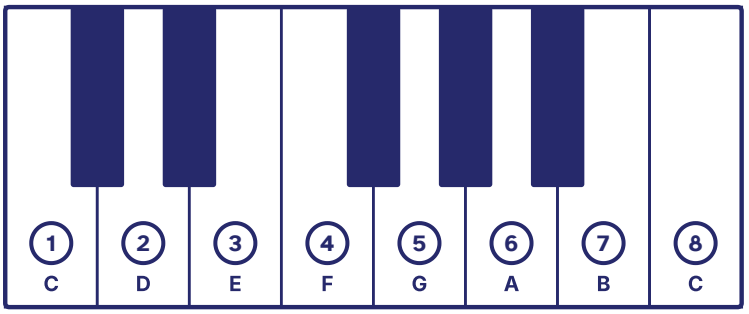
# Minor scales

Start from the major scale on the same key note and move three notes, the **3rd**, **6th** and **7th**, one key down. That is the **natural minor**: same home, darker colour.

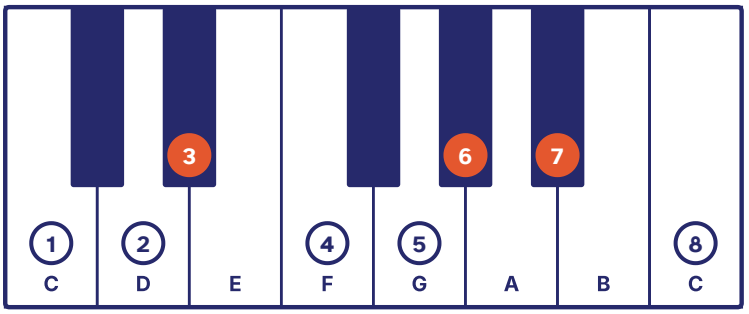
THE SECRET

**Minor = the major on the same note with 3, 6 and 7 one key lower.**

**C major**




**C minor: 3, 6 and 7 move one key left**
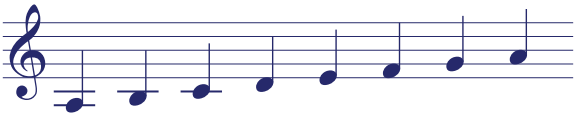



*The same home note · three notes one key lower*

THE RULE
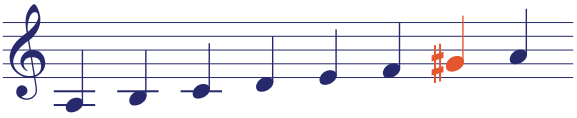
**Natural minor = the major with 3, 6 and 7 lowered one key. Harmonic minor = the natural minor with 7 raised again. Melodic minor = going up, 6 and 7 raised; coming down, the natural minor.**

## A minor in three forms
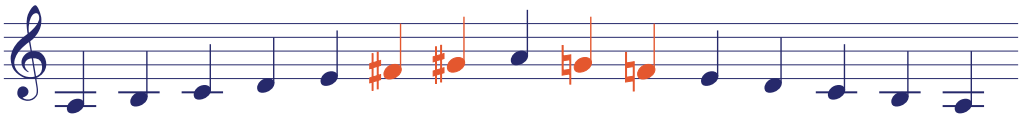


Natural: A major (F♯ C♯ G♯) with 3, 6 and 7 lowered: C, F and G. All white keys.

PART IV · SCALES AND KEYS · 2 · CONTINUED



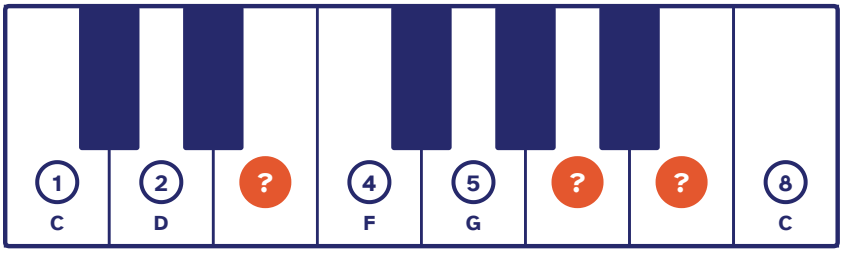
Harmonic: the 7th raised again, one key up: **G♯**.



Melodic: going up, 6 and 7 raised (**F♯ G♯**); coming down, back to **G F**.

EXAMPLE 1

### Build C natural minor
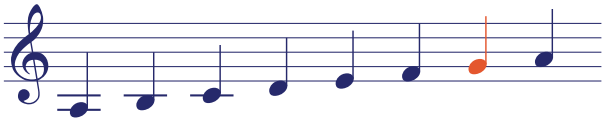



1. **Start from C major:** C D E F G A B C.
2. **The ? keys are 3, 6 and 7.** Move each one key left.
3. **Name the keys you land on.**

C natural minor: C D **E♭** F G **A♭** **B♭** C.

EXAMPLE 2

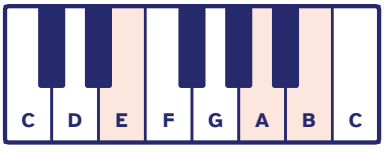
### Build A harmonic minor



1. **Start from A natural minor:** A B C D E F G A, all white.
2. **Take note 7** (the ?) and raise it one key.
3. **Name it.**

Note 7 becomes **G♯**: A harmonic minor is A B C D E F G♯ A.

TRY IT · 10 SECONDS



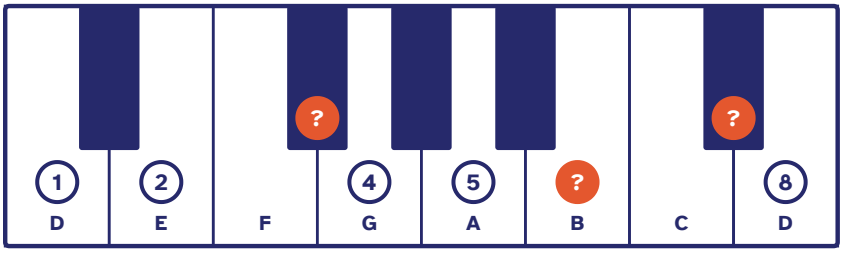

### Major, then minor.

Play C major, then play it again with E, A and B one key lower. Same home, new colour: bright, then dark.

PART IV · SCALES AND KEYS · 2 · CONTINUED

EXAMPLE 3

### Build D natural minor




1 **This is D major.** The ? keys are notes 3, 6 and 7.

2 **Move 3, 6 and 7** one key left each.

3 **Name the new keys.**

D natural minor: D E **F** G A **B♭** **C** D.

GAME

### Which minor?

1

**nothing raised**

2

**the 7th raised**

3
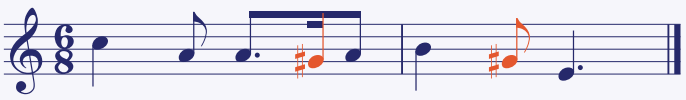
**6th and 7th raised going up, lowered coming down**

Three forms of the minor scale. Name each one.

Answer: 1 natural · 2 harmonic · 3 melodic.

YOU KNOW THIS

### Greensleeves



A minor, with a **G♯**: the raised 7th of the harmonic minor, leaning hard up to A.

**Try it.** Play it once with G♯, once with G: hear the pull disappear.

TRAP

- × Lowering the wrong notes: only 3, 6 and 7 move.
- × Playing the melodic minor's raised notes on the way down.

SEE ALSO

Major scales p. 93 · Keys and key signatures p. 99

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Scales

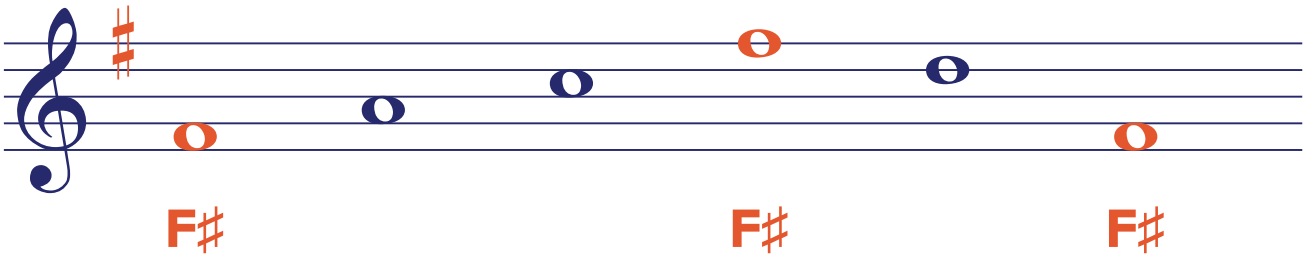
PART IV · SCALES AND KEYS · 3

# Keys and key signatures

Why write F♯ fifty times? The **key signature** at the start of each line says it once, for the whole piece. Its sharps or flats always line up in the same order.

THE SECRET

**The key signature says once, at the start, which notes are always sharp or flat.**




*One sharp at the start: every F plays F♯*



Sharps always come in this order: F C G D A E B.



Flats: B E A D G C F, the sharps backwards.

To remember the order: **F**ather **C**harles **G**oes **D**own **A**nd **E**nds **B**attle; backwards for the flats: **B**attle **E**nds **A**nd **D**own **G**oes **C**harles' **F**ather.

| KEY YOU BUILT | KEY SIGNATURE |
|---|---|
| C major | none |
| G major | 1♯: F♯ |
| D major | 2♯: F♯ C♯ |
| F major | 1♭: B♭ |

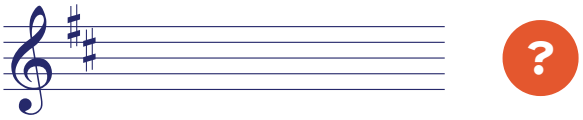
## One signature, two keys

Every key signature also belongs to a **minor** key: the **relative minor**. To find it, go **down a minor 3rd**: 2 keys between, then land. C major and A minor share every note: all white keys.

PART IV · SCALES AND KEYS · 3 · CONTINUED

EXAMPLE 1

### Which notes are sharp?

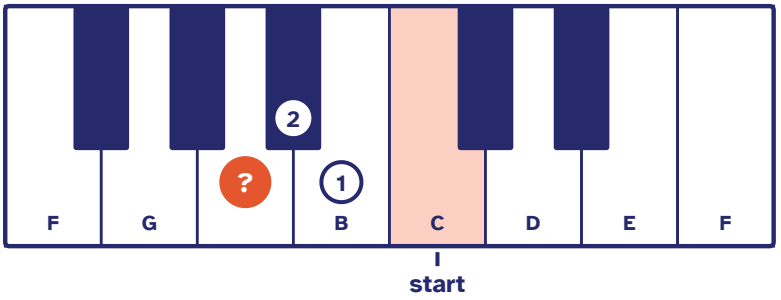



1. **Count the sharps.**
2. **Name them in order:** the first sharp is always F, then ...
3. **Which keys turn black** for the whole piece?

Two sharps, **F♯ and C♯**: every F and every C in the piece, high or low.

EXAMPLE 2

### Find the relative minor of C major

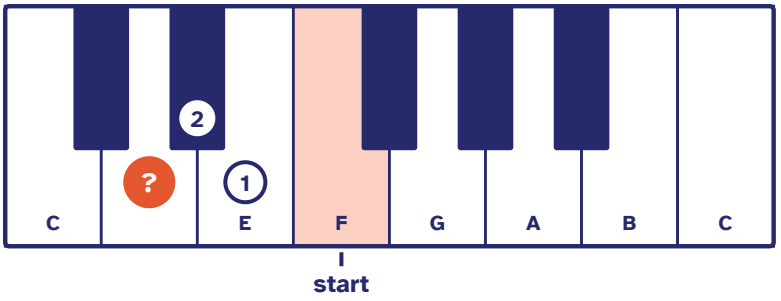



1. **Start on C**, the key note.
2. **Count 2 keys between, going down:** B is 1, B♭ is 2 ...
3. **The next key is the ?.**

You land on **A**: the relative minor of C major is **A minor**.

EXAMPLE 3

### Find the relative minor of F major




1. **Start on F.**
2. **Count 2 keys between, going down:** E is 1, E♭ is 2 ...
3. **The next key is the ?.**

The **?** is **D**: D minor, which shares F major's one flat, B♭.

PART IV · SCALES AND KEYS · 3 · CONTINUED

EXAMPLE 4

### Which notes are flat?



**1** **Count the flats.**

**2** **Name them in order:** the first flat is always B, then …

**3** **Which keys turn black?**

Three flats, **B♭ E♭ A♭**: every B, E and A in the piece.

GAME

### Which comes next?

**1**
sharps: F C G …

**2**
flats: B E …

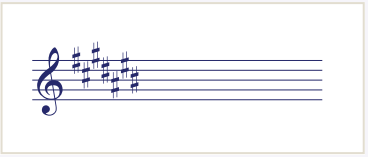
**3**
the 1st flat

**4**
the 2nd sharp

Use the orders of sharps and flats.

Answer: 1 D · 2 A · 3 B♭ · 4 C♯.

TRY IT · 10 SECONDS
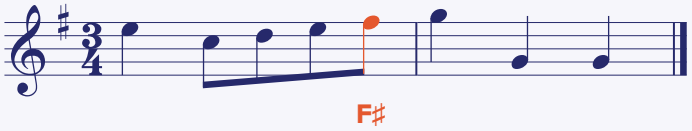


**Say the sharps in order.**

"Father Charles Goes Down And Ends Battle": F C G D A E B. Then backwards for the flats: B E A D G C F.

YOU KNOW THIS

### Minuet in G




One sharp in the key signature: every F is **F♯**, like the one just before the top G. One sharp, and the piece ends on G: **G major**.

**Try it.** Find every F in the piece and play it as F♯.

THINK OF IT A dress code on a party invitation: written once at the top, it applies to every guest all night. Only an accidental, a word to one guest, overrules it, and only until the bar line.

TRAP

- × Writing a key signature in the wrong order: sharps always F C G D A E B.
- × Reading it once and forgetting it: glance at the key signature at the start of every line.

SEE ALSO

Major scales p. 93 · Finding the key p. 102 · The circle of fifths p. 105

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Keys

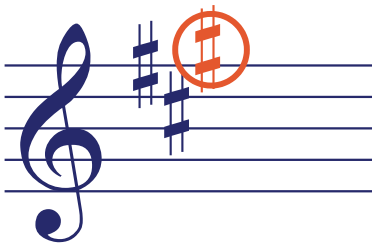
PART IV · SCALES AND KEYS · 4

# Finding the key

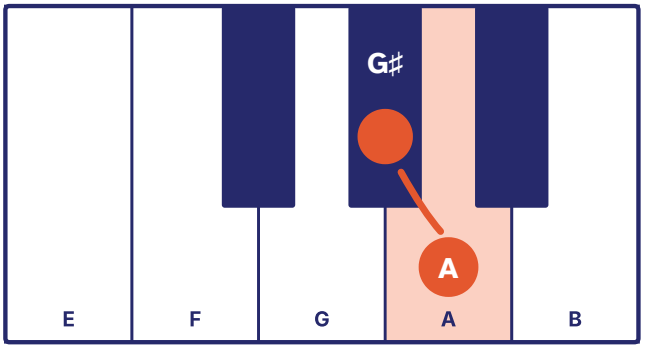
A key signature names two keys, a major and its relative minor. Two quick rules find the major; a minor 3rd down finds the minor; then the music itself tells you which of the two it is.

**THE SECRET**

**Last sharp, then the next key up. Second-last flat: that is the key.**



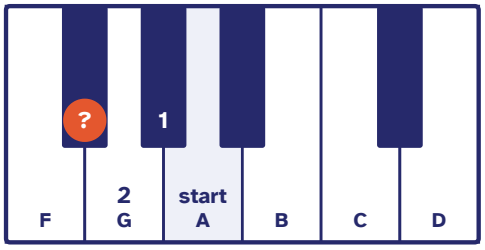
last sharp: G♯




*Sharps: the last sharp, then the very next key up. Three sharps: A major*



**Flats: the second-last flat is the key.**
B♭ E♭ A♭: the second-last is E♭, so **E♭ major**. One flat alone is F major.




**Major to minor: down a minor 3rd.**
From A: 2 keys between, G♯ and G, then land on **F♯**. Three sharps is A major **or** F♯ minor.

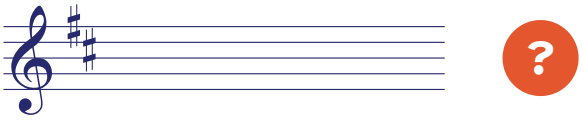
## Minor keys and their clue notes

| MINOR KEY | SIGNATURE | CLUE NOTE (THE RAISED 7TH) |
|---|---|---|
| A minor | none | G♯ |
| E minor | 1♯ | D♯ |
| D minor | 1♭ | C♯ |
| B minor | 2♯ | A♯ |
| G minor | 2♭ | F♯ |

PART IV · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 1

### Name the major key: sharps



1. **Find the last sharp,** the one furthest right.
2. **Name it** (F first, then C ...).
3. **Go to the very next key up.** That is the key.

The last sharp is C♯; the next key up is D: **D major**.

EXAMPLE 2

### Name the major key: flats
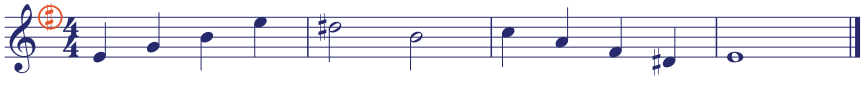


1. **Find the second-last flat.**
2. **Name it** (B first, then E ...).
3. **That flat names the key.**

B♭ E♭: the second-last is B♭: **B♭ major**.

EXAMPLE 3

### Major or minor?



1. **The key signature:** one sharp. Which major, and which minor a minor 3rd below it?
2. **Look for a sharp that is not in the signature.** A minor key raises its 7th: whose 7th is it?
3. **Look at the last note.**

One sharp, D♯ clues, ending on E: **E minor**, not G major.

GAME

### Name the major key

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| 2 sharps | 4 sharps | 1 flat | 3 flats |

Sharps: the last sharp, the very next key up. Flats: the second-last flat (one flat alone is F).

Answer: 1 D · 2 E · 3 F · 4 E♭.

PART IV · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 4

### Find a minor key from its clue



?

1. **Two flats:** B♭ major, or the minor a minor 3rd below it.
2. **Count 2 keys between going down** from B♭: A, A♭ … and land. Which minor key is it, and what is its 7th note?
3. **Raise that 7th**: if the piece is full of it, the key is minor.

Two flats with F♯ clues: **G minor** (its 7th, F, raised to F♯).

TRY IT · 10 SECONDS



### Three sharps, stop, step up.

Say “F C G”, stop: the last one is G♯. The next key up is A. Three sharps is A major. Try it with two and four.
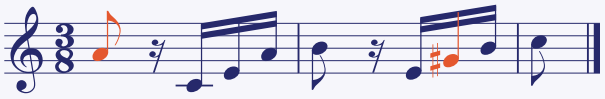
HOW TO

### Find the key of a new piece

1. **The key signature:** name the major key, and its relative minor a minor 3rd below.
2. **The clue notes:** look for the minor key’s raised 7th.
3. **The last bass note:** pieces nearly always end on the key note.

YOU KNOW THIS

### Für Elise (Beethoven)



No key signature, a G♯ as the clue, and the music keeps coming home to A: **A minor**.

**Try it.** Play A, then G♯ A: hear the G♯ lean into home.

THINK OF IT Detective work: the key signature narrows it to two suspects; the clue note and the last note name the culprit.

SEE ALSO
Keys and key signatures p. 99 · The circle of fifths p. 105

PRACTISE IT
Workbook Scales and keys 1 · Online: Library › Keys

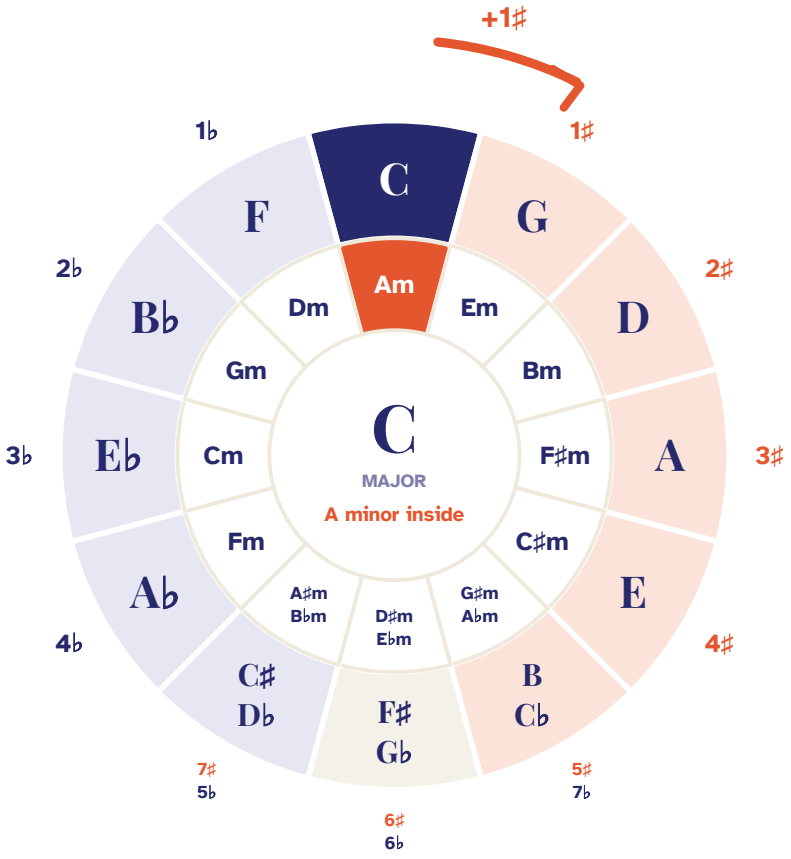
PART IV · SCALES AND KEYS · 5

# The circle of fifths

All the keys on one clock face. Go **right** round the circle and each key has **one more sharp**. Go **left** and each key has **one more flat**.

THE SECRET

**Go right: one more sharp. Go left: one more flat.**




*Right: one more sharp · left: one more flat*

## How to read the clock

**Outer ring:** the major keys. **Inner ring:** their relative minors. **Outside the circle:** how many sharps or flats.

## Neighbours are close

Keys side by side share all but one note, so music moves between them easily: from C major to G major or F major, or to the relative minor.

THE RULE

**Each step right is a 5th up: count the notes, C D E F G, and G is the next key round. That is why it is called the circle of fifths.**

PART IV · SCALES AND KEYS · 5 · CONTINUED

## The first steps round the circle

| STEPS | RIGHT (SHARPS) | LEFT (FLATS) |
|---|---|---|
| 1 | G major: F♯ | F major: B♭ |
| 2 | D major: F♯ C♯ | B♭ major: B♭ E♭ |
| 3 | A major: F♯ C♯ G♯ | E♭ major: B♭ E♭ A♭ |

EXAMPLE 1

### How many sharps in A major?




1. **Start at C**, at the top: no sharps.
2. **Go right** one step at a time: G, D …
3. **Count the steps** to A.

Three steps right: A major has **three sharps**, F♯ C♯ G♯.

EXAMPLE 2

### How many flats in E♭ major?

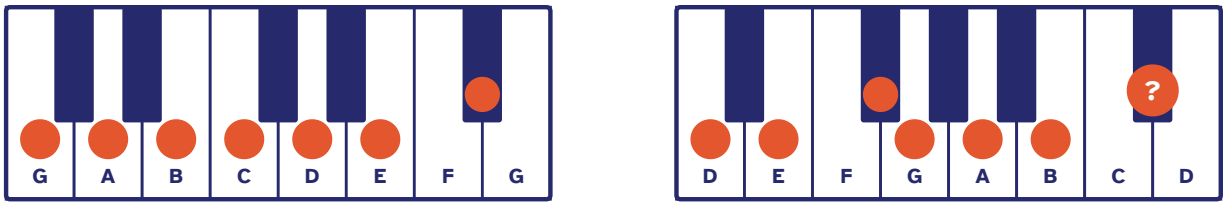



1. **Start at C** again.
2. **Go left:** F, B♭ …
3. **Count the steps** to E♭.

Three steps left: E♭ major has **three flats**, B♭ E♭ A♭.

EXAMPLE 3

### Why do G and D sound so close?



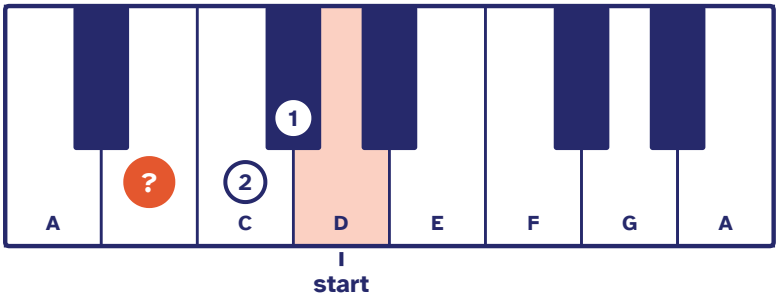

1. **G major** (left) and **D major** (right) are neighbours on the clock.
2. **Compare their notes:** which letters are the same?
3. **Which one letter changes?**

Six notes are the same; only C becomes **C♯**. One step round the circle changes just one note.

PART IV · SCALES AND KEYS · 5 · CONTINUED

EXAMPLE 4

### Find the minor inside D




1. **Start on D**, on the outer ring.
2. **Count 2 keys between, going down:** C♯ is 1, C is 2 …
3. **The next key is the ?.** It sits on the inner ring, just inside D.

The **?** is **B**: B minor, D major's relative minor. Both have two sharps.

GAME

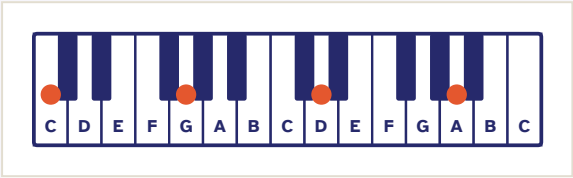
### Neighbours on the clock

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| right of D | left of C | opposite C | inside G |

Right adds a sharp. Inside each major sits its relative minor.

Answer: 1 A · 2 F · 3 F♯ / G♭ · 4 E minor.

TRY IT · 10 SECONDS




### Walk the circle: C G D A.

Play each key note with one finger, a 5th up each time. You just walked three steps right round the circle.

HOW TO

### Learn the circle by heart

1. **Right, out loud:** C G D A E B F♯.
2. **Left:** C F B♭ E♭ A♭ D♭ G♭.
3. **Check yourself:** the number of steps from C is the number of sharps or flats.

TRAP

- × Going the wrong way round: right adds sharps, left adds flats.
- × Forgetting the relative minors on the inner ring.

SEE ALSO

Keys and key signatures p. 99 · Finding the key p. 102 · Chords in a key p. 116

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Related keys

PART IV · SCALES AND KEYS · TIPS

# When a note sounds wrong

You read the note right, yet it sounds sour. The slip is usually in the key, not the note. Name the slip first, then fix only that note.
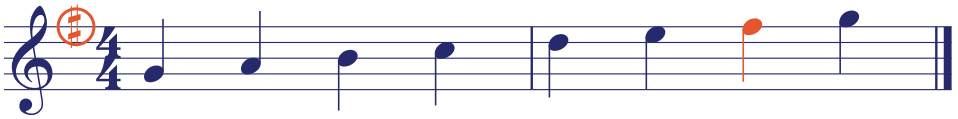
TIP

### What you notice, what to check

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| A note sounds sour, though you read it right | **the key signature**: is this note sharp or flat all the way through? | p. 99 |
| A key with many sharps or flats feels lost | **the circle**: go right for a sharp, left for a flat | p. 105 |
| A minor tune sounds wrong near the end | **the raised 7th**: harmonic and melodic minor raise it | p. 96 |
| A scale sounds wrong in the middle | **the short steps** (nothing between): 3–4 and 7–8 in every major scale | p. 93 |
| A sharp is missing from the key signature | **the order of sharps**: F C G D A E B | p. 99 |
| You are not sure which major key it is | **the last sharp**: the key is the very next key up | p. 102 |

EXAMPLE

### Find the slip



1. **Read the key signature:** one sharp, on the top line, F (ringed).
2. **The red note is an F**, so the sharp belongs to it too: every F, high or low.
3. **No natural cancels it**, so this F is F♯, here and all through the piece.

Play **F♯**, the black key. Plain F is the most common slip in G major.

HOW TO

### Three checks before you play

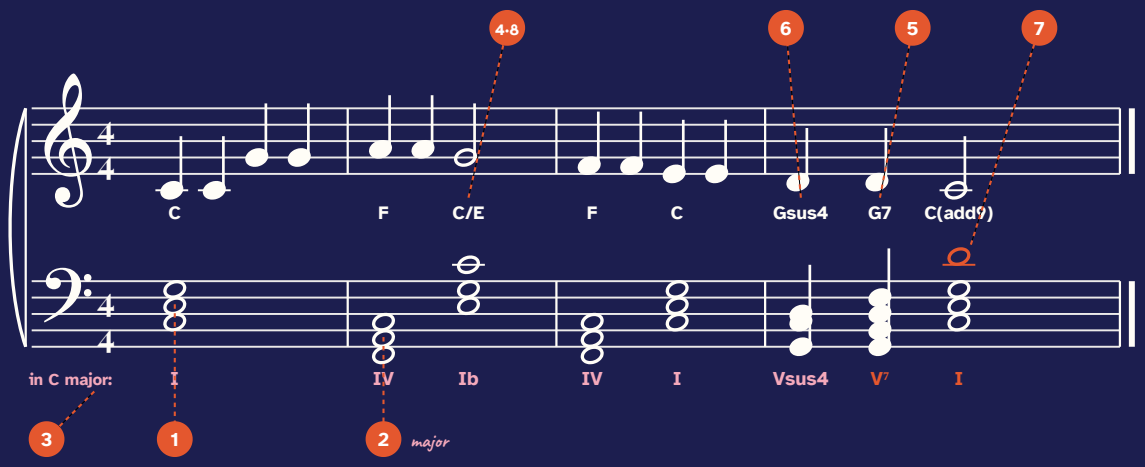
1. **The key signature:** say its sharps or flats aloud.
2. **The key:** name it from the last sharp or the second-last flat, then check the last note.
3. **The scale:** play it once, so your fingers know which black keys belong.

THINK OF IT Like proofreading: a spelling mistake jumps out once you know which words you usually get wrong. Keep a short list of your own slips, and check those first.

PART V · BUILD THEM

# Chords

Press three notes at once and you have a chord. Stack them, name them, turn them upside down, add layers on top. Every chapter of this part is in these four bars of Twinkle.




1 **Chords and triads** p. 110
three notes stacked like a layer cake

2 **Four kinds of triad** p. 113
major 3 + 2, minor 2 + 3

3 **Chords in a key** p. 116
Roman numerals: I, IV and V do most of the work

4 **Inversions** p. 119
the same notes, a different one at the bottom

5 **Seventh chords** p. 122
one more layer on top: G7

6 **Colour chords** p. 125
sus and add: a note moved or added

7 **The tower: 9, 11, 13** p. 128
keep skipping a letter

8 **Slash chords** p. 131
C/E: the chord, then the bass note

★ **Tips: when a chord sounds wrong** p. 134

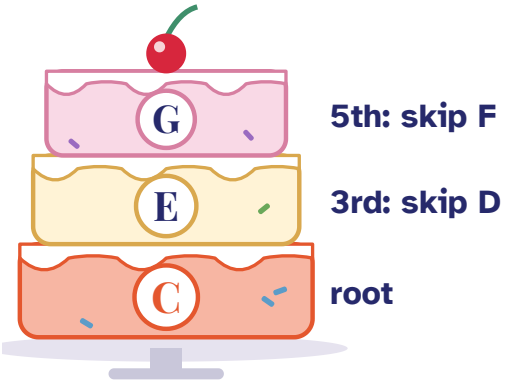
PART V · CHORDS · 1

# Chords and triads

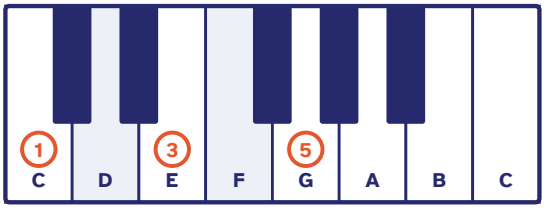
Press three or more notes at once and you have a **chord**. The favourite of all is the **triad**: three notes stacked like a layer cake.

THE SECRET

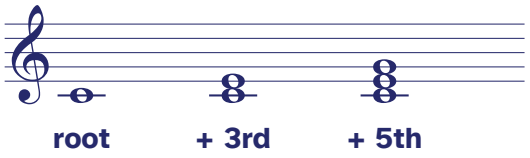
**Play one, skip one, play one: the cake rule.**




1. Pick a note: the **root**, the big layer at the bottom. It holds the chord up and gives it its name.
2. Skip one letter, take the next: **E**, the **3rd**.
3. Skip again: **G**, the **5th**. C E G is the triad of C.




On the keys: play one, skip one (shaded), play one, skip one, play one.




The cake on the staff: three lines in a row (or three spaces).

On the staff, each layer sits on the **next line up** (or the next space up): the root, a **3rd** above it, a **5th** above the root.

First met: Level ½ (the tonic triad of C, G and F major).

## All lines or all spaces

A stacked triad sits on **three lines in a row** or in **three spaces in a row**, in a neat stack. If one note breaks the pattern, it is not a stacked triad.




Lines, spaces, lines, spaces: every one a triad.

PART V · CHORDS · 1 · CONTINUED

## Solid and broken

Played together, a chord is **solid** (or blocked). Played one note at a time, it is **broken**. Spread up and down the keyboard, a broken chord becomes an **arpeggio**. Left-hand accompaniments, the arpeggios in technical work and the chord names over a song are all triads.

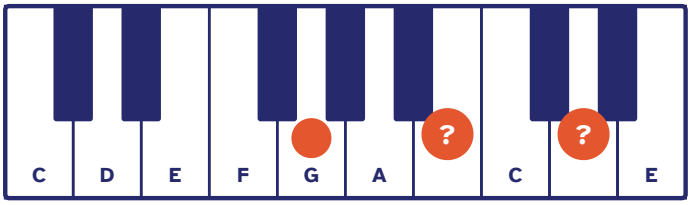
First met: Level 0; arpeggios from Level 1.




The same three notes, solid then broken.

EXAMPLE 1

### Build a triad on G

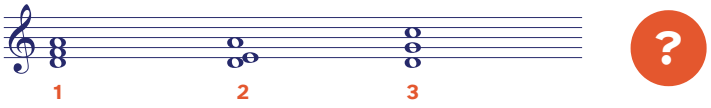



1. **Start on G** (the red dot): the root, the bottom layer.
2. **Skip one white key,** take the next: the first ?.
3. **Skip again,** take the next: the second ?.

G **B D**: the triad of **G**. Same shape as C E G, just moved up.

EXAMPLE 2

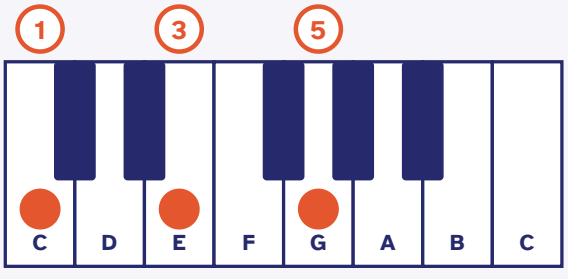
### Which one is a triad?




1. **Look at each chord:** are the three notes all on lines, or all in spaces?
2. **Count the notes** from each note to the next one up: is every gap a 3rd?
3. **Only one chord** passes both tests. Which?

**Chord 1** (D F A, three spaces in a row). Chord 2 has a 2nd (D–E); chord 3 is stacked in 4ths.

TRY IT · 10 SECONDS




**Thumb on C. Fingers 1, 3, 5 down together.**

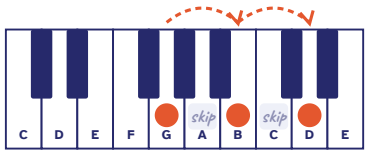
That is the C chord. Slide your hand to any white key and do it again: the cake shape always works.

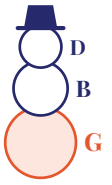
PART V · CHORDS · 1 · CONTINUED

GAME

## Build one yourself

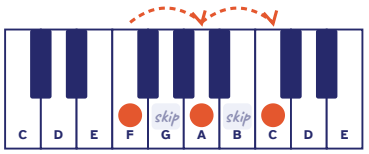
**G**

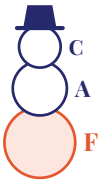



Start on G: **G B D**.

**F**

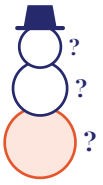



Start on F: **F A C**.
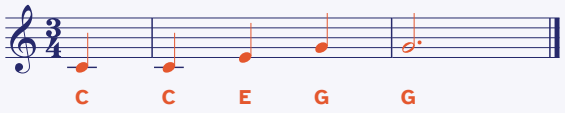
**?**




Your turn: start on **A**. Which three keys?

Build the cake on A: play one, skip one, play one, skip one, play one.

Answer: A C E: the triad on A.

YOU KNOW THIS

### The Blue Danube (Johann Strauss II), here in C




The most famous waltz tune simply climbs the **C triad**: **C, C E G, G**. Root, 3rd, 5th, then it holds the top: that is why it sounds so open.

**Try it.** Play C E G one at a time, then hum the waltz over it.

THINK OF IT Solid: three friends shout “Hello!” together. Broken: each says it in turn.

TRAP

- × Stacking 2nds or 4ths: a triad skips exactly one letter each time.
- × Naming a chord from its top note: the root names it (at the bottom when the notes are stacked in 3rds).

SEE ALSO
Counting intervals p. 42 · Major scales p. 93

PRACTISE IT
Workbook Harmony 1 · Online: Library › Chords

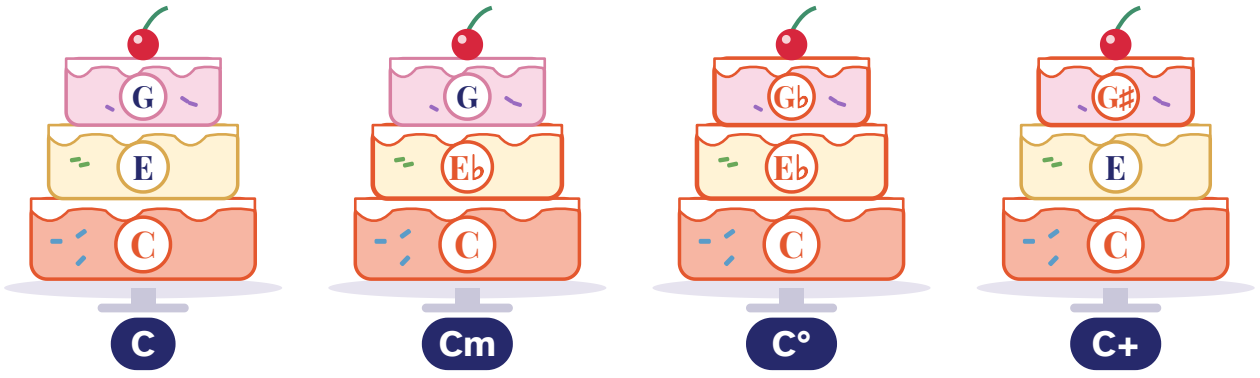
PART V · CHORDS · 2

# Four kinds of triad

Every triad is two 3rds stacked. To tell the kinds apart, count the keys **in between**, black and white, that you do not play: root to 3rd, then 3rd to 5th. Those two numbers name the kind, and its sound.

THE SECRET

**Count the keys between: 3 + 2 is major, 2 + 3 is minor.**

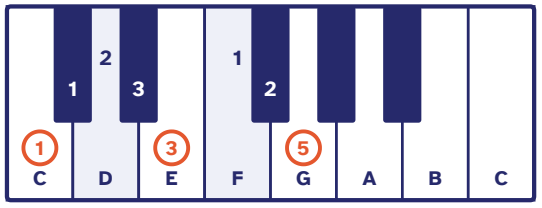



*Count the keys between: 3 + 2 is major, 2 + 3 is minor*

| KIND | KEYS BETWEEN | ON C | CHORD NAME | FIRST MET |
|---|---|---|---|---|
| **major** | **3 + 2** | C E G | C | Level ½ |
| **minor** | **2 + 3** | C E♭ G | Cm | Level 2 |
| **diminished** | **2 + 2** | C E♭ G♭ | C° or Cdim | Level 5 |
| **augmented** | **3 + 3** | C E G♯ | C+ or Caug | Level 6 |

Chord names: a letter alone means major (C); m means minor (Cm). You see them above the melody in songbooks and lead sheets: C | Am | F | G. In interval names, 3 keys between is a major 3rd and 2 is a minor 3rd (p. 45).

## Count the keys in between




Play C and E and count the keys you skip, black and white: C♯, D, D♯ is **3**. Then E to G: F, F♯ is **2**. **3 + 2 is major.** Any root works the same way: from D, skip 3 to F♯, then 2 to A: D F♯ A, D major.

PART V · CHORDS · 2 · CONTINUED

## The seven white-key triads, sorted



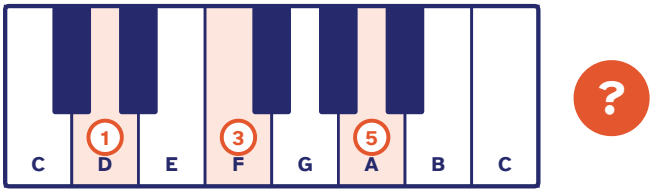

Always the same on the white keys: **three major** (C, F, G), **three minor** (D, E, A), **one diminished** (B), no augmented. Which ones belong to a key is the next chapter (p. 116).

## How they sound

| KIND | SOUNDS | HEARD IN |
|---|---|---|
| major | bright, settled, open | most cheerful tunes, fanfares |
| minor | darker, softer, sadder | lullabies, laments, film tension |
| diminished | tense, unstable, wants to move | suspense, spooky scenes |
| augmented | floating, dreamy, unsettled | dream scenes, a passing colour |

EXAMPLE 1

### Major or minor? D F A




1. **Root to 3rd:** count the keys between D and F, black and white.
2. **3rd to 5th:** count the keys between F and A.
3. **3 + 2 or 2 + 3?**

D to F has 2 between, F to A has 3: **2 + 3**, so D F A is **D minor** (Dm).

GAME

### Which kind?

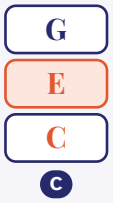
| 1 | 2 | 3 |
|---|---|---|
| G♭ | A | B♯ |
| E♭ | F♯ | G♯ |
| C | D | E |

Count the keys between: 3 + 2 major, 2 + 3 minor, 2 + 2 diminished, 3 + 3 augmented.

Answer: 1 diminished · 2 major · 3 augmented.

PART V · CHORDS · 2 · CONTINUED

TRY IT · 10 SECONDS

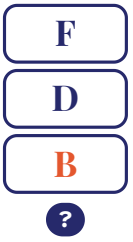



**C E G, then C E♭ G, then C E♭ G♭.**

Bright, dark, tense. Only the middle layer moves first, then the top one. Listen for the change each time.

EXAMPLE 2

**What kind is the triad on B?**
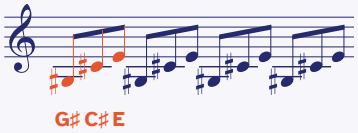



1. **Skip a letter:** B, D, F.
2. **Count the keys between** B and D, then D and F.
3. **Which kind has those two numbers?**

2 + 2: B D F is **B diminished** (B°), the only diminished triad on the white keys.

YOU KNOW THIS

**Beethoven, Moonlight Sonata (first movement)**




The famous slow triplets are the notes of a **C♯ minor triad**: C♯, E and G♯, rocking gently. Minor is why it sounds so dark and sad.

THINK OF IT Two gaps in the cake: a wide one (3 keys between) and a narrow one (2). Major: wide, then narrow, 3 + 2. Minor: narrow, then wide, 2 + 3. Diminished: two narrow, 2 + 2. Augmented: two wide, 3 + 3.

TRAP

- × Counting the keys you play: count only the keys in between, black and white.
- × Forgetting that in diminished and augmented triads the 5th changes too.
- × Spelling with the wrong letter: a triad skips one letter, so C minor is C E♭ G, never C D♯ G.

SEE ALSO
Chords and triads p. 110 · Major, minor and perfect p. 45

PRACTISE IT
Workbook Harmony 2 · Online: Library › Chords

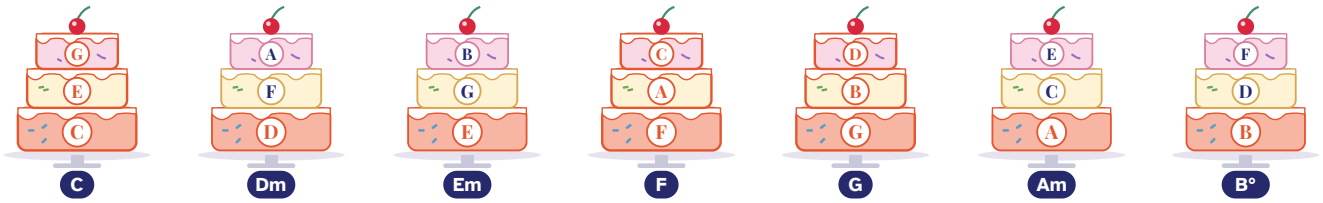
PART V · CHORDS · 3

# Chords in a key
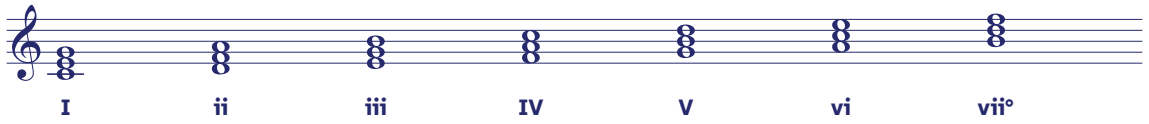
Build a triad on every note of a scale and you have the chords of that key. **Roman numerals** name each one by its place in the scale: I on the 1st note, IV on the 4th, V on the 5th. Capitals are major, small numerals are minor, and small with ° is diminished.

**THE SECRET**

**Build a cake on every note of the scale: I, IV and V are the strongest.**




*One cake on every note: I, IV and V do most of the work*




The seven chords of C major. Only white keys, so the kinds come out as on p. 114.

| NUMERAL | NAME | KIND | KEYS BETWEEN | IN C | IN G |
|---|---|---|---|---|---|
| **I** | tonic | major | 3 + 2 | C | G |
| ii | supertonic | minor | 2 + 3 | Dm | Am |
| iii | mediant | minor | 2 + 3 | Em | Bm |
| **IV** | subdominant | major | 3 + 2 | F | C |
| **V** | dominant | major | 3 + 2 | G | D |
| vi | submediant | minor | 2 + 3 | Am | Em |
| vii° | leading note | diminished | 2 + 2 | B° | F♯° |

The same in every major key: I, IV and V are major; ii, iii and vi are minor; vii° is diminished. First met: I, IV and V at Levels 1–2; the other chords from about Level 4.

## The three primary chords
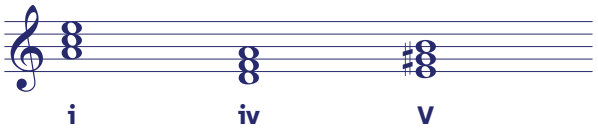
**I, IV and V** are the strongest chords of a key. Between them they hold **every note of the scale**: C E G, F A C, G B D. That is why so many tunes need only these three.

**THINK OF IT** I is the key's own chord. IV and V are its two big neighbours. Between the three of them, every note of the scale has a chord to sit in.

PART V · CHORDS · 3 · CONTINUED

## In a minor key

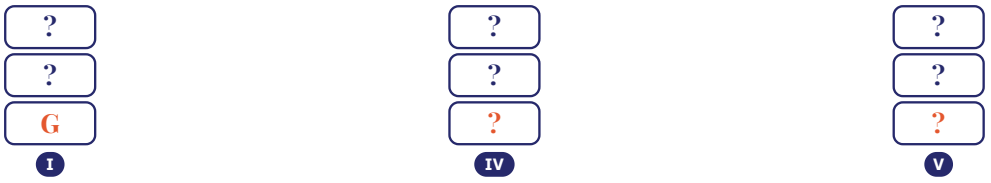
In A minor the three main chords are **i** (A minor), **iv** (D minor) and **V**: E G♯ B, made major by the **raised 7th** of the harmonic minor (p. 96), G♯. Check: E G♯ B is 3 + 2, major.




A minor: i, iv and V. The G♯ makes V major.

EXAMPLE 1

### Find I, IV and V in G major




1. **Count up the scale** from G: which notes are 1st, 4th and 5th?
2. **Build a cake on each,** using the key signature (one sharp: F♯).
3. **Check the keys between:** are all three 3 + 2?

In G major: **I = G** (G B D), **IV = C** (C E G), **V = D** (D F♯ A), all major.

GAME

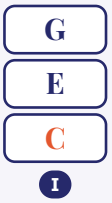
### Which numeral?




All in C major. Count up the scale from C to find each chord's number.

Answer: 1 IV · 2 V · 3 vi · 4 ii.

TRY IT · 10 SECONDS



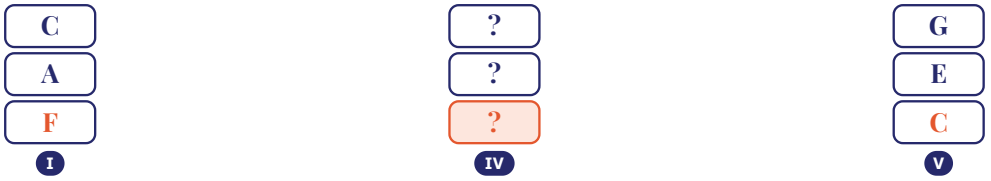

### Left hand: C, F, G.

Play the cakes of I, IV and V in C major, then sing any nursery tune over them: these three fit almost every note.

PART V · CHORDS · 3 · CONTINUED

EXAMPLE 2

## Find IV in F major




1. **Count up the scale** from F to the 4th note. Watch the key signature: one flat.
2. **Build the cake** on that note.
3. **Check the keys between.**

IV in F major is **B♭**: B♭ D F, 3 + 2, major.

EXAMPLE 3

## Find V in D minor




1. **The 5th note** of D minor is A: build A C E.
2. **The 7th note** of the D minor scale is C. In the harmonic minor it is raised: to what?
3. **Put it in the middle layer.** What kind is the chord now?

C becomes **C♯**: V in D minor is A C♯ E, **A major**.

EXAMPLE 4

## The chord on note 6 of C major




1. **Build the cake on A:** skip one, take the next, twice.
2. **Count the keys between:** major or minor?
3. **Note 6 of the scale:** which numeral, capital or small?

A C E, 2 + 3: **A minor**, numeral **vi** (small, because it is minor).

TRAP

- × Writing every numeral big: ii, iii and vi are minor, so they are written small.
- × Forgetting the raised 7th in a minor key: V in A minor is E G♯ B, not E G B.

SEE ALSO

Four kinds of triad p. 113 · Major scales p. 93 · Inversions p. 119

PRACTISE IT

Workbook Harmony 3 · Online: Library › Chords

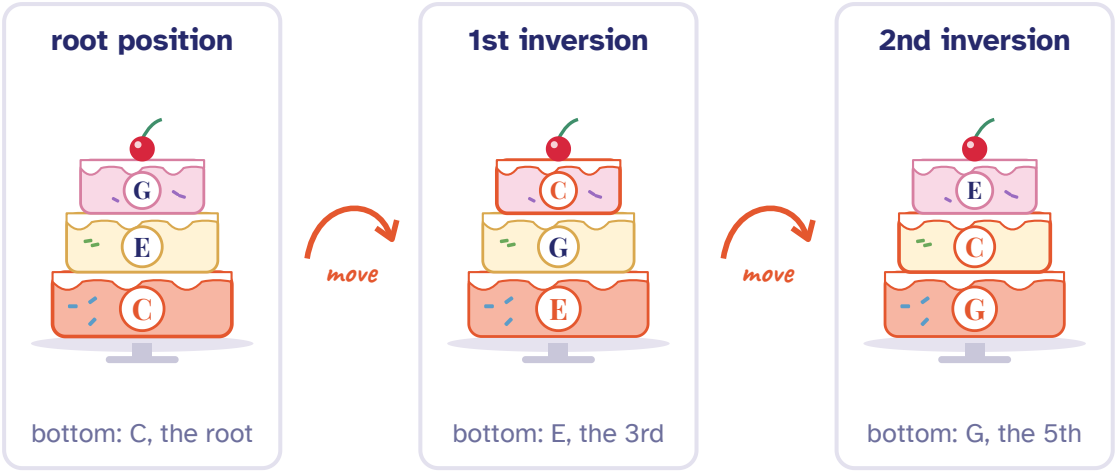
PART V · CHORDS · 4

# Inversions
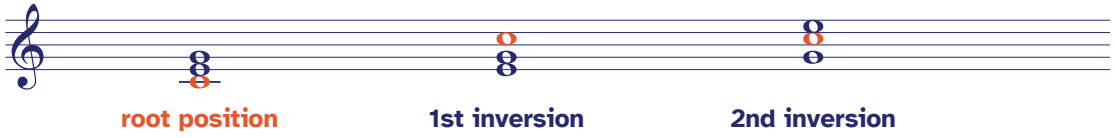
Same three notes, a different one at the bottom. The chord keeps its name; only the bass changes. The red layer is the root, C: watch it travel.

**THE SECRET**

**Move the bottom note to the top: same chord, new bass.**




1. Start with the cake **C E G**: root at the bottom.
2. Move the bottom layer to the top: **E G C**. That is the 1st inversion.
3. Move it again: **G C E**, the 2nd inversion. Still **C major**.




The same three chords on the staff; the red note is the root, C. To find the root, restack the notes every other letter (like C E G): then the bottom layer is the root.

## Why inversions?

They let your hand **move as little as possible**: keep any note two chords share, and move the others to the nearest note of the next chord. Left-hand accompaniments and hymn parts are written this way.

**GAME**

### Which position?

| 1 | 2 | 3 |
|---|---|---|
| C E G | E G C | G C E |

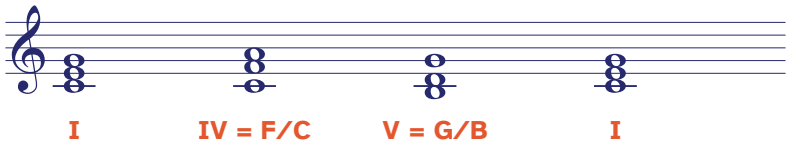
Same three notes. Look at the bottom one: root, 3rd or 5th?

Answer: 1 root position · 2 first inversion (E at the bottom) · 3 second inversion (G at the bottom).

PART V · CHORDS · 4 · CONTINUED

EXAMPLE 1

### Move as little as possible: C, F, G, C




1. **I:** C E G. **IV:** keep C, move E up to F and G up to A: C F A.
2. **V:** no shared note. C steps down to B, F down to D, A down to G: B D G.
3. **I again:** B up to C, D up to E, G stays: C E G.

F stands on C, G stands on B: **inversions** keep the hand in one place.

TRY IT · 10 SECONDS




### Climb the chord.

Play C E G, then E G C, then G C E, then C E G an octave up. Same cake, changing its bottom layer as it climbs.

EXAMPLE 2

### Write F major in first inversion




1. **Build F major** in root position first.
2. **First inversion:** which layer goes to the bottom, the 3rd or the 5th?
3. **Stack the other two** above it, in order.

The 3rd, A, goes to the bottom: **A C F**.

EXAMPLE 3

### Find the second inversion of G




1. **Build G major:** skip one, take the next, from G.
2. **Second inversion:** the 5th goes to the bottom.
3. **Stack the other two** above it.

The 5th, D, at the bottom: **D G B**.

PART V · CHORDS · 4 · CONTINUED

EXAMPLE 4

### Find the root of E G C




1. **Restack the notes** so that each skips one letter: try starting on each note.
2. **Which order works?** C E G, E G C or G C E?
3. **The bottom layer** of that stack is the root.

C E G works: the root is **C**. E G C is C major, 1st inversion.

EXAMPLE 5

### Which position? A D F
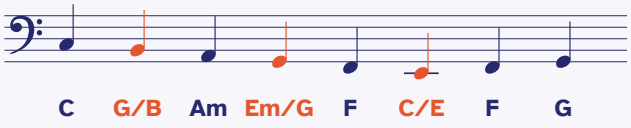



1. **Restack it** so each note skips one letter.
2. **Find the root** (the bottom layer of the stack) and the kind.
3. **Which layer is at the bottom** of A D F: root, 3rd or 5th?

Restacked: D F A, **D minor**. A, the 5th, is at the bottom: **2nd inversion**.

YOU KNOW THIS

### Pachelbel's Canon, the bass (here in C)




In the popular version the bass walks **straight down the scale**: C B A G F E. It can, because some chords are **inversions**: G stands on B, Em on G, C on E.

**Try it.** Play the bass slowly and hum the famous violin tune over it.

TRAP

- × Naming a chord from its bass note: in an inversion the bass is not the root. Restack it in 3rds first.
- × Thinking an inversion is a new chord: E G C is still C major.

SEE ALSO

Chords and triads p. 110 · Chords in a key p. 116 · Slash chords p. 131

PRACTISE IT

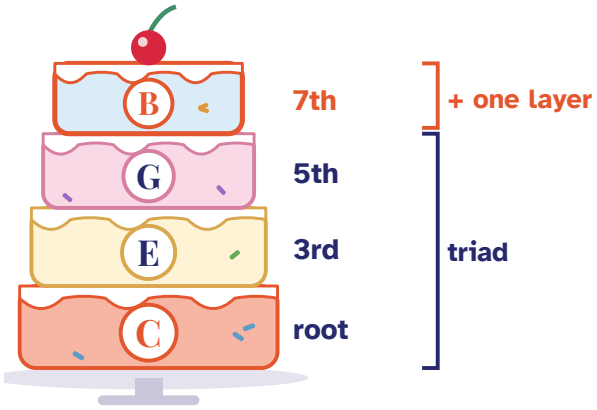
Workbook Harmony 4 · Online: Library › Chords

PART V · CHORDS · 5

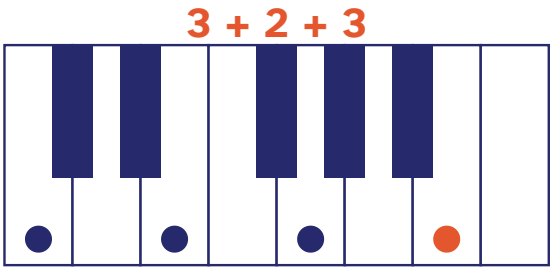
# Seventh chords

Add one more layer to the cake and the chord changes colour: dreamy, bluesy, spooky. Same rule as before: skip one note, stack the next. The new note is the **7th**.

**THE SECRET**
**One more layer on the cake and the chord changes colour.**

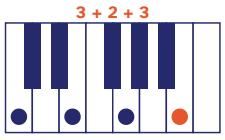



1. Build the triad: **C E G**.
2. Skip one, add the next: **B**.
3. Count the keys between: **3 + 2 + 3**. That is **Cmaj7**.



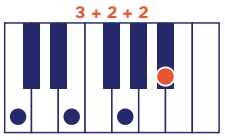

## Five flavours, all on C

**Cmaj7**



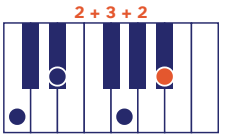

C E G B
**dreamy**
bossa nova, film

**C7**



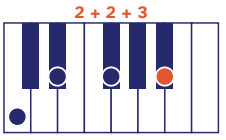

C E G B♭
**bluesy**
the blues

**Cm7**



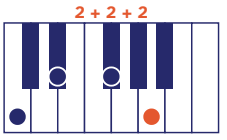

C E♭ G B♭
**mellow**
soul, jazz ballads

**Cø7**




C E♭ G♭ B♭
**searching**
jazz, film noir
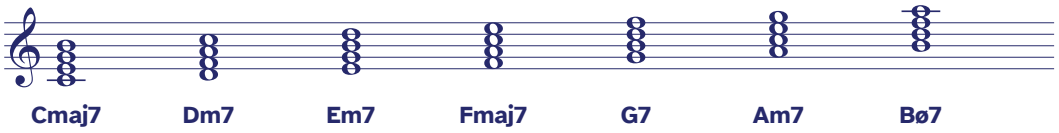
**C°7**




C E♭ G♭ B𝄫
**spooky**
old film villains

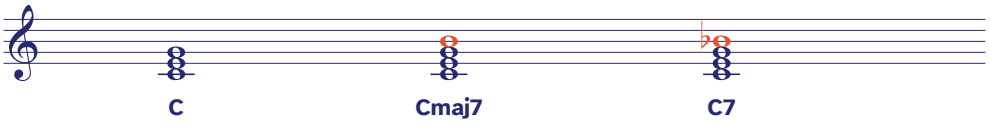
How the names work: major triad + major 7th = **maj7** · major triad + minor 7th = **7** · minor triad + minor 7th = **m7** · diminished triad + minor 7th = **ø7** · diminished triad + diminished 7th = **°7**.

## A seventh on every note of C major




Only white keys: count the keys between to name each one.

PART V · CHORDS · 5 · CONTINUED

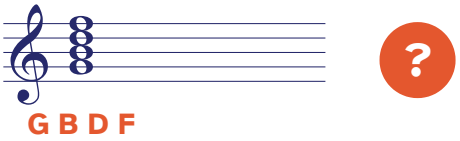



One note decides: B makes Cmaj7, B♭ makes C7. Everything below stays the same.

## Spot the difference

EXAMPLE 1

**Which seventh is it? G B D F**




1. **Count the keys between** each layer: G to B, B to D, D to F.
2. **Which of the five flavours** has those three numbers?
3. **The root is G.** Name the chord.

3 + 2 + 2, the bluesy one, on G: **G7**.

GAME

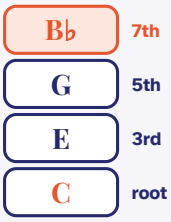
**Which seventh?**

| 1 | 2 | 3 |
|---|---|---|
| F A C E | C E G B♭ | D F A C |

A cake with one more layer. Name each chord.

Answer: 1 Fmaj7 · 2 C7 · 3 Dm7.

TRY IT · 10 SECONDS




**Play C7, then Cmaj7.**

C E G B♭, then C E G B. One key changes the top layer: bluesy, then dreamy.

PART V · CHORDS · 5 · CONTINUED

EXAMPLE 2

### Cmaj7 or C7: which note changes?

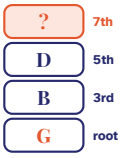



1. **Both start with C major:** C E G.
2. **Compare the top layers.** Which one note is different?
3. **Count the keys between** G and the top layer in each.

Only the top layer changes: **B** (3 between) is Cmaj7, **B♭** (2 between) is C7.

EXAMPLE 3

### Build G7

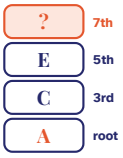



1. **Build G major:** G B D.
2. **Skip one more letter** above D: which?
3. **Count the keys between** D and it: 2 (the bluesy 7) or 3 (maj7)?

G B D **F**: 3 + 2 + 2, so it is **G7**.

EXAMPLE 4

### Build Am7




1. **Build A minor:** A C E (2 + 3).
2. **Add the next letter** but one above E.
3. **Count the keys between** E and the new layer.

A C E **G**: 2 + 3 + 2, so it is **Am7**.

TRAP

- × Writing C7 when you mean Cmaj7: "7" alone means the bluesy one, with B♭.
- × Stacking the 7th from the wrong note: count a 7th up from the root, not from the 5th.

SEE ALSO
Four kinds of triad p. 113 · Chords in a key p. 116

PRACTISE IT
Workbook Harmony 5 · Online: Library › Chords

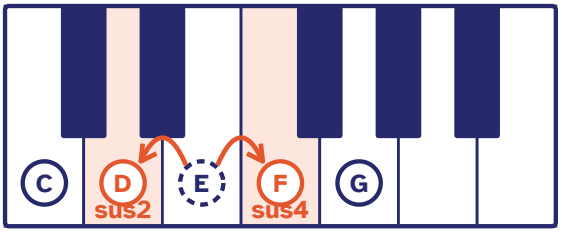
PART V · CHORDS · 6

# Colour chords

The 3rd is the chord's mood: major or minor, happy or sad. Slide it aside, take it away, or add one note, and a plain chord takes on a new colour. Five colours, all from C E G.

**THE SECRET**

**Move the 3rd, take it away, or add one note: five new colours.**

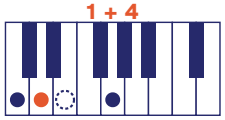



1. Start from **C E G**. The **E** (the 3rd) makes it major.
2. Slide the E up to the next letter, **F**: **Csus4**.
3. Or down to the letter below, **D**: **Csus2**. With no 3rd, the chord is neither major nor minor: it **hangs in the air**.

## Five colours, all on C

### Csus2

1 + 4



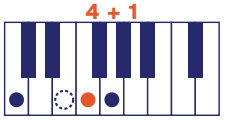
C D G

**open**

the 3rd steps down

### Csus4

4 + 1



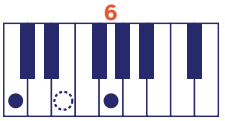
C F G

**floating**

the 3rd steps up

### C5

6



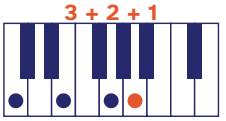
C G

**strong**

no 3rd at all

### C6

3 + 2 + 1



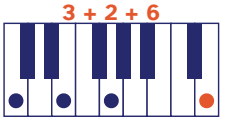
C E G A

**warm**

add the 6th

### Cadd9

3 + 2 + 6

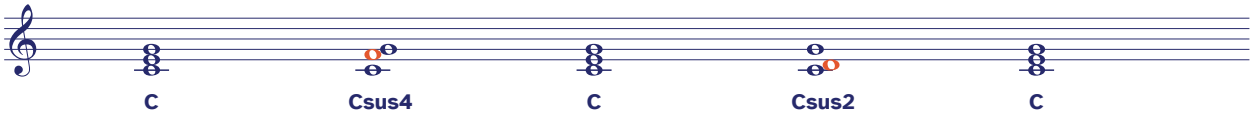


C E G D

**sparkly**

add the 9th on top

## Lean, then slide back




The red note leans, then slides back to E: F steps down, D steps up. Play it slowly: C, Csus4, C, Csus2, C. Guitarists love this shimmer.

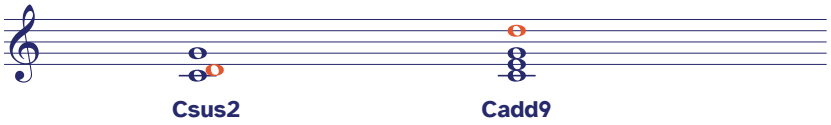
## On any root, the counts stay the same

| COLOUR | KEYS BETWEEN | ON C | ON G | ON F | ON D |
|---|---|---|---|---|---|
| **sus2** | 1 + 4 | C D G | G A D | F G C | D E A |
| **sus4** | 4 + 1 | C F G | G C D | F B♭ C | D G A |
| **5** | 6 | C G | G D | F C | D A |
| **6** | 3 + 2 + 1 | C E G **A** | G B D **E** | F A C **D** | D F♯ A **B** |
| **add9** | 3 + 2 + 6 | C E G **D** | G B D **A** | F A C **G** | D F♯ A **E** |

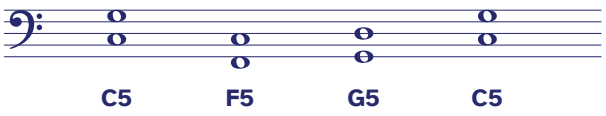
PART V · CHORDS · 6 · CONTINUED

**sus**: the 3rd held up or down · **5**: no 3rd at all · **6** and **add9**: one note added on top. Csus means Csus4.

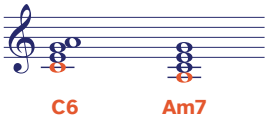
## Spot the difference




The same red D, two jobs. Csus2 throws the E out; Cadd9 keeps the E and puts the D on top.




**The power chord:** just the root and the 5th, played low in the left hand. With no 3rd it is neither happy nor sad, only **strong**: the sound of rock guitar. Play C5, F5, G5, C5 and drum the rhythm.




**C6 or Am7?** The same four notes, C E G A. The red bottom note decides: C at the bottom is **C6**; A at the bottom is **Am7**. More on the bass note in Slash chords, p. 131.

EXAMPLE 1

### Which colour is it? G C D




1. **Count the keys between:** G to C, then C to D.
2. **Find the card** with the same two numbers.
3. **Is there a 3rd (B)?** What has taken its place?

4 + 1, no B: G C D is **Gsus4**. Swap the C for **B** and you have G major.

GAME

### sus2, sus4 or add9?

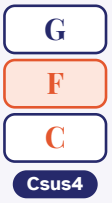
| 1 | 2 | 3 |
|---|---|---|
| C D G | C F G | C E G D |

In a sus chord the 3rd has moved; in an add chord nothing moved and a note was added.

Answer: 1 Csus2 · 2 Csus4 · 3 Cadd9.

PART V · CHORDS · 6 · CONTINUED

TRY IT · 10 SECONDS

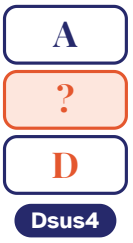



**Csus4, then C.**

Play C F G, then C E G. No 3rd, then a 3rd: floating, then bright.

EXAMPLE 2

## Build Dsus4

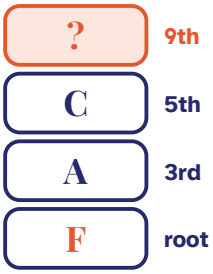



1. **Start with D major:** D F♯ A.
2. **sus4:** move the middle layer up to the next letter.
3. **Which key is it?**

F♯ moves up to **G**: Dsus4 is **D G A**.

EXAMPLE 3

## Build Fadd9




1. **Start with F major:** F A C, and keep every note.
2. **The 9th** is the 2nd, an octave up: which letter comes just after F?
3. **Put it on top.**

Fadd9: F A C **G**. Nothing moved; one note was added.

THINK OF IT Decorating the cake: a ribbon (the 6th) or a candle (the 9th) and it is still the same cake. Sus moves its middle layer; a power chord takes the middle layer out.

TRAP

× Calling Cadd9 a ninth chord: C9 also has the 7th (B♭). Add9 is just the triad plus the 9th (D on C), nothing else (The tower, p. 128).

SEE ALSO
Four kinds of triad p. 113 · Seventh chords p. 122

PRACTISE IT
Workbook Harmony 6 · Online: Library › Chords

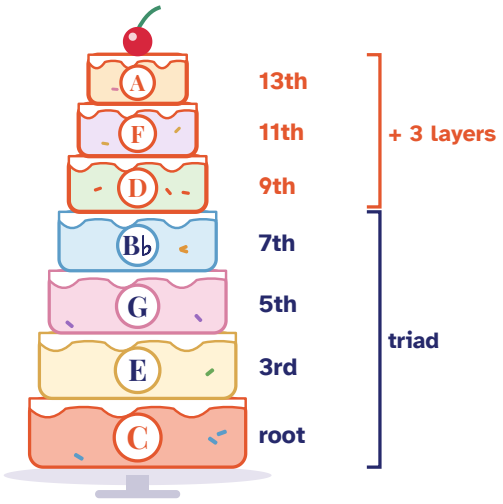
PART V · CHORDS · 7

# The tower: 9, 11, 13

Why stop at the 7th? Keep stacking 3rds and the cake grows three more layers: the 9th, the 11th and the 13th. At the top you have used **all seven letters, A to G**.

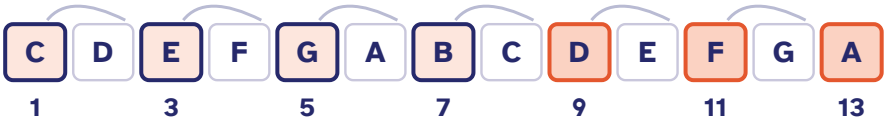
**THE SECRET**

**Keep skipping letters above the 7th: 9, 11, 13 are 2, 4, 6 an octave up.**



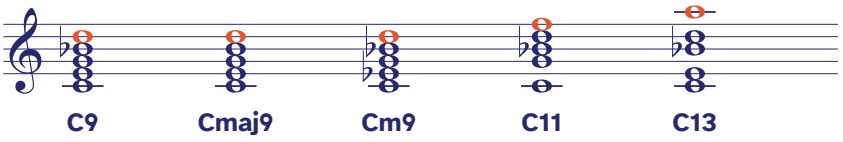

1. Build the seventh chord: **C E G B♭** (C7).
2. Keep skipping one letter: **D**, **F**, **A**. Those layers are the **9th, 11th and 13th**.
3. Take 7 away to find the letter: 9 → 2nd (D), 11 → 4th (F), 13 → 6th (A). Same letters, an octave up.

## Skip a letter, all the way up




Start on C and take every other letter: C E G B D F A (letters only: in C7 the B is B♭). After the 13th comes C again: the tower is full.

## Five tall chords on C

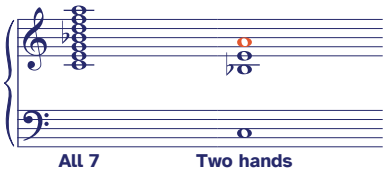



The red note is the top layer that gives each chord its name. C11 and C13 leave some notes out: see p. 129.

| CHORD | BUILT AS | NOTES | SOUNDS | HEARD IN |
|---|---|---|---|---|
| **C9** | C7 + 9th | C E G B♭ **D** | soulful | funk, soul |
| **Cmaj9** | Cmaj7 + 9th | C E G B **D** | dreamy | film, ballads |
| **Cm9** | Cm7 + 9th | C E♭ G B♭ **D** | smooth | neo-soul, R&B |
| **C11** | C9 + 11th, no 3rd | C G B♭ D **F** | airy | gospel, pop |
| **C13** | C7 + 13th | C E B♭ (D) **A** | rich | jazz, blues endings |

PART V · CHORDS · 7 · CONTINUED

## Too many notes? Leave some out




All seven notes sound muddy. Two hands play C13 with four: C in the left hand, B♭ E A in the right.

Ten fingers, seven layers: you choose. **Keep** the 3rd and the 7th (they say what kind of chord it is) and the red top note (its colour). **Leave out** the 5th first. The right hand can drop the root too, when the left hand or a bass player plays it.

With a major 3rd, the 11th (F) rubs against E. So C13 leaves out the F. C11 does the opposite: it keeps the F and leaves out the E.

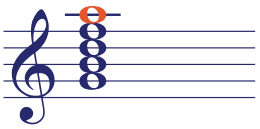
## 6 or 13? add9 or 9?

| SAME TOP NOTE | WITHOUT THE 7TH | WITH THE 7TH (B♭) |
|---|---|---|
| A on C | **C6**: C E G A | **C13**: C E B♭ A |
| D on C | **Cadd9**: C E G D | **C9**: C E G B♭ D |

Look for the 7th. No 7th: a colour chord (p. 125). A 7th: a tower chord.

EXAMPLE 1

### Name this tall chord: G B D F A




G B D F A

1. **Find the root:** each note skips one letter, so the bottom note is the root.
2. **The first four layers:** count the keys between. Which seventh?
3. **The red top layer:** which number is it above G?

G7 (3 + 2 + 2) with A, the 9th, on top: **G9**.

GAME

### Which floor?

1 — the 9th of C

2 — the 11th of C

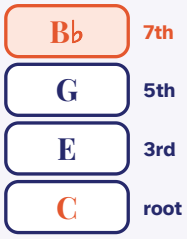
3 — the 13th of C

Build C7 (C E G B♭), then keep skipping one letter above the B♭.

Answer: 1 D · 2 F · 3 A.

PART V · CHORDS · 7 · CONTINUED

TRY IT · 10 SECONDS

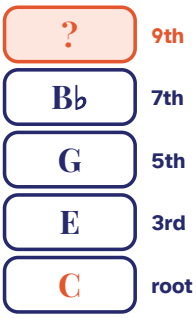



**Play C7, then add one more.**

Left hand C E G, right hand B♭, then add the next letter but one above it. You just built a tall chord.

EXAMPLE 2

**Build C9**




1. **Build C7:** C E G B♭.
2. **Skip one more letter** above B♭: which letter?
3. **Five notes:** share them between the hands if you need to.

C9: C E G B♭ **D**, one layer above the 7th.

HOW TO

**Build any tall chord**

1. **Start on the root** and skip one letter each time: 1 3 5 7 9 11 13.
2. **Stop** at the number in the name.
3. **Leave out** the 5th first; in a 13th chord, the 11th too, when it is too many notes for your hands.

THINK OF IT You don't need every layer. Keep the 3rd and the 7th, which say what kind of chord it is, and the top layer, which gives it its colour.

TRAP

- × Hunting far up for the 9th: it is just the 2nd, an octave higher (D on C).
- × Calling C13 a C6: the 13th chord has the 7th (B♭) as well.
- × Playing F and E together in C13 or C11: they clash. C13 drops the F; C11 drops the E.

SEE ALSO
Seventh chords p. 122 · Colour chords p. 125

PRACTISE IT
Workbook Harmony 7 · Online: Library › Chords

PART V · CHORDS · 8

# Slash chords

C/E means one chord with a chosen bass note: play a C chord, with E as the lowest note. It is not a fraction, and not two chords.

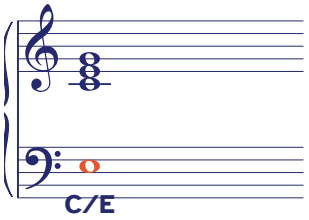
THE SECRET
**Chord before the slash, bass note after it.**

C/E

**the chord**
right hand

**the bass**
left hand, lowest note

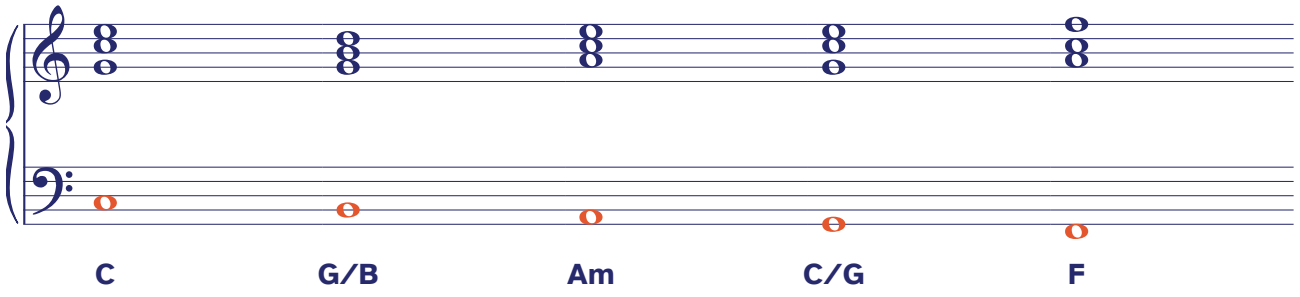
1. Before the slash: the **chord**. C means C E G, in the right hand.
2. After the slash: the **bass**. E goes in the left hand, below everything.
3. Play them together: **C/E**, “C over E”.




## Four jobs for a slash chord

| JOB | EXAMPLE | THE BASS NOTE IS | WHAT YOU HEAR |
|---|---|---|---|
| **Inversion** | C/E · C/G | a note of the chord | smoother changes (p. 119) |
| **Walk-down** | C · G/B · Am | one step down each time | a bass line that walks down |
| **Pedal** | F/C · G/C | the same note, held | the chords change above it |
| **Colour** | F/G · D/C | not in the chord | open, modern: pop, gospel |

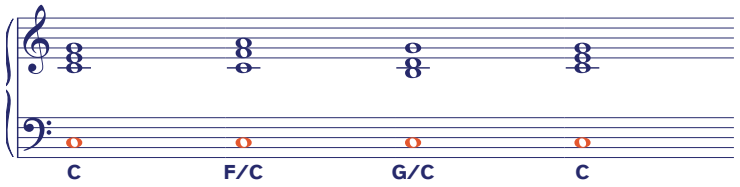
## Job 2: a bass that walks down




The red bass walks straight down, C B A G F, while the right hand moves a step at most. G/B and C/G make the steps possible.

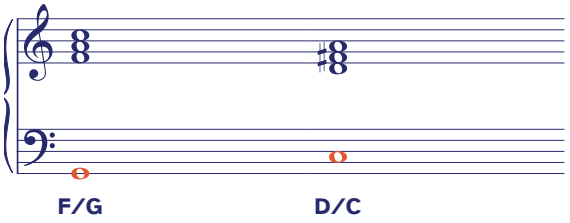
PART V · CHORDS · 8 · CONTINUED

## Job 3: a pedal, one bass under everything




The bass holds C while the chords change above it: C, F/C, G/C, C. The chords change while the C stays. (It is called a pedal after the organ's foot pedals, which hold long low notes.)

## Job 4: a bass from outside the chord

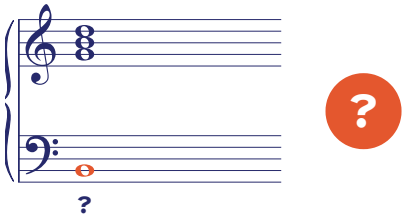



**F/G**: F A C over a G. G is not in the F chord, so it floats, open and unfinished. Pop and gospel players love it.

**D/C**: D F♯ A over a C. The C is not in the D chord, so it rubs against the chord.

EXAMPLE 1

### Name the slash chord




1. **The right hand:** which chord is G B D?
2. **The lowest note:** which note is in the left hand?
3. **Write chord, slash, bass.** Is the bass in the chord?

It is **G/B**, "G over B": B is in the G chord, so it is the 1st inversion.

GAME

### Read the slash

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| G/B | C/G | F/A | Am/C |

Before the slash: the chord. After it: the bass note. Which note goes in the left hand?

Answer: 1 B · 2 G · 3 A · 4 C.

PART V · CHORDS · 8 · CONTINUED

TRY IT · 10 SECONDS

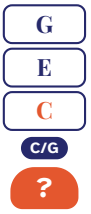



### Right hand C, left hand E.

Play C E G in the right hand and a low E in the left: C/E. Now try a low A under it: what would you call that?

EXAMPLE 2

### C/G: which note goes in the left hand?

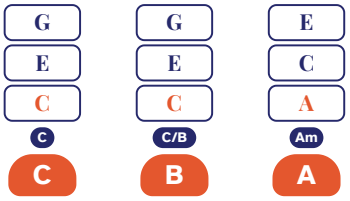



1. **Before the slash:** the chord. Which three notes go in the right hand?
2. **After the slash:** the bass. Which note goes low in the left hand?
3. **Play them together.**

Right hand C E G, left hand a low **G**: the C chord standing on G.

EXAMPLE 3

### Walk the bass down: C, C/B, Am




1. **Right hand** holds C E G for two chords.
2. **Left hand** walks down: C, then B.
3. **Then A**, under A C E: Am.

The bass walks **C B A**, one step at a time, while the chords hardly move.

THINK OF IT A cake and the plate it stands on: the slash says which cake (the chord) and which plate (the bass). Same cake, new plate, new view.

TRAP

- × Writing it back to front: the chord comes first, the bass second. E/C would be an E chord over C.
- × Letting the right hand dip below the bass: the letter after the slash must be the lowest note you play.
- × Forgetting the chord itself: in C/E the right hand still plays the whole C chord, C E G, not just C.

SEE ALSO
Inversions p. 119 · Colour chords p. 125

PRACTISE IT
Workbook Harmony 8 · Online: Library › Chords

PART V · CHORDS · TIPS

# When a chord sounds wrong

A chord that sounds wrong usually has one small slip. Name the slip, fix that one note, and the chord comes right.

TIP

### What you notice, what to check

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| A chord sounds dark when it should be bright | **the kind**: keys between, 3 + 2 or 2 + 3 | p. 113 |
| Your hand jumps all over | **an inversion**: keep the shared note still | p. 119 |
| A big chord sounds muddy | **leave notes out**: drop the 5th first | p. 128 |
| A seventh chord sounds wrong | **the 7th**: G7 adds F, a 7th above G (9 keys between) | p. 122 |
| A sus chord sounds unfinished | **the 4th**: the middle layer moved up a letter | p. 125 |
| The bass note sounds odd | **the slash**: the letter after it is the lowest note | p. 131 |

## Four ways to learn chords fast

1. **Build, then play**
   Say the cake (C E G) before your fingers go down.

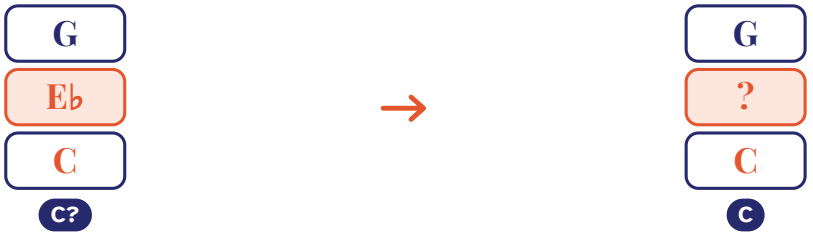
2. **Hands alone first**
   Left hand chords alone, then the tune alone, then together.

3. **One cake, three keys**
   Build the same kind of chord on C, G and F.

4. **Listen for the kind**
   Play major and minor side by side: bright, then dark.

EXAMPLE

### Fix a dark chord




1. **It sounds dark**, but the music wants a bright C chord.
2. **Count the keys between:** C to E♭, then E♭ to G. Which kind is that?
3. **Move one note** so it becomes 3 + 2. Which layer, and to which key?

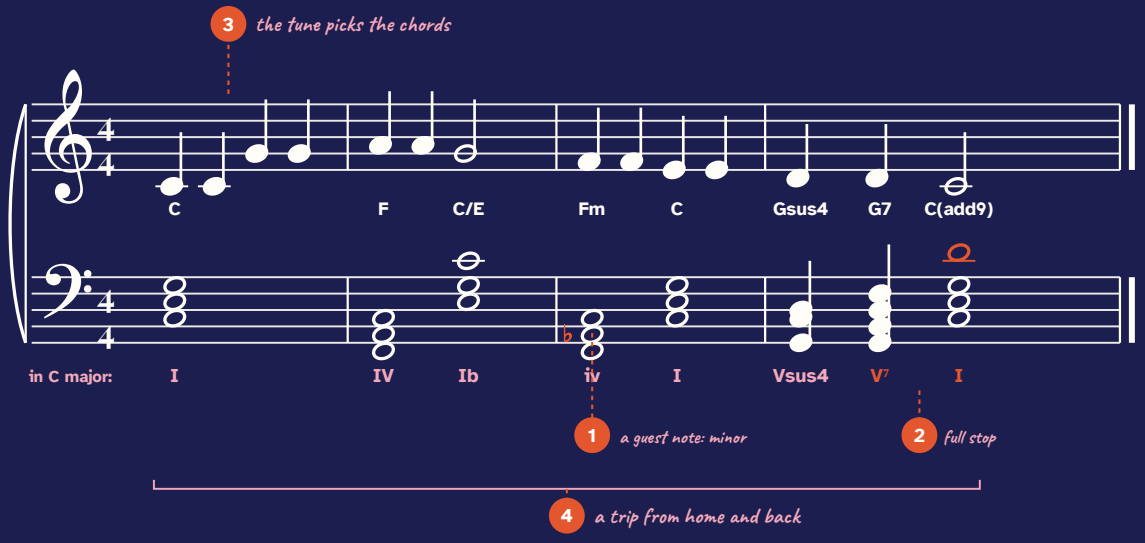
Move the middle layer from E♭ up to **E**: 3 + 2, bright C major again.

THINK OF IT Like a mechanic: one odd sound points to one part. Fix that part and the engine runs smoothly.

PART VI · PUT THEM TO WORK

# Harmony in music

A melody is the singer; harmony is the band behind it. Here the chords start moving: guests from outside the key, endings, chords under a tune, and the trips every song takes. All of it is in the same four bars.





1 **Chords from outside the key** p. 136
a guest note: A♭ in C major

2 **Cadences** p. 139
V to I ends a phrase like a full stop

3 **Chords under a melody** p. 142
the tune's main notes pick the chord

4 **Progressions everywhere** p. 144
the same few trips, in every song

★ **Tips: when harmony goes wrong** p. 146

PART VI · HARMONY IN MUSIC · 1

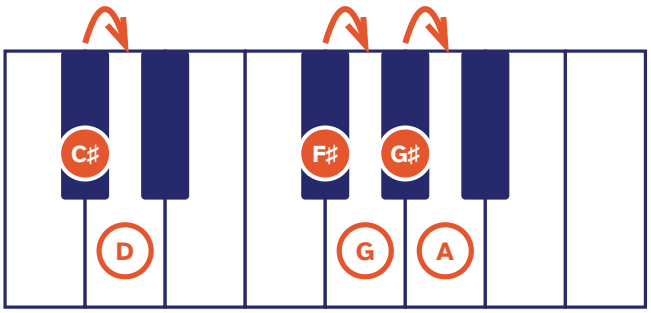
# Chords from outside the key

In C major, home is the white keys, so a black key in a chord is a **guest**. Guests never stay: each one moves to the **very next key** (no key between), and shows you where the music goes next.

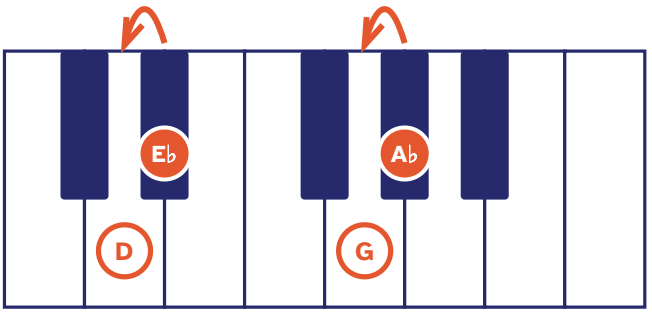
**THE SECRET**

**Sharp guests push up, flat guests sigh down: follow them one key and you find the next chord.**

### Sharp guests push up

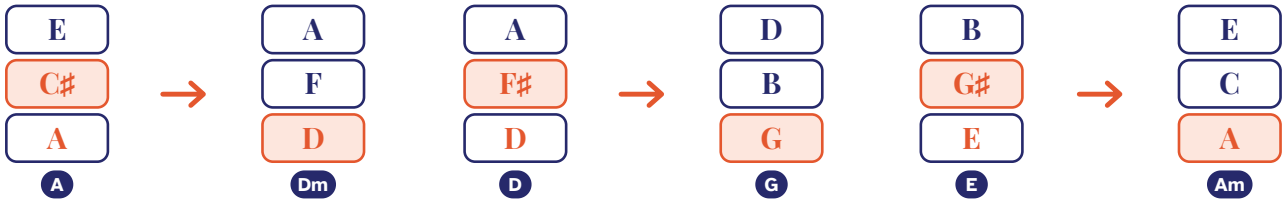



### Flat guests sigh down

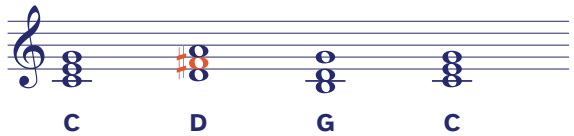



1. Find the **black key** in the chord: that is the guest.
2. A **sharp**? Go **up** to the very next key. A **flat**? Usually **down** to the very next key.
3. You have found a note of the **next chord**. Play it and hear it land.

## Sharp guests: up one key names the next chord

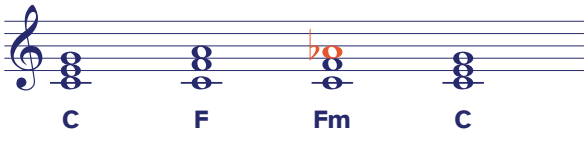



C♯ goes up to **D**, so the next chord is D minor. F♯ goes up to **G**: G. G♯ goes up to **A**: A minor.




Try it in a song: C, D, G, C. The **F♯** pushes up into G, then G leads home to C.

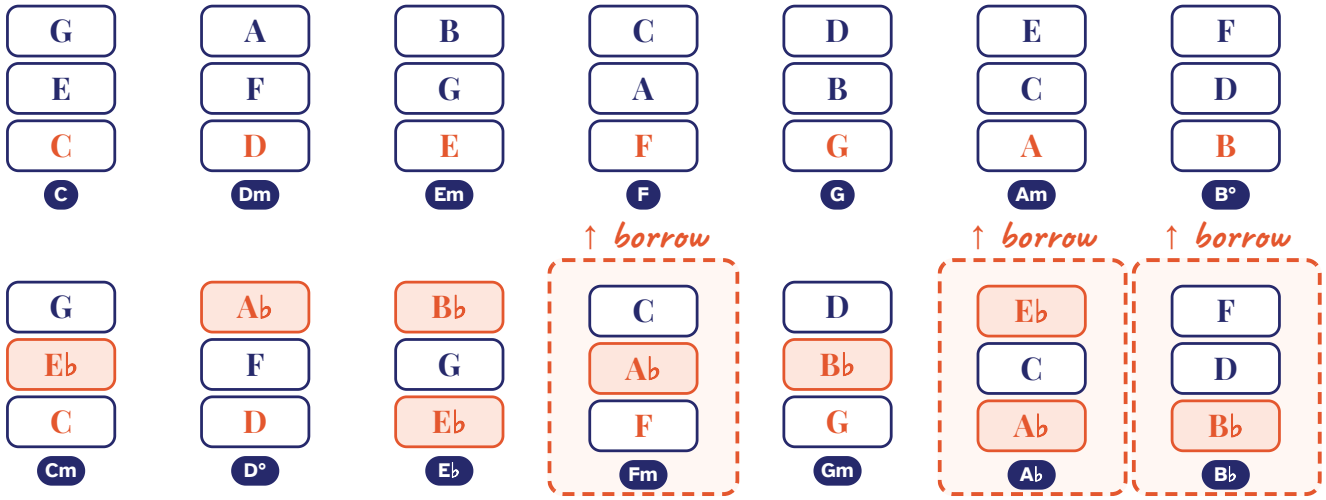
## Flat guests: borrowed from the twin, they sigh down




C major and C minor are **twins** (the parallel minor): the same home note, C, in different clothes. A flat guest is one piece of the twin's outfit. In C, F, Fm, C the **A♭** sighs down to G.

PART VI · HARMONY IN MUSIC · 1 · CONTINUED

## C major *at home*




## C minor *the twin: same home note, darker clothes*

### Their names, for later

| GUEST | NOTES | MOVES | NEXT CHORD | ITS NAME |
|---|---|---|---|---|
| **sharp guest** | D F♯ A | F♯ up to G | G | V of V |
| **sharp guest** | A C♯ E | C♯ up to D | Dm | V of ii |
| **sharp guest** | E G♯ B | G♯ up to A | Am | V of vi |
| **flat guest** | F A♭ C | A♭ down to G | C | borrowed iv |
| **flat guest** | A♭ C E♭ | A♭ down to G | G or C | borrowed ♭VI |
| **flat guest** | B♭ D F | B♭ on to C | C | borrowed ♭VII |

B♭ is the flat guest that walks **up** instead, on to C (one key between: a tone): the rock and film ending B♭ → C.

EXAMPLE

**Where do the guests go? B D♯ F♯ in C major**




1. **Find the black keys:** there are two guests here.
2. **Both are sharps:** move each up to the very next key. Where do they land?
3. **Which cake in C major** holds both landing notes?

D♯ goes up to E, F♯ up to G: both live in **E minor** (E G B). B D♯ F♯ leads to Em.

PART VI · HARMONY IN MUSIC · 1 · CONTINUED

GAME

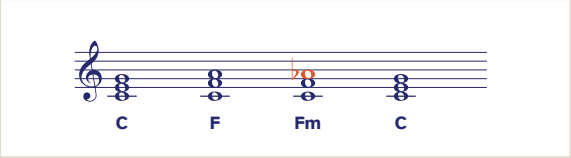
## Where does the guest go?

| 1 | 2 | 3 |
|---|---|---|
| E | E♭ | A |
| C♯ | C | F♯ |
| A | A♭ | D |
| A | A♭ | D |

In C major: find each guest, follow it one key, and write the chord that comes next.

Answer: 1 C♯ up to D: D minor · 2 A♭ down to G, E♭ down to D: G · 3 F♯ up to G: G.

TRY IT · 10 SECONDS




### Play C, F, Fm, C slowly.

Hear the A♭ sigh down to G? Now play C, D, G, C: the F♯ pushes up instead. One sighs, one pushes.

THINK OF IT A surprise guest at dinner: one new face changes the whole conversation, and then everyone goes home.

TRAP

- × Following a sharp downwards: sharps push **up**, flats usually sigh **down**.
- × Calling every black key a guest: in G major F♯ is at home. A guest is a note from outside the key you are in.
- × Learning the names first: play it, hear where the guest goes, then name it.
- × Mixing up the relatives: C minor is C major's twin (same home note). A minor is its neighbour (same key signature). We borrow from the twin.

SEE ALSO

Chords in a key p. 116 · Slash chords p. 131 · Cadences p. 139

PRACTISE IT

Workbook Harmony 9 · Online: Library › Chromatic chords

PART VI · HARMONY IN MUSIC · 2

# Cadences

A phrase is a musical sentence, and it needs punctuation. The last two chords of a phrase are its **cadence**. They tell you: finished, taking a breath, or surprise!

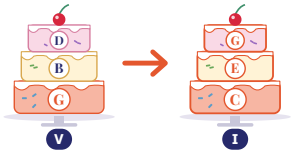
THE SECRET

**Music has full stops and commas too: listen to the last two chords of a phrase.**

.

**Perfect**

V → I

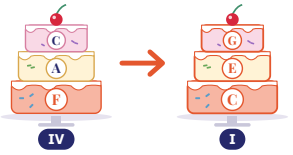



*Home. The end.*

Amen

**Plagal**

IV → I

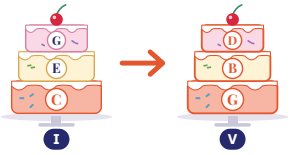



*A soft, gentle end.*

,

**Imperfect**

… → V

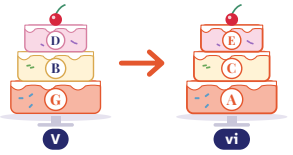



*To be continued…*

?!

**Interrupted**

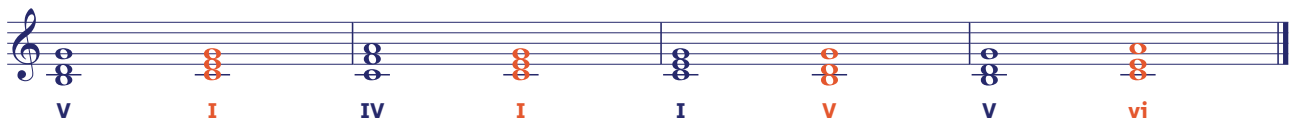
V → vi




*Surprise! Not home.*

1. Find the end of a phrase: often a long note, or a breath.
2. Name its **last two chords** with numerals: V I, IV I, anything then V, or V vi.
3. Match the punctuation: full stop ( . ), Amen, comma ( , ) or ?!

## The four cadences in C

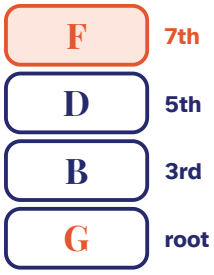



Perfect, plagal, imperfect, interrupted. The red chord is where each one lands.

| CADENCE | CHORDS | FEELS LIKE | ALSO CALLED |
|---|---|---|---|
| **Perfect** | V → I | a full stop: finished | authentic, full close |
| **Plagal** | IV → I | the “A-men” sung at the end of a song in church | Amen cadence |
| **Imperfect** | any → V | a comma: waiting | half close, half cadence |
| **Interrupted** | V → vi | a surprise: not home yet | deceptive |

PART VI · HARMONY IN MUSIC · 2 · CONTINUED

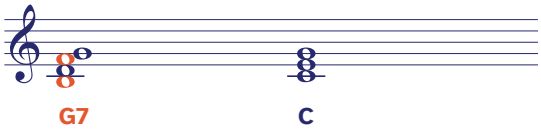
## Why G7 pulls home




Add the 7th to V and the full stop gets stronger.

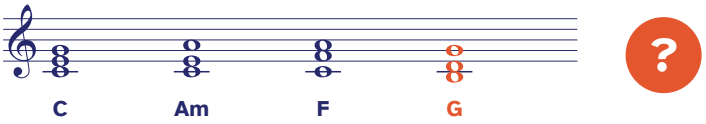
In **G7** two notes lean towards C major. The **B** rises to **C**, the **F** falls to **E**: each to the very next key, no key between. Two tiny steps, and you are home (Seventh chords, p. 122).

It works like a guest: B pushes up, F sighs down (p. 136).




EXAMPLE 1

### How does this phrase end? C Am F G

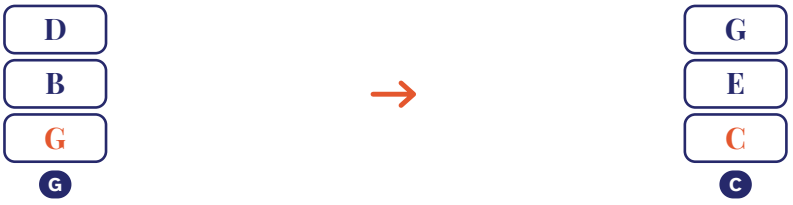



1. **Find the last two chords** of the phrase.
2. **Turn them into numerals** in C major.
3. **Which punctuation card** ends that way?

F then G is IV → V: it lands on V, the comma. An **imperfect cadence**.

EXAMPLE 2

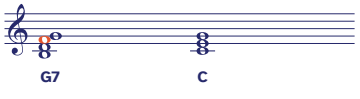
### How does Twinkle end?




1. **The last two chords** of Twinkle in C are G, then C.
2. **Turn them into numerals.**
3. **Which card** goes from that chord to that chord?

V → I: a **perfect cadence**, the full stop. That is why Twinkle sounds finished.

TRY IT · 10 SECONDS




### Play G7, then C. The end!

Now play G7, then A minor: “huh?!” That is the interrupted cadence. Then play F, then C, and sing “A-men”.

PART VI · HARMONY IN MUSIC · 2 · CONTINUED

EXAMPLE 3

### Surprise! G7, then A minor

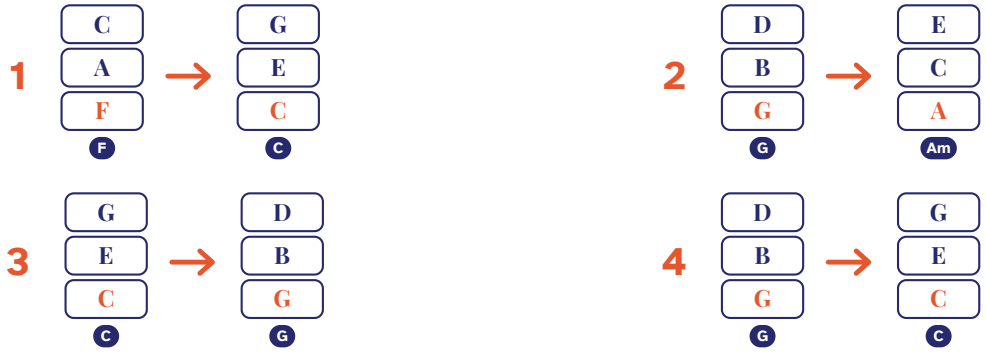



1. **G7 leans towards home**: you expect C.
2. **But the next chord is A minor.** Which numeral is it in C?
3. **V → that numeral:** which card?

V → vi: an **interrupted cadence**. The music was heading home, then turned the corner.

GAME

### Punctuate the phrases




Each pair ends a phrase in C major. Turn the letters into numerals first (C is I, F is IV, G is V, Am is vi), then write full stop, comma, Amen or ?! beside each.

Answer: 1 Amen (plagal), 2 ?! (interrupted), 3 comma (imperfect), 4 full stop (perfect).

THINK OF IT Punctuation in a sentence: a full stop ends it, a comma makes you wait, and a surprise makes you gasp. Music ends its phrases the same way.

TRAP

- × Calling every ending on I (C) perfect: a perfect cadence comes **from V**. F to C is plagal.
- × Mixing up imperfect and perfect: imperfect **ends on** V; perfect **starts on** V and lands on I.

SEE ALSO

Chords in a key p. 116 · Seventh chords p. 122 · Chords under a melody p. 142

PRACTISE IT

Workbook Harmony 10 · Online: Library › Cadences

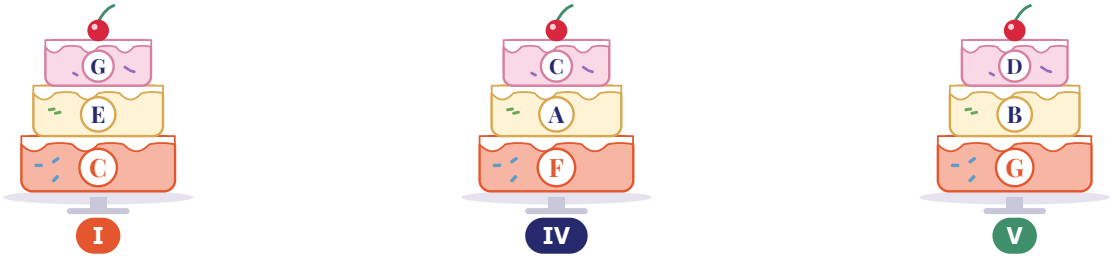
PART VI · HARMONY IN MUSIC · 3

# Chords under a melody

The melody is the singer; the chords are the band. To choose a chord for each bar, look at the melody's main notes and pick the cake holding them.

**THE SECRET**

**Look at the melody's main notes, then pick the cake that holds them.**




| C | D | E | F | G | A | B |
|---|---|---|---|---|---|---|
| I, IV | V | I | IV | I, V | IV | V |

*Which cake holds the tune's main notes?*

Every note of C major lives in I, IV or V. C and G live in two: let your ear choose.
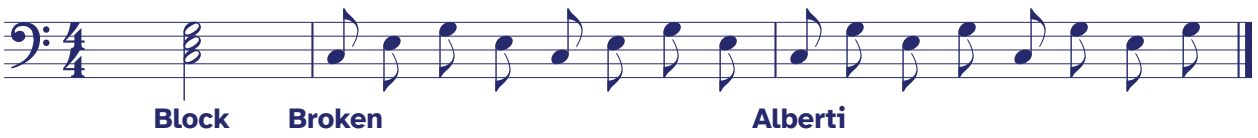
1. Look at the **main notes** of the bar: the long ones, and the ones on beats 1 and 3.
2. Find the cake that holds them: **I, IV or V**.
3. Play that chord in the left hand while the right hand plays the tune.

## Twinkle, Twinkle, with its chords




C C G G sit in I (C); A A in IV (F), then G in I; F F in IV, E E in I; D D in V (G), and the last C in I. The chord changes halfway through bars 2, 3 and 4.

## Three ways to play the chord in the left hand




Block: all three at once. Broken: one at a time, up and down. Alberti: low, high, middle, high, named after Domenico Alberti, who loved it.

PART VI · HARMONY IN MUSIC · 3 · CONTINUED

EXAMPLE

**Choose the chord: E D C D | E E E**

E D C D | E E E

red = beats 1 and 3

1. **Bar 1:** which notes fall on beats 1 and 3 (red)?
2. **Which cake** holds both of them: I, IV or V?
3. **Bar 2:** E E E. Which cake holds E?

E and C both live in **I**, and so does E: both bars take **I** (C). That is the start of Mary Had a Little Lamb.

GAME

**Pick the cake**
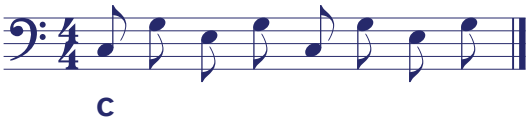
F A F A  D B G B  G E C E

1 2 3

Each bar is in C major. Which cake holds its notes: I, IV or V?

Answer: 1 IV (F A C) · 2 V (G B D) · 3 I (C E G).

TRY IT · 10 SECONDS




**Left hand: C G E G, over and over.**

That is the Alberti bass on the C chord. Keep it going while you hum Twinkle, then try it on F (F C A C) and G (G D B D).

YOU KNOW THIS

**Mozart, Sonata in C, K. 545: the left hand**




The opening left hand is an **Alberti bass**: C G E G is just the C chord, broken into low, high, middle, high; D G F G is G7 (G B D F, without the B). Mozart wrote it an octave higher, starting on middle C.

**Try it.** Play this left hand alone, then add the famous tune: C, E G, B, C D C.

THINK OF IT Like picking a coat for the weather: when the tune changes, change the chord.

TRAP

- × Choosing a chord for a passing note: use the main notes (strong beats, long notes).
- × Playing the left hand louder than the tune: the melody is the star, the chords are the stage.

SEE ALSO

Chords in a key p. 116 · Cadences p. 139

PRACTISE IT

Workbook Harmony 11 · Online: Library › Chords

PART VI · HARMONY IN MUSIC · 4

# Progressions everywhere

A **progression** is a trip from chord to chord. A handful of trips turn up in song after song. Learn them by their numbers and you can play them in any key.

**THE SECRET**
**Most songs take the same few trips: home, away, a pull, and home again.**

I home · vi iii home's cousins · IV ii away · V the pull back home

**The pop four**
countless pop songs and ballads

I C → V G → vi Am → IV F

**The fifties**
old rock 'n' roll and doo-wop

I C → vi Am → IV F → V G

**The jazz three-step**
jazz standards (as ii7, V7, Imaj7)

ii Dm → V G → I C

**The canon**
Pachelbel's Canon and its many children

I C → V G → vi Am → iii Em → IV F → I C → IV F → V G

1. Learn the trip by **numbers**: I V vi IV.
2. Pick a key and turn the numbers into chords: in C, **C G Am F**.
3. Play it round and round. Change key: the numbers stay the same.

## Same trip, any key

| The pop four | I | V | vi | IV |
|---|---|---|---|---|
| **in C** | C | G | Am | F |
| **in G** | G | D | Em | C |
| **in F** | F | C | Dm | B♭ |
| **in D** | D | A | Bm | G |

## The twelve-bar blues

| | | | |
|---|---|---|---|
| I C | I C | I C | I C |
| IV F | IV F | I C | I C |
| V G | IV F | I C | V G |

The most famous trip of all is twelve bars long and uses only **I, IV and V**. Almost every blues, and a lot of rock 'n' roll, follows it. Play it in C with one chord per bar, four beats each.

Make it bluesy: turn every chord into a seventh (C7, F7, G7, p. 122).

PART VI · HARMONY IN MUSIC · 4 · CONTINUED

EXAMPLE

### Play the pop four in A major

I ? → V ? → vi ? → IV ?

1. **Write the A major scale** (three sharps) and number the notes.
2. **Find notes 1, 5, 6 and 4.**
3. **Build each cake:** which one is minor?

A, E, F♯m, D: **the pop four in A.** Same trip, new key.

GAME

### Name the trip

1 C · Am · F · G

2 Dm · G · C

3 C · G · Am · F

All in C major. Turn each into numbers, then name the trip: the pop four, the fifties or the jazz three-step.

Answer: 1 the fifties (I vi IV V) · 2 the jazz three-step (ii V I) · 3 the pop four (I V vi IV).

TRY IT · 10 SECONDS

I C → V G → vi Am → IV F

### Play C, G, Am, F, four beats each.

Go round four times without stopping, left hand plays the chord's name note (C for C, A for Am), right hand the whole cake. Hum any tune you like over it.

TRAP

× Mixing up the pop four and the fifties: same four chords, different order. Listen to which chord comes second.

SEE ALSO

Chords in a key p. 116 · Cadences p. 139 · Chords under a melody p. 142

PRACTISE IT

Workbook Harmony 12 · Online: Library › Chords

PART VI · HARMONY IN MUSIC · TIPS

# When the harmony goes wrong

Every chord can be right and the music still sound wrong: the ending will not close, or the left hand fights the tune. The slip is in how the chords move. Find which move it is, then fix that one.

TIP

### What you notice, what to check

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| The end does not feel finished | **the cadence**: end with V → I | p. 139 |
| A sharp or flat surprises you | **a guest**: sharps go up, flats go down, one key | p. 136 |
| The left hand fights the tune | **the main notes**: pick the cake holding them | p. 142 |
| A new song feels hard to learn | **the trip**: find I, IV, V and vi by number | p. 144 |

## Four ways to hear the harmony

**1 Hum the bass**
Sing the lowest notes: they show where the chords move.

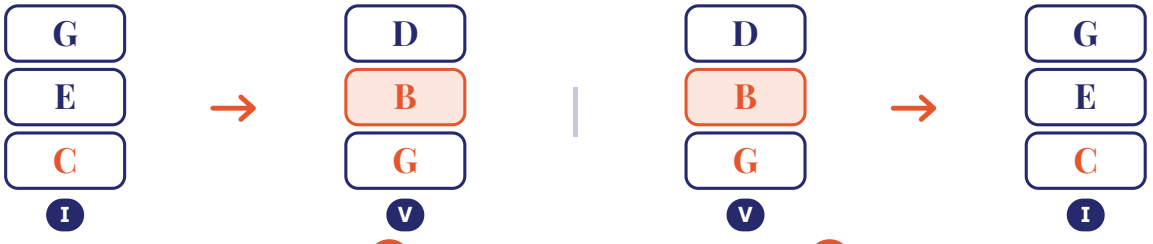
**2 Tick the full stops**
Mark every V → I in your piece with a red tick.

**3 One trip, three keys**
Play I–V–vi–IV in C, G and F: same numbers, new letters.

**4 Chord a tune you know**
Pick I, IV or V for each bar of Twinkle or Happy Birthday.

EXAMPLE

### Fix the ending




**1 The piece stops, but it does not feel finished.** Look at the last two chords.

**2 Name them:** C then G is I → V. That is a comma, not a full stop.

**3 Swap the order:** G then C is V → I. The red B climbs one key to C, and the music lands.

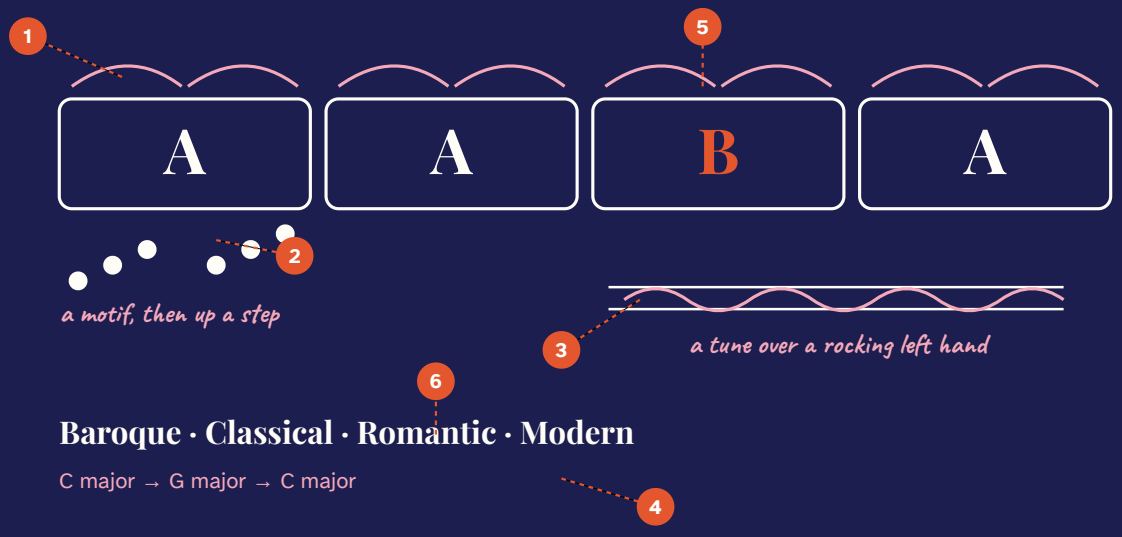
A strong ending needs **V → I** (IV → I for a soft Amen): the last two chords are the full stop.

THINK OF IT Like a road trip: once you know where home is, every detour makes sense, and every road leads back to I.

PART VII · HOW MUSIC IS BUILT

# Shape and style

Step back from the notes and see the building: small ideas, repeated and moved, in sections that come and go, in the style of their time. Everything in this part is on this map.





1 **Motifs and phrases** p. 148
small ideas build sentences: question and answer

2 **Repetition and sequence** p. 150
say it again, higher, in another voice, or something new

3 **Texture** p. 152
one line, a tune with chords, or tunes woven together

4 **Changing key** p. 154
a new sharp that keeps coming back: a new key

5 **Form** p. 156
the map of a piece: AB, ABA, ABACA, variations

6 **Styles through time** p. 159
Baroque, Classical, Romantic and Modern

★ **Tips: when a piece goes nowhere** p. 161

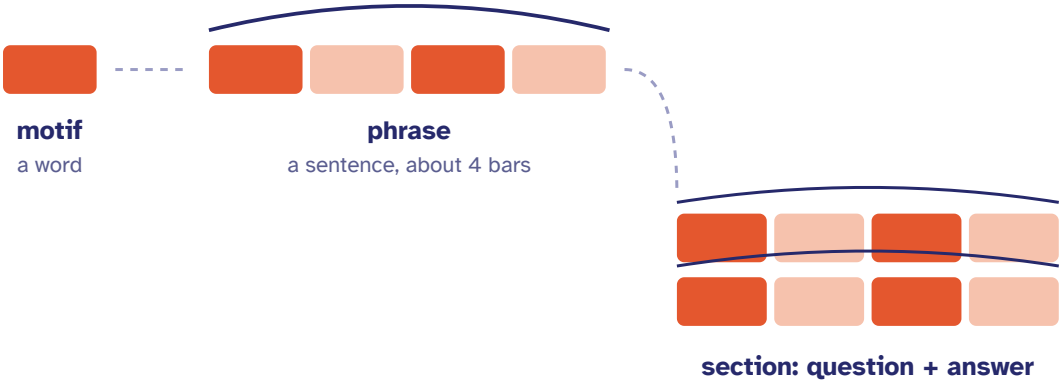
PART VII · SHAPE AND STYLE · 1

# Motifs and phrases

A tune is built like a story. A tiny idea, the **motif**, grows into a **phrase**, a musical sentence; two phrases make a question and its answer. Spot the building blocks and a page of notes turns into a few ideas.
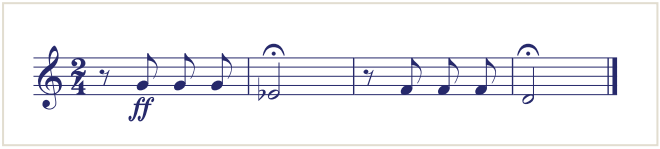
THE SECRET

**A motif is a word, a phrase a sentence: a question ends away from home, an answer comes home.**
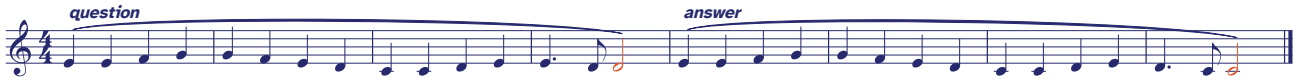



*Words, sentences, paragraphs*

## A motif




Four notes, short-short-short-long: Beethoven builds the whole first movement of his Fifth Symphony from them. A motif is short (2 to 6 notes) and easy to recognise: its **rhythm** and its **shape** come back again and again.

## Phrases: question and answer




Ode to Joy: two 4-bar phrases. The first ends on D, away from home (a question); the second comes home to C (an answer).
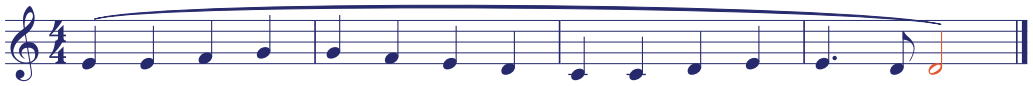
THE RULE

**Most phrases are 4 bars long. A question phrase ends away from the key note; the answer comes home to it.**

PART VII · SHAPE AND STYLE · 1 · CONTINUED

EXAMPLE 1

### Question or answer?
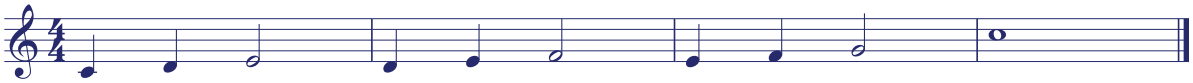


1. **The key is C major.** What is the home note?
2. **The last note** of this phrase is…?
3. **Is it home?**

It ends on D, not C: a **question**. The answer phrase that follows ends on C.

EXAMPLE 2

### Find the motif



1. **Look at bar 1:** what shape and rhythm?
2. **Bars 2 and 3:** the same shape?
3. **Where** does each start?

Step, step, long: three notes. It comes **three times**, each a step higher.

GAME
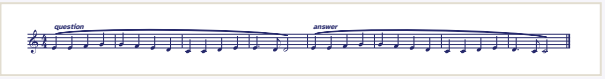
### How many phrases?

| 1 | 2 | 3 |
|---|---|---|
| 8 bars | 12 bars | 16 bars |

Each phrase is 4 bars long. How many phrases in a tune of…

Answer: 1 two · 2 three · 3 four.

TRY IT · 10 SECONDS
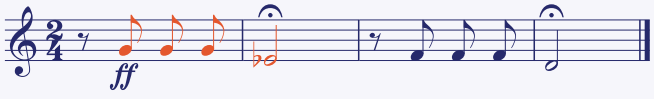



**Sing a question, then an answer.**

Sing the question and stop on the D: it hangs in the air. Now sing the answer and land on C.

YOU KNOW THIS

### Beethoven, Symphony No. 5




"Da-da-da-**DUM**": the most famous motif ever written. Listen to the first minute and count how many times it comes back.

**Try it.** Tap the rhythm on the lid, then sing it a step lower.

TRAP

× Playing every note the same: each phrase has one high point to grow towards.

SEE ALSO

Slurs and phrases p. 80 · Cadences p. 139

PRACTISE IT

Online: Library › Phrases

PART VII · SHAPE AND STYLE · 2

# Repetition and sequence

Composers make one idea go a long way. They say it again, say it again higher or lower, let another hand copy it, or say something completely new.
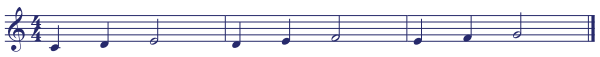
**THE SECRET**

**Again, again higher, again in the other hand, or something new.**
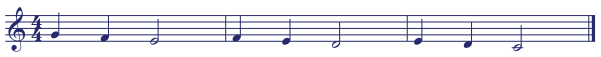



*Four ways to use one idea*

| NAME | WHAT HAPPENS |
|---|---|
| **repetition** | the same notes again |
| **sequence** | the same shape, starting higher or lower |
| **imitation** | another hand or voice copies the idea |
| **contrast** | something new: a different shape or rhythm |



A rising sequence: C D E, D E F, E F G.

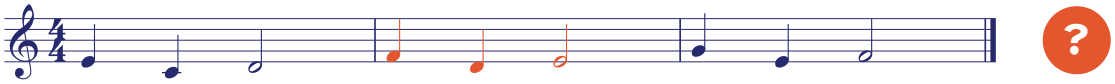


A falling sequence: each step one note lower.

**TWO-MINUTE DRILL** Take any three-note motif and play it as a sequence, starting on every white key from C up to the next C, then back down.

EXAMPLE 1

**Repetition or sequence?**
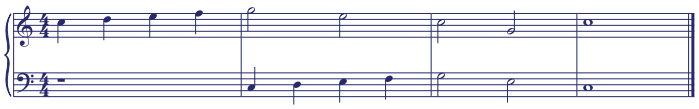


1. **Bar 1:** which notes?
2. **Bar 2:** the same notes, or the same shape on new notes?
3. **Bar 3?**

Same shape, a step higher each bar: a **sequence**.

PART VII · SHAPE AND STYLE · 2 · CONTINUED

EXAMPLE 2

**Which hand copies?**
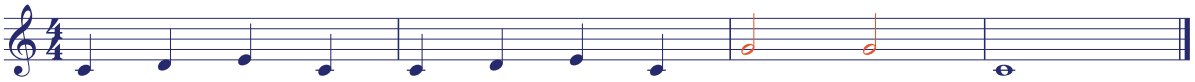


1. **The right hand** starts with C D E F G E C.
2. **The left hand** waits a bar. Then?
3. **Same notes**, how many octaves lower?

The left hand **imitates** the right one bar later, two octaves down.

EXAMPLE 3

**Where is the contrast?**



1. **Bars 1 and 2:** same or different?
2. **Bar 3:** what changes, the notes or the rhythm?
3. **Bar 4?**

Bars 1–2 repeat; **bar 3** brings something new: long notes on G.

GAME

**Which way?**

1 same notes

2 one step up

3 left hand copies

4 new rhythm

Name each one.

Answer: 1 repetition · 2 sequence · 3 imitation · 4 contrast.

TRY IT · 10 SECONDS
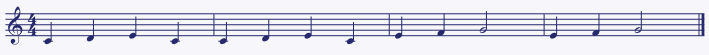


**Make a sequence.**

Play C D E, then D E F, then E F G with fingers 1 2 3. Now go down: G F E, F E D, E D C.

YOU KNOW THIS

**Frère Jacques**



Every bar is said **twice**: repetition. Sung as a round, each voice enters two bars after the last: imitation.

**Try it.** Sing it as a round with a friend.

TRAP

× Calling any repeat a sequence: a sequence moves to new notes.

SEE ALSO
Motifs and phrases p. 148 · Texture p. 152

PRACTISE IT
Online: Library › Patterns

PART VII · SHAPE AND STYLE · 3

# Texture

Texture is how many lines sound at once, and how they fit together: one line alone, a tune with chords, or several tunes woven together like threads in cloth.

**THE SECRET**

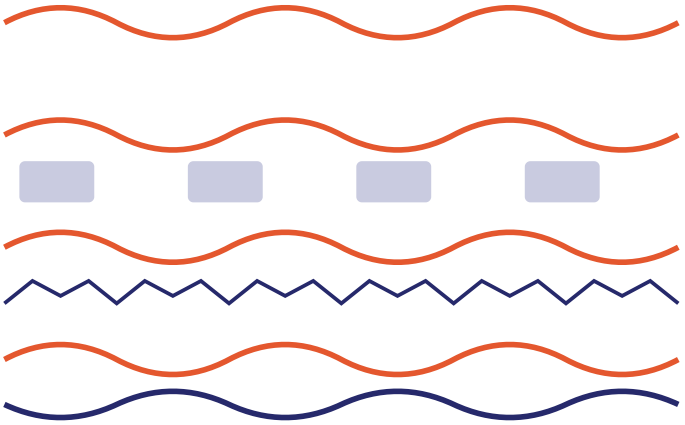
**Count the lines: one alone, a tune with chords, or tunes woven together.**

**one line**
monophony

**a tune with chords**
melody and accompaniment
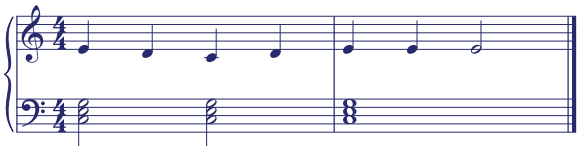
**a tune over broken chords**
Alberti bass, arpeggios

**two tunes at once**
polyphony
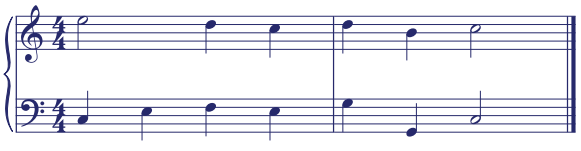


*Thin or thick: count the threads*

| WORD | WHAT YOU HEAR |
|---|---|
| **monophonic** | one line alone, or in octaves (unison) |
| **homophonic** | a tune with chords underneath: melody and accompaniment |
| **polyphonic** (contrapuntal) | two or more tunes at once, each its own |
| **Alberti bass** | a broken chord rocking low–high–middle–high |
| **block chords** | all the chord notes struck together |



Melody and block chords.



Polyphony: two tunes, each with its own rhythm.

GAME

**One line or more?**

1 **a tune alone** 2 **a tune and chords** 3 **two tunes at once**

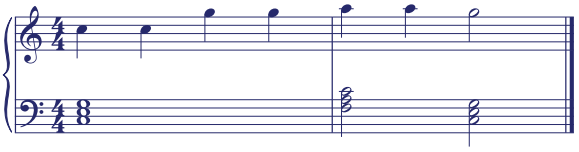
Name the texture.

Answer: 1 monophonic · 2 homophonic · 3 polyphonic.

PART VII · SHAPE AND STYLE · 3 · CONTINUED

EXAMPLE 1

### Name the texture
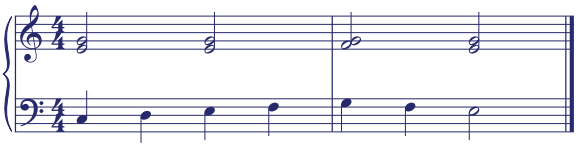


1. **The right hand:** a tune or chords?
2. **The left hand:** a tune or chords?
3. **So** which word?

A tune over chords: **homophonic**, melody and accompaniment.

EXAMPLE 2

### Which hand has the tune?
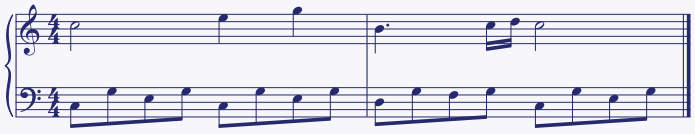


1. **The right hand** repeats two-note chords.
2. **The left hand** moves step by step.
3. **Which one** should sing out?

The **left hand** has the tune: play it louder, keep the right-hand chords soft.

YOU KNOW THIS

### Mozart, Sonata in C, K. 545



The right hand sings; the left hand rocks **C G E G**: an Alberti bass, the sound of the Classical style.

**Try it.** Play only the left hand, softly and evenly.

TRAP

- × Letting the accompaniment cover the tune: tune louder, chords softer.
- × Losing one of two tunes: practise each hand alone and sing it.

SEE ALSO
Chords under a melody p. 142 · Styles p. 159

PRACTISE IT
Online: Library › Texture

PART VII · SHAPE AND STYLE · 4

# Changing key

Music can travel to a new key and come home again: a **modulation**. The signpost is a new sharp, flat or natural that keeps coming back, and a cadence that lands in the new key.

THE SECRET

**A new sharp or flat that keeps coming back, then a cadence: you are in a new key.**

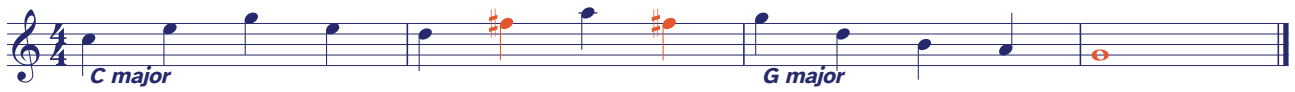



*A trip away, and home again*

## The nearest keys

A piece in C major usually visits the keys next door on the circle of fifths (p. 105): only one note changes.
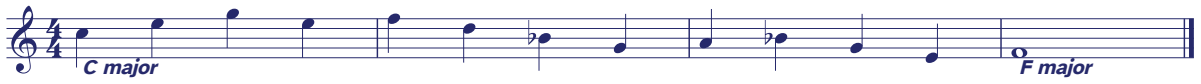
| FROM C MAJOR TO | THE SIGNPOST | WHY IT IS CLOSE |
|---|---|---|
| **G major** | F♯ appears | the key a 5th up (one more sharp) |
| **F major** | B♭ appears | the key a 4th up (one more flat) |
| **A minor** | G♯ appears | the relative minor (same key signature) |




Bar 1 in C major; F♯ twice in bar 2, then the phrase lands on G: now G major.

EXAMPLE 1

**Where does the key change?**




1. **Find the accidental.**
2. **Which major key** has one flat, B♭?
3. **Where** does the phrase land?

B♭ in bar 2 and again in bar 3, landing on F: **F major**.

PART VII · SHAPE AND STYLE · 4 · CONTINUED

EXAMPLE 2

**Which nearby key?**

## In G major, a C♯ keeps appearing ?

1. **G major** has one sharp, F♯.
2. **C♯ is one more sharp.**
3. **Which key** has F♯ and C♯?

**D major**, the key a 5th up: one more sharp.

EXAMPLE 3

**A new key, or just passing?**

## One F♯ in a bar of C major, then F♮ everywhere after ?

1. **Does the F♯ come back?**
2. **Is there a cadence** in G?
3. **So**, a new key?

No: one passing F♯ is a decoration. A modulation **keeps** its new note and lands with a cadence.

GAME
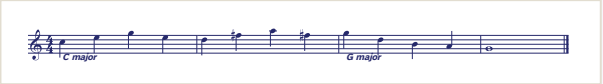
**Where to next?**

1. C major + F♯
2. C major + B♭
3. G major + C♯
4. F major + E♭

Which nearby key has the music moved to?

Answer: 1 G major · 2 F major · 3 D major · 4 B♭ major.

TRY IT · 10 SECONDS




**Hear the pull.**

Play the four bars twice: once with F♮ in bar 2, once with F♯. With the F♯, the G at the end sounds like a new home.

THINK OF IT A modulation is a holiday: new scenery for a while, but the piece usually comes home at the end.

TRAP

- × Taking one passing accidental as a new key: it must keep coming back.
- × Missing the way home: most pieces end in the key they began in, so watch for the old note returning.

SEE ALSO

The circle of fifths p. 105 · Cadences p. 139

PRACTISE IT

Online: Library › Keys

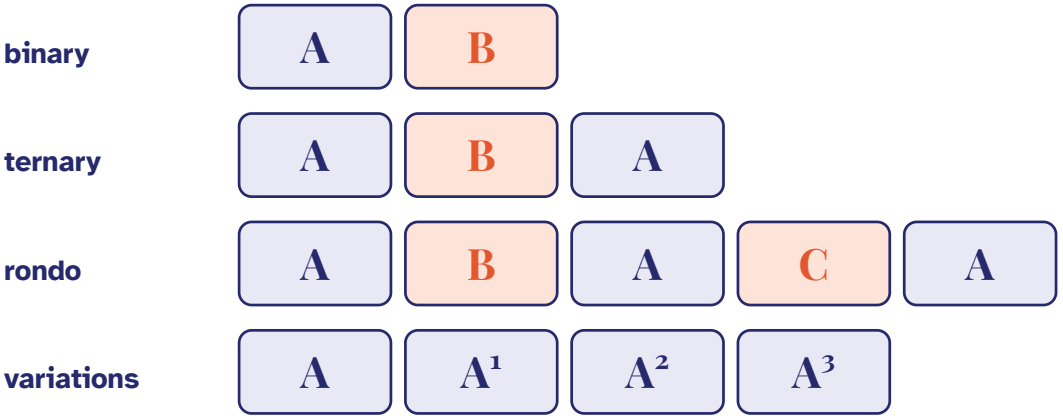
PART VII · SHAPE AND STYLE · 5

# Form

Form is the map of a whole piece: which sections come, in what order, and which come back. Letters name them: **A** for the first idea, **B** for the next new one, and so on.

THE SECRET

**Same music, same letter. New music, next letter.**




*The map of a piece, in letters*

| FORM | LETTERS | SHAPE | YOU KNOW IT FROM |
|---|---|---|---|
| **binary** | A B | two halves, each often repeated | Baroque dances: minuets, gigues |
| **ternary** | A B A | out, away, and back | many short Romantic pieces |
| **rondo** | A B A C A | A keeps coming back between new ideas | Beethoven, Für Elise |
| **theme and variations** | A $A^1$ $A^2$ $A^3$ | one tune, dressed differently each time | Mozart's variations on "Twinkle" |
| **strophic** | A A A | the same music for every verse | hymns, folk songs |
| **verse and chorus** | V C V C | the chorus words stay the same | pop songs |

THE RULE

**Same music, same letter. New music, the next letter. A little changed: add a small number ($A^1$).**

LISTEN FOR IT

**Petzold**
Minuet in G (Notebook for Anna Magdalena Bach; once thought to be Bach's)
binary: two halves, each repeated

**Beethoven**
Für Elise
rondo: A B A C A

**Mozart**
Twelve Variations on "Ah vous dirai-je, Maman"
theme and variations, on the tune of Twinkle

PART VII · SHAPE AND STYLE · 5 · CONTINUED

EXTRA · GOING FURTHER

### Sonata form

The big first movements of sonatas and symphonies: an **exposition** (two themes, the second in a new key), a **development** (the themes taken apart and moved through keys), and a **recapitulation** (both themes again, now at home).

EXAMPLE 1

**Which letter makes it ternary?**




1. **A B** alone: two sections.
2. **Ternary** goes out, away and...?
3. **Which letter** fills the box?

**A**. A B is binary; A B **A** is ternary.

EXAMPLE 2

**Which form?**
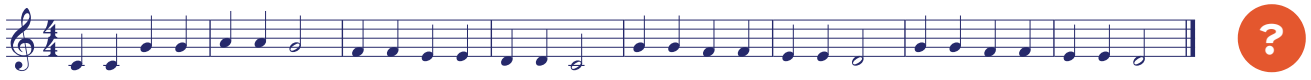



1. **Count the sections.**
2. **Which one keeps coming back?**
3. **What comes between?**

A returns between two new ideas: a **rondo**.

EXAMPLE 3

**Map Twinkle**




1. **Bars 1–4:** call them A.
2. **Bars 5–8:** new or the same?
3. **The last four bars** (not shown) repeat bars 1–4.

Bars 5–8 are new: B. Then A again: **A B A**, ternary.

TRY IT · 10 SECONDS



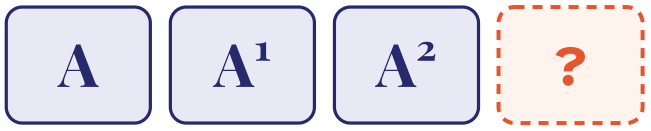

**Map a piece you play.**

Give every 4-bar phrase a letter. Same music, same letter. What form is it?

PART VII · SHAPE AND STYLE · 5 · CONTINUED

EXAMPLE 4

**What comes next?**




1. **A** is the theme.
2. **A¹ and A²**: the same tune, dressed differently.
3. **The next one** is called...?

**A³**: a third variation. Theme and variations keep one tune and change its clothes.

GAME

**Name the form**

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| A B | A B A | A B A C A | A A¹ A² A³ |

Name each map.

Answer: 1 binary · 2 ternary · 3 rondo · 4 theme and variations.

HOW TO

**Learn a piece by its map**

1. **Letter the sections** in pencil.
2. **Learn A first**: it comes back, so it pays twice.
3. **Then the new sections**, and practise the joins between them.

YOU KNOW THIS

**Beethoven, Für Elise**




The famous opening tune is **A**. It is heard three times, with two new sections in between: a rondo.

**Try it.** Listen and raise your hand every time A returns.

THINK OF IT Form is the floor plan of a house: once you know the rooms, you never get lost in it.

TRAP

- × Giving every phrase a new letter: same music, same letter.
- × Counting a repeat as a new section: A played twice is still A.

SEE ALSO

Repeats and road signs p. 71 · Changing key p. 154

PRACTISE IT

Online: Library › Form

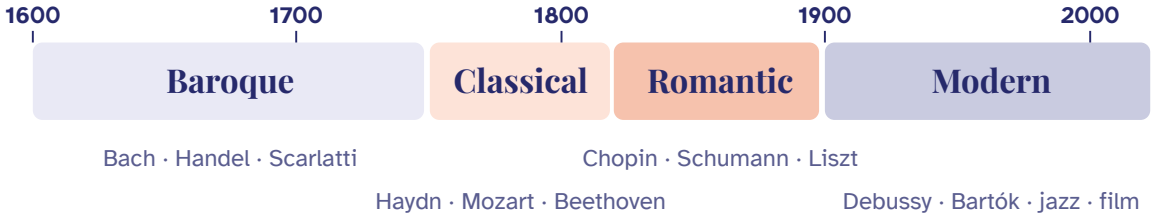
PART VII · SHAPE AND STYLE · 6

# Styles through time

Music changes with its time, like clothes and buildings. Four big periods cover most piano music. Each has a sound you can learn to recognise, and a way it likes to be played.

THE SECRET

**Listen for the texture, the ornaments and the pedal: they tell you the century.**




## *Four centuries, four sounds*

| PERIOD | LISTEN FOR |
|---|---|
| **Baroque** 1600–1750 | harpsichord and organ; two or more tunes woven together; many ornaments; loud or soft in blocks; a steady beat throughout |
| **Classical** 1750–1820 | the new piano; a clear tune with an Alberti bass; tidy 4-bar phrases; crescendos and contrasts |
| **Romantic** 1820–1900 | big feelings; rich chords and the pedal; rubato; the whole keyboard, from very soft to very loud |
| **Modern** 1900–today | new scales and harmonies, surprising rhythms; impressionism, jazz, minimalism, film music |

## Playing in style

| | PEDAL | TIME | TOUCH |
|---|---|---|---|
| **Baroque** | little or none | steady | clear, often detached |
| **Classical** | light | steady | neat, even, singing tune |
| **Romantic** | rich | rubato | deep, legato |
| **Modern** | as marked | as marked | whatever the music asks |

LISTEN FOR IT

**Notebook for Anna Magdalena Bach**
Minuet in G (Petzold)
two lines, a steady dance

**Mozart**
Sonata in C, K. 545
Alberti bass, clear phrases

**Chopin**
Nocturnes
a singing tune, rubato, pedal

PART VII · SHAPE AND STYLE · 6 · CONTINUED

EXAMPLE 1

**Which period?**

**A tune over an Alberti bass, tidy 4-bar phrases** ?

1. **Alberti bass:** which period did it belong to?
2. **Tidy phrases**, question and answer.
3. **So?**

**Classical**: think Mozart.

EXAMPLE 2

**Which period?**

**Two equal tunes, many trills, written for harpsichord** ?

1. **Two tunes woven together**: which texture?
2. **Harpsichord**, before the piano.
3. **So?**

**Baroque**: think Bach.

EXAMPLE 3

**Which period?**

**Rubato, rich pedalled chords, huge leaps across the keyboard** ?

1. **Rubato**: time stretched for feeling.
2. **Rich chords** held in the pedal.
3. **So?**

**Romantic**: think Chopin.

GAME

**Who belongs where?**

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| Bach | Mozart | Chopin | Debussy |

Match each composer to a period.

Answer: 1 Baroque · 2 Classical · 3 Romantic · 4 Modern.

TRAP

- × Pedalling Bach like Chopin: Baroque music needs little or no pedal.
- × Romantic rubato in Mozart: keep Classical time steady.

SEE ALSO
Ornaments p. 88 · The pedals p. 85

PRACTISE IT
Online: Library › Styles

PART VII · SHAPE AND STYLE · TIPS

# When a piece goes nowhere

Every note can be right and the piece still wander. The fix is to see the shape: where the phrases go, what comes back, and where the high point is.

TIP

### What you notice, what to check

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| Every bar sounds the same | **the phrases:** where does each one grow? | p. 148 |
| It feels like a list of notes | **the motif** and its sequences | p. 150 |
| The tune is lost | **the texture:** which line sings? | p. 152 |
| A passage suddenly feels strange | **a new key:** look for the new accidental | p. 154 |
| You lose your place | **the form:** letter the sections | p. 156 |
| It sounds out of style | **the period:** pedal, time and touch | p. 159 |

## Four ways to find the shape

1. **Pencil the phrases**
   Draw a curve over each phrase and mark its high point.
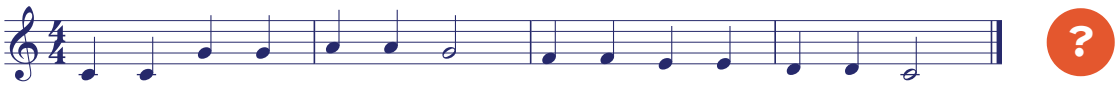
2. **Letter the sections**
   Write A, B, A in the margin. Practise each section on its own.

3. **Sing the tune**
   Sing the top line while you play the rest softly.

4. **Listen to a recording**
   Follow the score and hear the shape before you play it.

EXAMPLE

### Where is the high point?




1. **Sing it.**
2. **Which note** is highest?
3. **Grow to it**, then let the phrase fall.

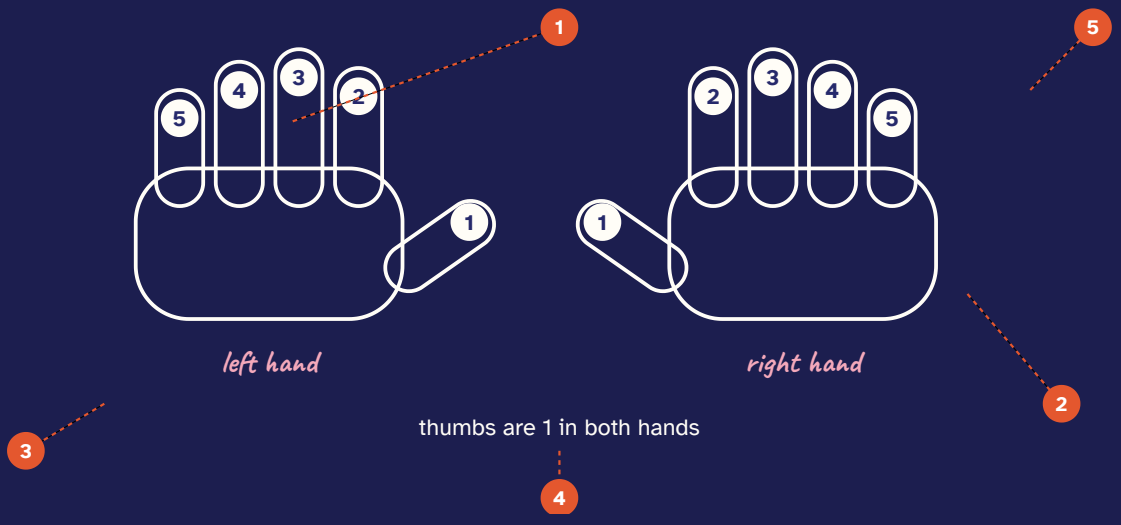
The **As in bar 2**: grow to them, then fall home to C.

THINK OF IT A map does not make the journey, but it stops you getting lost.

PART VIII · AT THE PIANO

# Playing and listening

Everything before this page is knowledge. This part turns it into playing: the right fingers, practice that works, reading at sight, a trained ear, and the calm to perform.




1 **Fingering** p. 163
finger numbers, five-finger positions, thumb under

2 **Practising well** p. 165
slow, small chunks, then put it back together

3 **Sight-reading** p. 167
scan first, keep going, read ahead

4 **Listening** p. 169
clap it back, high or low, major or minor

5 **Performing** p. 171
warm up, start calmly, keep going after a slip

★ **Tips: when practice isn't working** p. 173

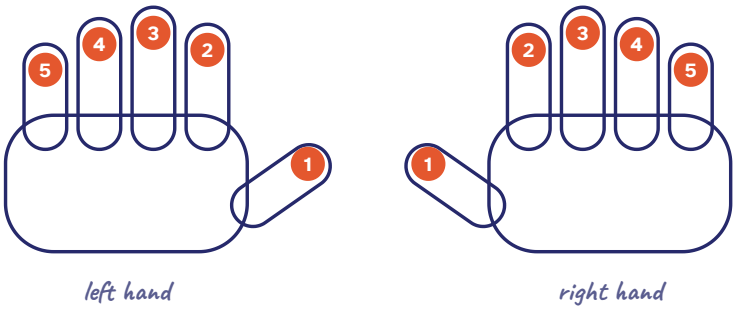
PART VIII · PLAYING AND LISTENING · 1

# Fingering

Each finger has a number, the same in both hands: the thumb is 1, the little finger 5. Good fingering is planned, not guessed: choose it once, write it in, and use it every time.
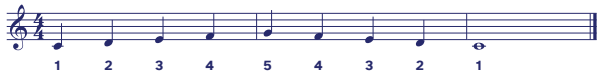
**THE SECRET**

**Thumbs are 1 in both hands. Choose a fingering once, write it in, keep it.**
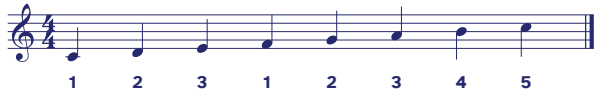



*Thumbs are 1 in both hands*

## Five-finger position
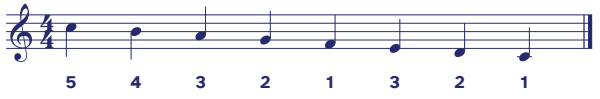



Thumb on C, one finger on each key up to G: five notes without moving the hand. Many first pieces stay in one position.

## Thumb under, finger over




Up the scale: 1 2 3, thumb under to F, then 1 2 3 4 5.




Down: 5 4 3 2 1, then 3 crosses over the thumb.
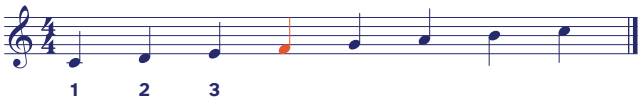
**TWO-MINUTE DRILL** Play C D E F G with each hand, saying the finger numbers: right 1 2 3 4 5 up, left 5 4 3 2 1 up. Then both hands together.

**Left hand, C major up:** 5 4 3 2 1, then 3 crosses over to A, then 2 1. **Down:** 1 2 3, thumb under to F, then 1 2 3 4 5.

PART VIII · PLAYING AND LISTENING · 1 · CONTINUED

EXAMPLE 1

**Which finger plays the F?**




?

1. **C D E** are 1 2 3.
2. **The F:** finger 4, or the thumb tucked under?
3. **Then** how do you reach the top C?

The **thumb** passes under to F; then 2 3 4 5 reach the top C.

EXAMPLE 2

**Which finger plays G?**

**Right hand, thumb on C** ?

1. **Thumb on C** is 1.
2. **Count up** one finger per key: D, E, F, G.
3. **So G** is…?

Finger **5**: C D E F G is 1 2 3 4 5.

GAME
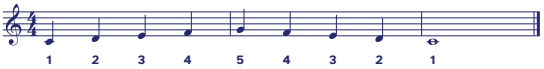
**Name the finger**

| | |
|---|---|
| 1 thumb | 2 index |
| 3 middle | 4 little |

Give the finger number.

Answer: 1 one · 2 two · 3 three · 4 five.

TRY IT · 10 SECONDS




**Five fingers, five keys.**

Right hand on C D E F G. Play up and down, saying the finger numbers aloud. Then the left hand: 5 4 3 2 1 from C up.

TRAP

- × Changing fingering every time you play: choose one, write it in, keep it.
- × Landing heavily on the thumb as it passes under: let it glide.

SEE ALSO

The keyboard p. 22 · Major scales p. 93

PRACTISE IT

Online: Scales with fingering

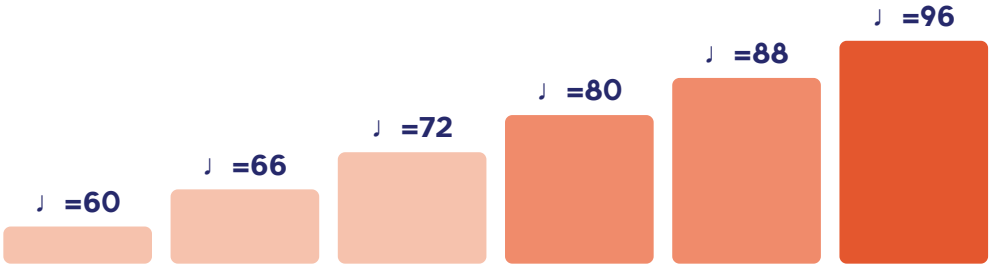
PART VIII · PLAYING AND LISTENING · 2

# Practising well

Practice is not playing a piece again and again from the top. It is fixing the hard bits, slowly and correctly, then joining them back together.

THE SECRET

**Slow, small, and correct; then a little faster; then join the pieces.**




three correct times at one speed, then one step up

*Slow and correct, one step at a time*

## A fifteen-minute plan

| TIME | WHAT YOU DO |
|---|---|
| **2 min** | warm up: a scale or five-finger exercise |
| **5 min** | the hardest bars, slowly, hands separately |
| **5 min** | join the bars, hands together, still slow |
| **3 min** | play the whole section; mark what slipped |

## Four tools

| TOOL | HOW |
|---|---|
| **small chunks** | one bar, or two, plus the first note of the next |
| **hands separately** | each hand alone until it is easy, then together |
| **slow, then faster** | the metronome ladder: three correct times, then one notch up |
| **count aloud** | say the beats: it keeps the rhythm honest |

EXTRA · GOING FURTHER
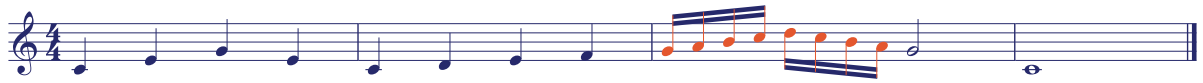
### Practise in rhythms

For a run of even quavers, play it long–short, then short–long, slowly. When you go back to even notes, the run is suddenly easier: every note has been the “long” one.

PART VIII · PLAYING AND LISTENING · 2 · CONTINUED

EXAMPLE 1

**Which bar first?**



1. **Play it through once.** Where did it slip?
2. **The semiquavers** in bar 3: hardest?
3. **Start there**, slowly.

Bar 3, the semiquavers: practise the **hardest bar first**, slowly, then join bars 2–4.

EXAMPLE 2

**When do you speed up?**

♩ =60: right, right, wrong ?

1. **How many correct** in a row do you need?
2. **After a mistake**, start counting again.
3. **Then** one notch up.

Not yet: three **correct** in a row at 60, then try 66.

GAME

**Good habit or not?**

| 1 | 2 |
|---|---|
| from the top | hardest first |
| 3 | 4 |
| faster to fix | slow, correct |

Which help?

Answer: 2 and 4 help; 1 and 3 waste time.

TRY IT · 10 SECONDS

**60 → 66 → 72**

**Climb the ladder.**

Pick one hard bar. Play it three times correctly at a slow speed, then one notch faster. Stop when it slips twice.

HOW TO

**Keep a practice log**

1. **Write the date** and what you practised.
2. **Note one thing** that got better and one that did not.
3. **Start tomorrow** with the one that did not.

THINK OF IT Practice is like building a wall: lay each brick straight and the wall stands on its own.

TRAP

- × Practising mistakes: playing fast and wrong teaches your fingers the wrong notes.
- × Only playing what you already know: spend most time on the hard bits.

SEE ALSO
Speed words p. 82 · Fingering p. 163

PRACTISE IT
Online: Metronome · Practice log
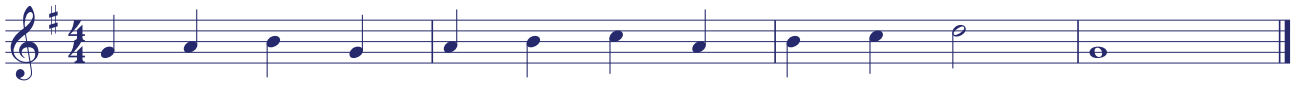
PART VIII · PLAYING AND LISTENING · 3

# Sight-reading

Sight-reading is playing music you have never seen. The secret is a short look first, then keeping the beat going whatever happens: a wrong note passes, a stop is heard by everyone.

**THE SECRET**

**Look before you play, then never stop: the beat matters more than any note.**



Thirty seconds first: one sharp (G major), 4/4, starts on G, steps up in a pattern.
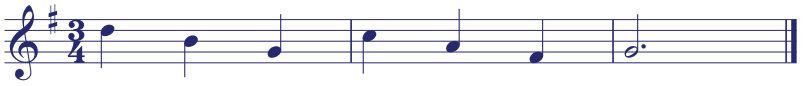
*Look first, then never stop*

## The thirty-second scan

| CHECK | LOOK FOR |
|---|---|
| **1 key** | sharps or flats at the start |
| **2 time** | the time signature: count a bar before you start |
| **3 clefs and notes** | the first note in each hand; the lowest and highest |
| **4 rhythm** | the trickiest rhythm: tap it |
| **5 patterns** | steps, skips, repeats, sequences |
| **6 marks** | dynamics, slurs, speed words |

EXAMPLE 1

**What do you check first?**



? 

1 **The key:** how many sharps? 2 **The time:** how many beats? 3 **The F in bar 2:** F or F♯?

G major, 3/4. The F in bar 2 is **F♯**, from the key signature. Count “1 2 3” and go.

EXAMPLE 2

**A wrong note in bar 2: now what?**

**Stop and fix it, or keep going?** ?
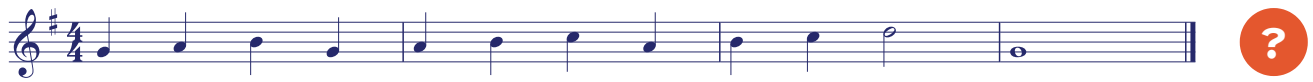
1 **Is the beat** still going? 2 **What does a listener** notice more? 3 **So?**

**Keep going**: catch the next beat. A stop is heard far more than a wrong note.

PART VIII · PLAYING AND LISTENING · 3 · CONTINUED

EXAMPLE 3

### Spot the pattern



1. **Bar 1:** what shape?
2. **Bar 2:** the same shape, where?
3. **So**, how much do you really need to read?

A **sequence**: bar 2 is bar 1 a step higher. Read the shape, not every note.

GAME

### Scan order

1 rhythm

2 key
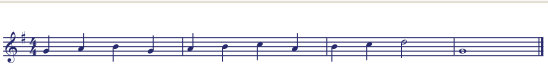
3 time

4 patterns

Put the four checks in the order of the scan.

Answer: key · time · rhythm · patterns.

TRY IT · 10 SECONDS



**One new piece a day.**

Pick something easy: scan for thirty seconds, then play it once, slowly, without stopping. Only then go back.

HOW TO

### Read ahead

1. **While you play** one bar, let your eyes move to the next.
2. **Practise it:** cover each bar with a card as you play it.
3. **Read shapes** (steps, skips, chords), not single notes.

THINK OF IT Sight-reading is like reading aloud: you look a few words ahead, and if you trip on one, the sentence goes on.

TRAP

- × Choosing a speed you cannot keep: sight-read slowly.
- × Looking down at your hands: keep your eyes on the music.
- × Skipping the scan: thirty seconds of looking saves many wrong notes.

SEE ALSO

Reading the notes p. 32 · Repetition and sequence p. 150

PRACTISE IT

Online: Sight-reading trainer

PART VIII · PLAYING AND LISTENING · 4

# Listening

Your ear is a musical instrument too. Train it in small steps: clap a rhythm back, sing a tune back, then hear whether music goes up or down, is major or minor, and whether it has finished.
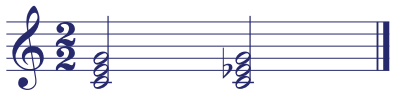
THE SECRET

**Sing it back first: if you can sing it, you can play it.**




*Questions for your ear*

## Major or minor?



C E G, then C E♭ G: bright, then darker.

Only the middle note changes. Major usually sounds bright and open, minor darker or sadder. Sing the middle note of each and hear it drop.

## Intervals by song

| INTERVAL | THE FIRST TWO NOTES OF… |
|---|---|
| major 2nd | Frère Jacques (Frè-re) |
| major 3rd | Oh When the Saints (Oh when) |
| perfect 4th | Amazing Grace (A-ma) |
| perfect 5th | Twinkle (twin-kle **TWIN**: the 2nd to the 3rd note) |
| major 6th | My Bonnie (My Bon-) |
| octave | Somewhere over the Rainbow (Some-where) |

HOW TO
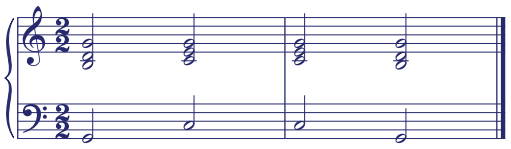
**Clap it back**

1. **Someone claps** a rhythm of two bars.
2. **You clap it back** straight away, keeping the beat.
3. **Then sing back** a short tune, then find it on the keys.

PART VIII · PLAYING AND LISTENING · 4 · CONTINUED

EXAMPLE 1

### Finished or not?

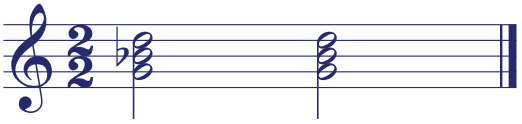


1 **Bar 1** ends on which chord? 2 **Bar 2** ends on which chord? 3 **Which ending** sounds finished?

Bar 1 ends **home** on C (a perfect cadence: finished); bar 2 stops on G (imperfect: not finished).

EXAMPLE 2

### Major or minor?



1 **Play the first chord.** Bright or dark? 2 **The second?** 3 **Which note** changed?

First **minor**, then **major**: only the middle note moved, B♭ up to B.

GAME

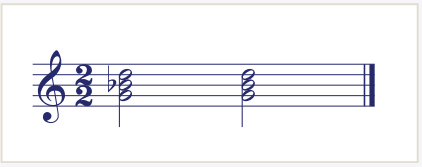
### Which song?

| 1 | 2 | 3 |
|---|---|---|
| 5th up | octave up | 4th up |

Name a song with that leap near the start.

Answer: 1 Twinkle · 2 Somewhere over the Rainbow · 3 Amazing Grace.

TRY IT · 10 SECONDS



### Bright or dark?

Ask someone to play major or minor chords in any order. Say “bright” or “dark” for each. Ten in a row?

TRAP

× Guessing from the hands: close your eyes and listen.

SEE ALSO

Intervals p. 45 · Cadences p. 139

PRACTISE IT

Online: Ear trainer

PART VIII · PLAYING AND LISTENING · 5

# Performing

Playing for others is a skill you can practise, like scales. Nerves are normal: they mean you care. A routine before you start and a plan for slips make them much smaller.

THE SECRET

**Breathe, feel a bar of the beat, then begin. After a slip, catch the next beat.**




*Four steps to the first note*

| WHEN | WHAT YOU DO |
|---|---|
| **Before** | warm up your hands; practise starting, not only the whole piece; play it to a friend or family first |
| **At the piano** | adjust the bench; feet flat; breathe out; feel one bar of the beat in your head |
| **If you slip** | keep the beat going; catch the next beat or the next phrase; never go back |
| **After** | hands in your lap for a moment, stand, bow, smile |

THE RULE

**The listener follows the beat and the mood. Keep both going and a slip is forgotten in a second.**

TWO-MINUTE DRILL Every day this week, play your piece once from the start without stopping, whatever happens, as if someone were listening.

EXTRA · GOING FURTHER

### Playing from memory

Learn it in sections and know where each starts. Memory works best when it is **four kinds at once**: the sound, the feel in your fingers, the look of the page, and knowing the chords and the form.

HOW TO

### Practise performing

1. **Choose three starting places** in your piece and practise starting from each.
2. **Play it through once a day** without stopping, whatever happens.
3. **Play to someone**: a friend, family, a phone camera.

PART VIII · PLAYING AND LISTENING · 5 · CONTINUED

EXAMPLE 1

**How do you start?**

**You sit down.**
**Your heart is racing.** ?

1. **The bench:** right height, right distance?
2. **Your breath:** in, then out slowly.
3. **The beat:** what do you hear first?

Adjust, breathe out, and **feel one bar** of the speed in your head. Then the first note.

EXAMPLE 2

**You slip in bar 5: what now?**

**Bar 5: a wrong note, and your hand freezes** ?

1. **Do you stop** and start again?
2. **Where is the beat** now?
3. **Where can you rejoin?**

Keep counting and come back in on the **next beat**, or at the start of the next phrase. Never go back.

GAME

**Help or not?**

1 going back

2 breathe first

3 keep the beat

4 rush to finish

Which help a performance?

Answer: 2 and 3 help; 1 and 4 do not.

TRY IT · 10 SECONDS

**breathe · beat · play · bow**

**A one-minute concert.**

Choose eight bars. Walk to the piano, sit, breathe, feel a bar, play, bow. Do it for someone today.

THINK OF IT The audience wants you to do well: they are on your side from the first note.

TRAP

- × Starting before you are ready: breathe and feel the beat first.
- × Apologising with your face after a slip: keep going and nobody minds.

SEE ALSO
Practising well p. 165 · Expression p. 76

PRACTISE IT
Online: Practice log

PART VIII · PLAYING AND LISTENING · TIPS

# When practice isn't working

You practise every day and it is not getting better. Usually the practice, not the player, needs fixing: too fast, too long, or always from the top.

TIP

**What you notice, what to check**

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| You stumble in the same place every time | **the fingering:** write one in and keep it | p. 163 |
| It is fine slowly, wrong at speed | **the ladder:** three correct, then one notch up | p. 165 |
| You play it from the top every time | **small chunks**, hardest first | p. 165 |
| New pieces take forever to learn | **the scan:** key, time, rhythm, patterns | p. 167 |
| You cannot hear your own mistakes | **listening:** record yourself, sing it | p. 169 |
| It falls apart in front of people | **a routine** and practising starts | p. 171 |

## Four ways to unstick practice

1. **Halve the speed**
   Whatever speed you use, try half. Correct and slow beats fast and wrong.

2. **Shorten the chunk**
   If two bars fail, try one. If one fails, try one beat.

3. **Take a break**
   Ten short sessions beat one long one. Your brain keeps learning between them.

4. **Ask a question**
   Write down what goes wrong, and take it to your teacher.

EXAMPLE

**Ten minutes left: what do you play?**

**One bar you keep missing, and a whole piece you know** ?

1. **Which needs the time more?**
2. **How many minutes** on the hard bar?
3. **And** the last minutes?

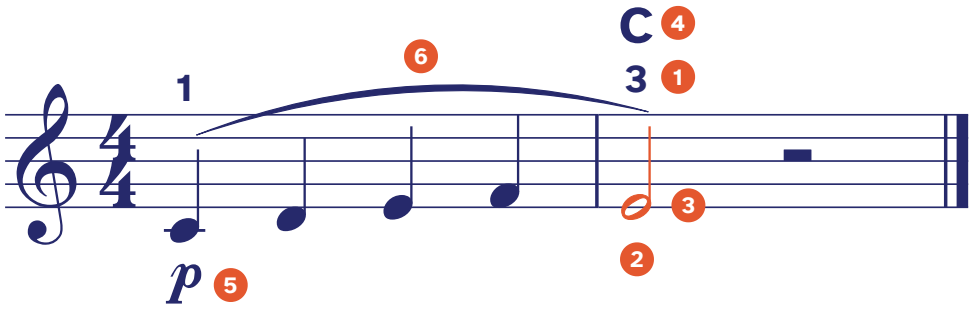
About **seven minutes** on the hard bar, slowly, then play the whole piece once at the end, for joy.

THINK OF IT Practice does not make perfect: perfect practice does. Slow and correct is the fastest way.

PUTTING IT TOGETHER · 1

# Everything one note tells you

Look at the red note. It carries six pieces of information at once, and each one can be read straight off the page. Read all six and the note is right; miss one and it is only nearly right.




**1 The finger number**

**3**: the middle finger plays it. Fingering is planned, not guessed.

PART VIII · PLAYING AND LISTENING · P. 163

**2 The beat**

Beat 1 of bar 2, and it is held for **two beats**.

PART II · TIME · P. 50

**3 The note name**

**E**: the white key just right of the two black keys.

PART I · PITCH · P. 22

**4 The chord**

The **C** above the staff: this E belongs to the C major chord.

PART V · CHORDS · P. 110

**5 The marks**

*p* and the slur: play it softly and smoothly.

PART III · EXPRESSION · P. 76

**6 The phrase**

The slur ends here, so the note lands gently, like the end of a sentence.

PART VII · SHAPE AND STYLE · P. 148

### THE SAME SIX, IN ANY MUSIC

**A CLASSICAL SCORE**
Most of it is written in: finger numbers, dynamics, slurs. Your job is to read every one.

**A POP LEAD SHEET**
A melody and chord names only. You choose the fingers, the marks and the shape yourself.

**A JAZZ CHART**
Chords and a style word such as Swing: the beat and the shape are yours to feel.

Try it on any piece: point to one note and find all six. Finger numbers count as much as the note name.

THINK OF IT A note is like a word said aloud: its spelling, its timing, its tone of voice and its place in the sentence all change what it means.

PUTTING IT TOGETHER · 2

# When a passage won't come right

Name the problem first, then fix that one thing. Most hard bars are hard for one reason, and the six things a note tells you point straight to it.

TIP

### What you notice, what to check

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| You stumble in the same place every time | **the finger number** | Part VIII · Playing and listening |
| It rushes, drags or the counting slips | **the beat** | Part II · Time |
| A note sounds wrong | **the note name** and the key | Part I · Pitch, Part IV · Scales and ke |
| A chord sounds odd or thin | **the chord** | Part V · Chords |
| It is correct but flat and dull | **the marks** | Part III · Expression |
| It does not seem to go anywhere | **the phrase** | Part VII · Shape and style |

## Six ways to fix a hard bar

1. **Slow it right down**
   Slow enough to be right three times in a row, then a little faster.
2. **Hands separately**
   Each hand on its own until it is easy, then together, slowly.
3. **Start from the end**
   Play the last beat, then the last two, then three: you always finish on what you know best.
4. **Practise the join**
   The last note of one bar and the first of the next: most slips live there.
5. **Climb the metronome**
   Up four clicks each time it is clean; down eight if it falls apart.
6. **Put it back**
   Play two bars before and two after, so the fixed bar lives inside the music again.

## Before you play a new piece: a thirty-second look

**Key signature**
Which sharps or flats, all the way through?

**Time signature**
How many beats in a bar, and which note is one beat?

**Tempo and mood**
The words at the top: how fast, what character?

**Patterns**
Bars that repeat or nearly repeat: learn them once.

THINK OF IT Like a doctor: name the symptom first, then treat that one thing. Practising everything harder rarely fixes the one bar that is wrong.

GLOSSARY

# Glossary

Every musical word in the Handbook, in one line, with the page where it is explained.

**8va (ottava)** play the notes under the dotted line an octave higher p.36

**a tempo** back to the speed before p.82

**accelerando (accel.)** gradually faster p.82

**accent** a > over a note: start it more strongly p.79

**acciaccatura** a tiny note with a slash, crushed in as fast as possible p.88

**accidental** a sharp, flat or natural written before a note; it lasts until the bar line p.39

**Adagio** slow, at ease p.82

**Alberti bass** a broken chord in the left hand: low, high, middle, high (C G E G) p.152

**Allegro** fast and lively p.82

**anacrusis** an upbeat: notes before the first full bar p.65

**Andante** at a walking pace p.82

**appoggiatura** a tiny note without a slash; it leans, taking half the main note's time p.88

**articulation** how notes are joined or separated: slurs, staccato, tenuto, accents p.79

**augmented triad** a triad with 3 + 3 keys between p.113

**bar** a group of beats between two bar lines p.50

**Baroque** music of about 1600–1750: Bach, Handel p.159

**bass clef** the F clef: its two dots hug the line for bass F p.27

**beam** a thick line joining quavers or semiquavers into a beat p.65

**beat** the steady pulse you count p.50

**binary form** two sections, A B p.156

**broken chord** the notes of a chord played one after another p.142

**cadence** the last two chords of a phrase: musical punctuation p.139

**cantabile** in a singing style p.79

**chord** three or more notes played together p.110

**circle of fifths** the keys in a circle: go right, one more sharp; go left, one more flat p.105

**Classical** music of about 1750–1820: Haydn, Mozart p.159

**clef** the sign at the start of the staff that names one line p.27

**coda** an ending section; jump to it at "To Coda" p.71

**compound time** each beat is a dotted note and divides into three (6/8, 9/8, 12/8) p.62

**crescendo (cresc.)** getting louder, little by little p.76

**crotchet** a note worth one beat in 4/4 p.53

**da capo (D.C.)** "from the head": go back to the beginning p.71

**dal segno (D.S.)** "from the sign": go back to the segno p.71

**diminished triad** a triad with 2 + 2 keys between p.113

**diminuendo (dim.)** getting softer, little by little p.76

**dolce** sweetly p.79

**dominant** the 5th note of the scale, and the chord built on it (V) p.116

**dot** adds half the note's value p.56

**duplet** two notes in the time of three p.65

**dynamics** the signs for loud and soft: pp p mp mf f ff p.76

GLOSSARY · CONTINUED

**enharmonic** two names for one key, such as F♯ and G♭ p.39

**fermata** a pause: hold the note longer than written p.82

**Fine** "the end": after a D.C. or D.S., stop here p.71

**five-finger position** one finger on each of five keys in a row p.163

**flat (♭)** the very next key to the left p.39

**form** the plan of a whole piece, in letters p.156

**forte (f)** loud p.76

**grand staff** treble and bass staves joined by a brace p.27

**hairpin** a crescendo or diminuendo sign that opens or closes p.76

**hemiola** two bars of three felt as three bars of two p.68

**homophonic** a tune with chords underneath p.152

**imitation** another hand or voice copies the idea p.150

**interval** the distance between two notes p.42

**inversion** a chord with a note other than the root at the bottom p.119

**key** the scale a piece is built on, and its home note p.99

**key signature** the sharps or flats at the start of every line p.99

**Largo** very slow and broad p.82

**leading note** the 7th note of the scale, one key below home p.93

**ledger line** a short line for a note above or below the staff p.36

**legato** smoothly joined p.79

**major scale** tone, tone, semitone, tone, tone, tone, semitone p.93

**major triad** a triad with 3 + 2 keys between p.113

**metronome mark** ♩ = 76: 76 beats in a minute p.82

**mezzo (m)** "half": mp half soft, mf half loud p.76

**Middle C** the C nearest the middle of the keyboard, on a ledger line between the staves p.22

**minim** a note worth two crotchets p.53

**minor scale** the major scale on the same note with 3, 6 and 7 one key lower p.96

**minor triad** a triad with 2 + 3 keys between p.113

**Moderato** at a moderate speed p.82

**modulation** changing key during a piece p.154

**monophonic** one line alone p.152

**mordent** the note, its neighbour above (or below), the note p.88

**motif** a short idea of 2 to 6 notes that comes back p.148

**natural (♮)** cancels a sharp or flat p.39

**octave** from one note to the next of the same name: 11 keys between p.45

**ornament** a small sign for a quick group of notes around the main note p.88

**pedal (sustain)** the right pedal: notes keep ringing until you lift it p.85

**perfect cadence** V to I: a full stop p.139

**phrase** a musical sentence, often 4 bars p.148

**pianissimo (pp)** very soft p.76

**piano (p)** soft p.76

**plagal cadence** IV to I: the "Amen" p.139

**polyphonic** two or more tunes at once p.152

**Presto** very fast p.82

**progression** a trip from chord to chord p.144

**pulse** the steady beat underneath the music p.50

**quaver** half a crotchet p.53

GLOSSARY · CONTINUED

**rallentando (rall.)** gradually slower p.82

**relative minor** the minor key with the same key signature, a minor 3rd below p.96

**repeat sign** two dots and a double bar: play the bars again p.71

**rest** a silence of a set length p.53

**ritardando (rit.)** gradually slower p.82

**Romantic** music of about 1820–1900: Chopin, Schumann p.159

**rondo** A B A C A: the first idea keeps coming back p.156

**root** the note a chord is built on and named after p.110

**rubato** time stretched and given back, for expression p.68

**scale** the notes of a key in order, up or down p.93

**semibreve** a whole note: four crotchets p.53

**semiquaver** half a quaver p.53

**semitone** the very next key: no key between p.22

**sequence** the same shape repeated higher or lower p.150

**seventh chord** a triad with one more 3rd on top p.122

**sforzando (sfz)** one note made suddenly strong p.76

**sharp (♯)** the very next key to the right p.39

**sight-reading** playing music at first sight p.167

**simple time** each beat divides into two (2/4, 3/4, 4/4) p.59

**slash chord** C/E: the chord, then the bass note p.131

**slur** a curve over different notes: join them p.79

**staccato** a dot: short, detached p.79

**staff** the five lines and four spaces music is written on p.27

**subdominant** the 4th note of the scale, and the chord built on it (IV) p.116

**swing** pairs of quavers played long–short p.68

**syncopation** the stress falls off the beat p.68

**tempo** the speed of the beat p.50

**tenuto** a line: hold the note its full length p.79

**ternary form** A B A: out, away and back p.156

**texture** how many lines sound at once, and how they fit p.152

**theme and variations** one tune, changed each time it returns p.156

**tie** a curve joining two notes of the same pitch: one longer sound p.56

**time signature** top: how many beats; bottom: which note gets one beat p.59

**tone** two semitones: one key between p.22

**tonic** the 1st note of the scale: home p.93

**treble clef** the G clef: it curls round the line for treble G p.27

**triad** three notes stacked: play one, skip one, play one p.110

**trill** the note and its neighbour above, back and forth quickly p.88

**triplet** three notes in the time of two p.65

**turn** above, the note, below, the note p.88

**twelve-bar blues** I I I I · IV IV I I · V IV I I p.144

**una corda (u.c.)** the left (soft) pedal p.85

**upbeat** see anacrusis p.65

INDEX

# Index

Where each word appears. The first page is where it is explained.


**8va (ottava)** 36, 37
**a tempo** 82, 83
**accelerando (accel.)** 82, 83
**accent** 79
**acciaccatura** 88
**accidental** 39, 155
**Adagio** 82
**Alberti bass** 143, 152, 159, 160
**Allegro** 82
**anacrusis** 65
**Andante** 82
**appoggiatura** 88, 89
**arpeggio** 110, 111
**articulation** 79
**augmented triad** 113
**bar** 50, 51, 53, 55, 57
**Baroque** 156, 159, 160
**bass clef** 27
**beam** 53, 65, 66
**beat** 50, 51, 53, 55, 56
**binary form** 156
**block chords** 152
**borrowed chord** 136
**broken chord** 142, 152
**C clef** 27
**cadence** 139, 140, 146
**cantabile** 79, 80
**chord** 110, 111, 113, 116, 117
**circle of fifths** 105
**Classical** 159, 160
**clef** 27, 33, 35, 48
**coda** 71, 73
**common time** 59
**compound time** 62, 63
**crescendo (cresc.)** 76, 77, 159
**crotchet** 53, 55, 57, 59, 60
**cut common time** 59
**da capo (D.C.)** 71
**dal segno (D.S.)** 71
**degree names** 93
**diminished triad** 113
**diminuendo (dim.)** 76, 77
**dolce** 79, 80
**dominant** 116
**dot** 56, 57
**duplet** 65
**dynamics** 76, 91
**enharmonic** 39, 41
**fermata** 82, 83, 91
**Fine** 71, 72, 173
**first and second time bars** 71
**five-finger position** 163
**flat (♭)** 39, 40, 41, 48
**form** 156, 157, 161
**forte (f)** 76, 77
**grand staff** 27
**hairpin** 76, 77
**half pedal** 85, 86
**harmonic minor** 96, 97
**hemiola** 68
**homophonic** 152, 153
**imitation** 150
**imperfect cadence** 139, 140
**interrupted cadence** 139, 140
**interval** 42, 43, 48
**inversion** 119, 120, 131, 134
**irregular time** 62
**key** 93, 94, 95, 97, 99
**key signature** 39, 99, 104, 108, 154
**Largo** 82
**leading note** 93, 116
**ledger line** 36, 37, 48
**legato** 79, 80
**major scale** 93, 95, 145
**major triad** 113
**melodic minor** 96
**metronome** 50, 83, 165
**metronome mark** 50, 82, 83
**mezzo (m)** 76
**Middle C** 22, 23, 24, 25, 27
**minim** 53, 55, 57, 59, 60
**minor scale** 96
**minor triad** 113, 114
**Moderato** 82
**modulation** 154
**monophonic** 152
**mood words** 79, 80
**mordent** 88, 89
**motif** 148, 149, 161
**natural (♮)** 39, 41
**octave** 23, 37, 45, 48
**ornament** 88, 89, 91
**pedal (sustain)** 85, 86, 91
**perfect cadence** 139, 140
**performing** 171, 172
**phrase** 148, 149, 159, 160, 161
**phrase mark** 79, 80
**pianissimo (pp)** 76
**piano (p)** 29, 76, 77, 159, 171
**plagal cadence** 139

INDEX · CONTINUED

**polyphonic** 152
**power chord** 125, 126
**practising** 165
**Presto** 82
**progression** 144
**pulse** 50, 74
**quaver** 53, 57, 59, 62, 63
**rallentando (rall.)** 82, 83
**relative minor** 96, 99, 100, 154
**repeat sign** 50, 71, 72
**rest** 53, 55, 60, 74
**ritardando (rit.)** 82, 83
**Roman numerals** 116
**Romantic** 156, 159, 160
**rondo** 156, 157
**root** 110, 114, 119, 120, 123
**rubato** 68, 69
**scale** 93, 95, 96, 108
**semibreve** 53
**semiquaver** 53, 59, 66, 166
**semitone** 22, 39, 42
**sequence** 150, 151
**seventh chord** 122, 134
**sforzando (sfz)** 76, 77
**sharp (♯)** 39, 40, 48
**sight-reading** 167
**simple time** 59, 62
**slash chord** 131, 132
**slur** 79, 80, 91
**sonata form** 156, 157
**sostenuto** 85, 86
**staccato** 79, 91
**staff** 27, 28, 29, 32, 34
**strophic** 156
**subdominant** 116
**sus chord** 125, 134
**swing** 68, 69
**syncopation** 68
**tempo** 50
**tenuto** 79
**ternary form** 156
**texture** 152, 153, 161
**theme and variations** 156
**thumb under** 163
**tie** 56, 57
**time signature** 50, 51, 59, 60, 62
**tone** 22, 42
**tonic** 93, 116
**treble clef** 27
**triad** 110, 111, 113, 114
**trill** 88, 89
**triplet** 65, 66
**turn** 88, 89, 91
**twelve-bar blues** 144, 145
**two-note slur** 79, 80
**una corda (u.c.)** 85
**verse and chorus** 156